



OM-D (US)

Revision 7
Effective Date 07.01.2026

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For Reference Only

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Approval Sheet
for



Issuance No.: 0
Revision No.: 7
Dated: 08.01.2026

Reference Pages: see List of Effective Articles

Approval Sheet:
This sheet is applicable for

Date/Signature	Date/Signature
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Record of Revision (RoR)

Version	Date	Entered by	Highlight
R7	07.01.2026 12:24	Zachary Leggett	
R6	29.09.2025 21:49	Paul Russo	
R5	26.03.2025 10:08	Zachary Leggett	
R4	02.01.2025 12:48	Zachary Leggett	
R3	20.11.2024 10:20	Johann Sweitzer	
R2	01.11.2024 12:04	Zachary Leggett	
R1	10.06.2024 20:25	Paul Russo	
R0	27.03.2023 11:36	Marc Scoville	

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List of affected Articles

	Subject	Modified By	Modify Date
NEW	2.3.1 Line Experience Module (LEM)		02.12.2025
NEW	2.3.1.1 Purpose		18.12.2025
NEW	2.3.1.2 Objectives		02.12.2025
NEW	2.3.1.3 Scheduling and Applicability		18.12.2025
NEW	2.3.1.4 Ground Training Elements		18.12.2025
NEW	2.3.1.5 Flight Training Elements		18.12.2025
NEW	2.3.1.6 Complexity		02.12.2025
NEW	2.3.1.7 LEM Structure		18.12.2025
NEW	2.3.1.8 Scenario Development		18.12.2025
NEW	2.3.1.8.1 Examples of LEM Scenarios		02.12.2025
NEW	2.3.1.8.1.1 Scenario Template A – Offshore Hoist Operation		02.12.2025
NEW	2.3.1.8.1.2 Scenario Template B – Class D Load to Vessel		02.12.2025
NEW	2.3.1.8.1.3 Scenario Template C – Night SAR Mission (NVG)		02.12.2025
NEW	2.3.1.8.1.4 Scenario Template D – Offshore IFR Day/Night Crew Change		02.12.2025
NEW	2.3.1.9 Evaluation and Feedback		02.12.2025
NEW	2.3.1.10 Pilot Training Recordkeeping		02.12.2025
NEW	3.8.3 Differences Training Single-Pilot vs. Dual-Pilot Operations		20.11.2025
NEW	3.8.3.1 Module 1 – Regulatory Overview and Flight Crew Roles		20.11.2025
NEW	3.8.3.2 Module 2 – Crew Resource Management (CRM) and Communication		20.11.2025
NEW	3.8.3.3 Module 3 – Normal Procedures Comparison		20.11.2025
NEW	3.8.3.4 Module 4 – Abnormal and Emergency Procedures		20.11.2025
NEW	3.8.3.5 Module 5 – Aircraft Systems and Equipment Considerations		20.11.2025
NEW	3.8.3.6 Module 6 – Flight Scenarios		18.12.2025



NEW	3.8.3.7 Module 7 – Evaluation		20.11.2025
MODIFIED	6.1.3 TF 03 - Evaluation Form		07.01.2026

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For Reference Only



0 Administration and Control

0.1 Log of revision / document versions

[14 CFR § 135.325(a)(1) and 135.325]

HeliService USA maintains all it's manuals on Fleetplan, a digital manual writing and editing system.

1. Current revision:
 - a. The published manuals in Fleetplan shall be the most current revision approved by the CHDO. The CP and/or DO are the only ones with the ability to publish an update to the OM-D in Fleetplan.
 - b. From the digital version, all revision history and changes can be seen.
 - c. All FAA inspectors assigned to the certificate will be given a password and access to Fleetplan in order to view all manuals. If requested a PDF version shall be provided. See below for distribution of PDF versions incase of browser or connectivity issues.
2. Editing & Revision:
 - a. If requested, HeliService USA shall always send a PDF copy of manual revisions to the CHDO for approval.
 - b. Additionally, the Fleetplan system allows for regulatory oversight and revision approval in the Fleetplan manual system.
 - c. This method is as follows and is the preferred method.
 - i. All CHDO personnel have a password to access Fleetplan.
 - ii. They are part of the approval chain internally to Fleetplan.
 - iii. The DO shall submit the changes, which arrives in the CHDO's fleetplan inbox.
 - iv. They can either approve or decline the revisions made.
 - v. The CHDO personnel may add comments to any rejected revisions.
 - vi. If disapproved, they are returned to the CP/DO's inbox where the appropriate adjustments can be made.
 - vii. This process continues until acceptable revisions have been approved.

Note: Due to the certificate transfer and extensive manual revision and incorporating updated operating procedures, training programs and updated helicopter operations, all previous manuals written by Scottsdale Helicopters have been superceded by HeliService USA manuals beginning at Rev 0.

0.2 Approval

[14 CFR § 135.325(a)(1) and 135.325]

IAW 14 CFR § 135.323(a)(1) and 135.325, HeliService USA will obtain initial and final FAA approval for the training program and subsequent revisions. No training shall be conducted until approval is granted. The current FAA approval documentation will be kept at the front of this manual.

0.3 Abbreviations, terms, and definitions

The following terms are used throughout this training manual and are defined as follows:

Training Program:

A system of instruction which includes curricula, facilities, instructors, check pilots, courseware, instructional delivery methods and testing and checking procedures. This system must satisfy the training program requirements of FAR Part 135 and ensure that each crewmember remains adequately trained for each aircraft, duty position and kind of operation in which the person serves.

Curriculum:

A complete training agenda specific to an aircraft type and a crewmember duty position. Each curriculum consists of several curriculum segments.

Curriculum Segment:

An integral phase of a curriculum which can be separately evaluated and individually approved but by itself does not qualify a person for a crewmember duty position. Each curriculum segment consists of one or more training modules.

Training Module:

An integral part of a curriculum segment, which contains descriptive information, elements, or events which, relate to a specific subject. For example, a ground training curriculum segment must have training modules (composed of "elements") pertaining to aircraft systems (hydraulic, pneumatic, electrical, etc.). As another example, a flight training curriculum segment must have training modules (composed of "events") pertaining to normal, abnormal and emergency procedures. A training module includes the outline, appropriate courseware and the instructional delivery methods. It is usually completed in a single training session.

Courseware:

Instructional material developed for each curriculum. This is information in lesson plans, instructor guides, computer software programs, audiovisual programs, workbooks, aircraft operating manuals and handouts. Courseware must accurately reflect curriculum requirements, be effectively organized and properly integrate with instructional delivery methods.

Instructional Delivery Methods:

Methodology for conveying information to a student. For example, this may include lectures, demonstrations, audiovisual presentations, home study assignments, workshops and drills. Training devices, simulators, aircraft and computer work stations are also considered instructional delivery methods.

Eligibility Period:

Eligibility Period (Base Month): The three (3) calendar month period defining Training/Checking Months. HeliService USA will further define this period as:

1. Early Grace Month-The calendar month before the training/checking month.



2. Base Month -The training/checking month.
3. Late Grace Month - The calendar month after the training/checking month.

Note: During this period an HSUSA pilot must receive recurrent training, a flight check, or a competency check, to remain in a qualified status. Training or checking completed during the eligibility period, is considered to be completed during the "training/checking month."

Categories of training:

Courses of training which provide the necessary training and checking or testing for various types of crewmembers who have not previously qualified (or have or will become unqualified) to serve unsupervised in specific duty positions. Each category of training consists of one or more curriculums.

Initial New-Hire:

This training category is for Pilots who have no previous experience with HeliService USA (newly hired). Initial New-Hire Training includes basic indoctrination training and training for a specific duty position and aircraft type. Except for a basic indoctrination curriculum segment, the regulatory requirements for initial new-hire (INH) and initial equipment (IE) training are the same. Since initial new-hire training is usually the employee's first exposure to specific HSUSA methods, systems, and procedures, it is the most comprehensive of the six categories of training. For this reason, initial new-hire training is a distinct, separate category of training and should not be confused with initial equipment training. Initial equipment training is a separate category of training.

Transition Training:

This category of training is for personnel who have been previously trained and qualified for a specific duty position by the operator and are being re-assigned to the same duty position on a different aircraft type.

Recurrent Training:

This category of training is for personnel who have been trained and qualified by the operator and will continue to serve in the same duty position and aircraft type and must receive recurring training or checking within an appropriate eligibility period to maintain currency.

Re-qualification Training:

This category of training is for personnel who have been trained and qualified by the operator, but have become unqualified to serve in a particular duty position and/or aircraft, due to the following:

1. Not receiving recurrent training and/or, a required flight or competency check, within the appropriate eligibility period.
2. An unsatisfactory check flight.

Initial Equipment Training: This category of training is for personnel who have been previously trained and qualified in a duty position by the operator in an aircraft and are being reassigned to another duty position on a different aircraft type which they were not previously trained and qualified by the operator for that duty position and aircraft type.



1 General

1.1 Introduction

This OM-D describes and sets forth the requirements and standards for the HeliService USA Part 135 Training Program. All references within this manual to HeliService USA, HSUSA, Certificate Holder, Operator or the Company are deemed to mean HeliService USA LLC. This manual is regulatory and directive in nature and compliance by HeliService USA personnel is mandatory.

1.2 Applicability

This document sets forth the standards and requirements for the establishment and maintenance of an approved training program for crew members, check airmen, flight instructors and other applicable operations personnel employed by or under contract to HeliService USA. All crewmembers shall complete all required ground, flight, and CRM training listed in this manual prior to being assigned duties.

All training will be conducted at the direction of the Chief Pilot. The Chief Pilot can, if he so chooses, name a Check Airmen as a designated representative / Training Manager. The Chief Pilot or his designated representative will be responsible for the administration of all tests, written or oral.

1.3 Training Program Overview

[14 CFR § 135.293(e)]

Upon completion of the program, by correctly answering 80 percent of the questions (corrected to 100 percent) on written and/or oral tests, pilots will be able to successfully demonstrate their knowledge of the regulations, policies, and procedures applicable to 14 CFR Part 135 operations. Pilots will be able to fly the aircraft/simulator in a manner that confirms they are obviously the master of the aircraft, with the successful outcome of each maneuver never in doubt. They will be able to safely operate the aircraft in assigned airspace.

1.4 How to Use This Manual

1. Find relevant training curriculum for a given student in Chapter 3 of this manual.
 - a. Confirm the appropriate training objective/curriculum are correct.
 - b. Confirm prerequisites are met.
2. Reference training curriculum table for applicable curriculum segments.
 - a. Located in this table is the total planned hours for each curriculum segment.
 - b. Also located in the table is the required modules for each curriculum segment.
 - c. Follow the sequence of the table (top segment first, etc.).
3. Reference Chapter 4 of this manual for appropriate modules.
 - a. Utilize chart in Chapter 1.5 and select best instructional delivery method for each module.



- b. Use modules to guide course of instruction and ensure each of the required subjects listed in the module are covered.
- c. Log completion of each module on the appropriate form. Signing off of each module is confirming each subject in the module has been covered in its entirety.
- d. Signing off the entire curriculum segment is confirming the total number of programmed training hours were completed. If proficiency was attained with less hours than required, annotate the total hours completed.

1.5 Facilities, Courseware, and Instructional Delivery Methods

1. HeliService USA classrooms are equipped for presentation of audiovisual media. Additionally, while facilities are under construction or renovation, HeliService USA will as a lessee of FlightLevel Aviation, have access to their conference rooms for instruction or another facility that provides a conducive learning environment.
2. Instructional delivery methods are used in various combinations to convey information to trainees during Ground and Flight Training. Instructional delivery methods include lectures, audiovisual presentations, home study assignments, procedures training in cockpit mock ups, demonstrations, performance of drills, computer based training (CTS, AVSTAR, etc.) and the use of simulators and/or aircraft.
3. Courseware is used in various combinations to aid the presentation of information during Ground and Flight Training. Courseware includes overhead viewgraphs, slide presentations, various handouts/other training material referenced within this manual, videos, and lesson plans / syllabuses.

1.5.1 Distance Learning / Computer Based Training (CBT)

1. The FAA has defined distance learning as training accomplished by any training method not including an instructor and a gathering of trainees collocated in a traditional classroom. (Distance learning is known by other terms such as E-learning, home study, self-guided training, virtual classroom, distributed training, computer-based training (CBT), Web-based training, and others). HeliService USA will utilize these methods.
2. HeliService USA will also utilize CBT to train selected training modules within the HSUSA Training Program.
 - a. When approved by the FAA CH, HSUSA will contract with Computer Based Training (CBT) vendors to support all categories of training. Companies such as Computer Training Systems, Inc. (CTS) and Avstar are companies that may be hired to provide the training.
3. Once approved, HeliService USA Personnel will receive a unique ID and Password to log on to the required training session. Each module will consist of two components, learning and testing. The computer based system will track scores and maintains each pilot record. The Director of Operations or Chief Pilot, will be responsible for CBT content selection while HSUSA Check Pilot and Instructors will be responsible to certify training module qualification. Products from CTS and Avstar provide comprehensive tracking and record tools.



4. Curriculum Segment Modules that are based on CBT will be identified with a notation: (CBT). CBT Modules may be taught using traditional classroom methods as an approved alternative.

1.5.2 Instructional Delivery Methods/Courseware Table

Note: Instructors will use the most suitable instructional delivery methods and Courseware available to conduct training. A specific list of the applicable instructional delivery methods and courseware used for each curriculum segment is shown in the following table.

Curriculum Segment	IDM	CW
Basic Indoctrination Training - Operator Specific	A,B,C,G,E	1,2,3,5
Basic Indoctrination Training - Airman Specific	A,B,C,G	1,2,3,5
Crew Resource Management (Human Factors)	A,B,C,G	1,2,3,5
Aircraft Specific - Emergency Drill Training	A,E,F,I,G	3,5
Emergency Situation Training	A,B,C,E,G	1,2,3,5
Hazardous Material Training	A,B,C,G	1,3,5
Special Segment Training	A,B,C,G,H,I	1,2,3,4,5
Aircraft Specific - General Operational Subjects Training	A,C,G	3,5
Aircraft Specific - Aircraft Systems Training	A,C,D,G	1,2,3,5
Aircraft Specific - Systems Integration Training	A,D,H,I	3,5
Aircraft Specific - Flight Training	A,E,F,H,I	3,5
Aircraft Specific - Differences Training	A,C,D,G,H,I	3,5
Instructional Delivery Methods Legend		Courseware Legend
A. Lecture	1. Overhead Viewgraphs	
B. Audiovisual	2. Slide Presentations	
C. Home Study	3. Handout/Training Manuals	
D. Cockpit Mock up	4. Video	
E. Demonstration	5. Lesson Plans/Syllabi	
F. Performance of Drills		
G. Computer Based Training		
H. Simulators		
I. Aircraft		

1.5.3 Authorized Instructor Credit



A HeliService USA Instructor or Check Pilot who conducts a classroom subject within a ground curriculum segment, training module, a complete course of ground training, or emergency drills required within this approved training program will be considered to have completed that subject, course, or drill for their own training requirement. Another instructor must still certify their proficiency.

1.5.4 Adequate Number of Instructors and Check Pilots

[14 CFR § 135.323(a)(2)(4)]

HeliService USA will provide an adequate number of qualified instructors and check pilots to meet the requirements of the training program as dictated in 14 CFR § 135.323(a)(2) and (4). The analysis of required staffing will be an ongoing process undertaken by the Managing Director, CP, DO, and Training Manager.

1.5.5 Flight Training

For purposes of this training program, Flight Training means training conducted in a flight training device, flight simulator, aircraft, or in a combination of the above. Any training or checking unable to be completed in a flight training device or simulator may be accomplished in an aircraft. Training may be conducted by HeliService USA or the contract training partner.

1.5.6 Aircraft Training Using Flight Simulator Modules

When aircraft training is conducted, flight simulator modules will be considered flight training modules. Flight training events deemed unsafe or impractical if attempted in actual flight will be simulated in the aircraft to the extent possible.

All simulators used for training or checking must meet the requirements of 14 CFR § 135.335(b):

1. It must be specifically approved for:
 - a. The certificate holder; and
 - b. The particular maneuver, procedure, or crewmember function involved.
2. It must maintain the performance, functional, and other characteristics that are required for approval.
3. Additionally, for aircraft simulators, it must be:
 - a. Approved for the type aircraft and, if applicable, the particular variation within type for which the training or check is being conducted; and
 - b. Modified to conform with any modification to the aircraft being simulated that changes the performance, functional, or other characteristics required for approval.

1.5.6.1 Approved Simulators & Differences



1.5.6.1.1 Assessment of difference

HeliService USA will utilize the following simulators as part of their training program. The statement of qualification from the FAA national simulator program (NSP) shall be submitted and the simulator approved for usage in HeliService USA's training program by the CHDO prior to use.

The following shall be evaluated and if applicable, submitted to the CHDO along with the NSP statement of qualification:

1. Any differences between the simulator to be utilized and HeliService USA's aircraft.
2. Any limitations in performing required maneuvers IAW the 8900.1 Ch 19 V3 Sec 6 Table 3-68.

1.5.6.1.2 AW169

1. Leonardo-Philadelphia Training Academy, 3050 Red Lion Rd. Philadelphia, PA 19114
 - a. Aircraft: AW169
 - b. Device number: 1772
 - c. Qualification Level: Helo FFS - D
 - d. Expiration date: 4/30/25
 - e. Differences: None

1.6 Programmed Training Hours

This training program uses planned training hours to allow for flexible training based upon demonstrated pilot proficiency and knowledge. Completion of a module, drill, subject, or any other training is dictated by an instructor certifying proficiency to the required standard, not a specific amount of training hours having been completed.

As such, a reduction in planned training hours is allowed and may occur at any time when the instructor believes that the individual has reached the required level of proficiency during his or her scheduled training, provided that all elements for ground training and all events for flight training have been successfully trained. Conversely, if a pilot fails to meet the training objectives within the planned training hours, additional time may be added until the pilot can demonstrate the required pilot proficiency and knowledge.

1.7 Certification of Checking and Training

Following completion of any particular ground training subject, segment of flight training, course of training, flight check, competence check, or observation, the responsible instructor or check pilot shall endorse the pilot's record as such to certify proficiency and knowledge.

This requirement for ALL pilots, instructors, and check pilots is met by endorsing the HeliService USA training forms in section 6 of this manual.



1.8 Qualification Requirements

[14 CFR § 135.337(b)(c), and 135.338(b)(c)]

1. 14 CFR § 135.337 - Check Airmen (Simulator) and (Aircraft):
Check Airmen used in this training program must meet all requirements of 14 CFR § 135.337(b) or 135.337(c).
2. 14 CFR § 135.338 - Flight Instructors (Simulator) and (Aircraft):
Flight Instructors used in this training program must meet all requirements of 14 CFR § 135.338(b) or 135.338(c).
3. 14 CFR § 135.338 - Ground instructors:
Ground Instructors used in this training program must be nominated and approved by the Chief Pilot or Director of Operations. The following are automatically considered nominated and approved by the Director of Operations / Chief Pilot.
 - a. Managing Director
 - b. Director of Operations (Director of Operations)
 - c. Chief Pilot (Chief Pilot)
 - d. HSUSA Check Pilots
 - e. HSUSA Instructors
 - f. HR Personnel (Basic Indoc modules 1 & 2 only)

Note 1: A list of current personnel approved as ground instructors by the Director of Operations and/or Chief Pilot, including those listed above, will be maintained by HeliService USA.

Note 2: Aircraft ground instruction will only be conducted by instructors or check airmen (HSUSA or contract) who are rated in the aircraft instruction is being provided for.

1.8.1 Instructor Pilot

An applicant for Instructor Pilot shall:

1. Hold a CFI/II (H) certificate & ATP(H) .
2. Have at least 2,500 hours total flying experience.
3. Have at least 100 hours PIC on appropriate type.
4. FAA Class II Medical issued within the past 12 months (aircraft only).
5. Have at least two years offshore / or similar experience (or relevant role experience if not engaged in offshore operations).
6. Exceptions to the above-mentioned minimum experience can be made in individual cases Chief Pilot, especially with the introduction of a new helicopter type in the HeliService USA fleet.

Note: The Director of Operations or Chief Pilot may waive certain of the preceding requirements as long as the pilot meets 14 CFR § 135.338(b) or 135.338(c).

1.8.2 Check Airmen

An applicant for a Check Airmen certificate shall:



1. Hold a CFI/II (H) certificate & ATP(H).
2. Have at least 2,500 hours total flying experience.
3. Have at least 100 hours PIC on appropriate type.
4. Have completed 1,500 hours of flight as a pilot on multi-pilot helicopters, of which at least 500 hours shall be as PIC.
5. Completed 50 hours of flight instruction as an instructor in the applicable type or an FSTD representing that type.
6. FAA Class II Medical issued within the past 12 months (aircraft only).

Note: The Director of Operations or Chief Pilot may waive certain of the preceding requirements as long as the pilot meets 14 CFR § 135.337(b) or 135.337(c).

1.8.3 Contract Instructors and Check Pilots

Contract Agencies

This document represents the primary source, but not the only source of training. Contractual Agency Instruction is designed to match and reinforce HSUSA's training programs. Contract agencies will be required to submit a course outline, which will be reviewed by the Chief pilot to determine applicability to the HSUSA Training Program. Final approval must be obtained from the companies Principal Operations Inspector and Operations Specification A031 issued.

HSUSA will permit and facilitate access to its aircraft and cockpits by employees of the training organization(s) listed in OpSpec A031 for the purpose of maintaining their line-performance and line-observation currency as contract instructors and/or contract check pilots.

Contract Training and Checking

Contractor training and checking may be provided by companies operating under the same part or 14 CFR 142 training center, IAW the specifications set forth in this training program and have met the requirements of 135.337 through 135.340. The following must be adhered to:

1. There is a preexisting contract or other arrangement.
2. The Contract Agencies have been audited IAW the requirements in Operations Specification A031.
3. Training provided by these contractors will be conducted in accordance with this approved training program.
4. Authority may be delegated to a contractor for training subjects; however, HSUSA may not delegate the responsibility for the conduct of contracted training. The FAA Principal Operations Inspector must approve each contractual arrangement.

Upon completion of the Contract Instructor and Check Pilot Module the appropriate training form will be completed and held in the Contractors training folder.

1.8.4 Record Keeping & PRIA



1.8.4.1 Pilot Records

[14 CFR § 135.63(a)(4)(i)(ii)(iii)(iv)(v)(vi)(vii)(viii)(x)]

HeliService USA will keep all training records required by 14 CFR § 135.63 at its principle base of operations for no less than 12 months:

1. Full name of crewmember
2. Copy of certificate type and number
3. Record of pilots aeronautical experience
4. Current duties and date of assignment to those duties
5. Copy of medical certificate
6. Date and results of required checks
7. Pilot flight and duty time
8. Date of completion of Initial and Recurrent training

HeliService USA must record each action taken to release any flight crewmember from employment due to physical or professional disqualification.

1.8.4.2 Crew Records

1.8.4.3 PRIA

HeliService USA will comply with the requirements of The Pilot Records Improvement Act of 1996. Portions of this legislation require that record be kept of individual instances of PRIA requests and that a pilot's company records be maintained for at least five years. These records will be maintained in accordance with 14 CFR 111.255.

Additionally, no individual may begin service as a pilot operating under Part 135 unless the pilot's information in the PRD has been reviewed by HeliService USA.

1.8.4.4 Pilots Record Database (PRD)

1.8.4.4.1 General

The PRD Act requires the PRD to contain pilot records dating from August 1, 2005 for FAA certification events, enforcement records, pilot employment history, training data, and drug and alcohol testing records. The PRD Act also requires that the PRD replace PRIA after a period of transition. Until the PRD is fully implemented, PRIA remains in effect. Implementation of the PRD will begin June 2023.

1.8.4.4.1.1 Records contained in PRD

1.8.4.4.1.1.1 Airman Certification Information

Reviewing entities should use airmen certification information the FAA maintains to validate a pilot's qualifications. The PRD provides an individual's current airman certificates, associated ratings, and any



limitations to the certificate or ratings. The date and certificate grade sought for any failed attempt to pass a practical test required to obtain a certificate or type rating under part 61 (since August 2010) is included in the airman certification information. Also, the PRD contains a pilot's latest medical certificate, including its class and any limitations.

1.8.4.4.1.1.2 Accident/Incident Information

Summaries of accidents and incidents that occurred on or after October 1, 2015, are included in the PRD to provide a reviewing entity the opportunity to review any negative data or helpful information that displays a pilot's decision-making skills. Records will be shown when the pilot was listed as either first pilot (pilot in command (PIC)) or second pilot (second in command (SIC)). It is important to note that the pilot may not have been acting as PIC even though there is a report associated with that pilot.

1.8.4.4.1.1.3 Enforcement Information

The PRD includes summaries of legal enforcement actions resulting in a finding by the Administrator of a violation of 49 U.S.C. or a regulation prescribed or order issued under 49 U.S.C. that was not subsequently overturned dated on or after August 1, 2010. These records are limited to enforcement action resulting in a suspension, revocation, or civil penalty.

1.8.4.4.1.1.4 Relevant Drug and Alcohol Testing Information

The PRD includes records related to pre-employment drug or alcohol testing violations for a pilot that was not hired, and records of drug or alcohol testing violations with an employer regulated by another Department of Transportation (DOT) agency where the individual was employed as a pilot. These records are uploaded by the FAA on behalf of the regulated employer or employer regulated under another DOT agency.

1.8.4.4.2 PRD Roles

1.8.4.4.2.1 Responsible Person

The PRD Responsible Person (RP) is an individual assigned responsibilities in accordance with 116.69(a).

1. The RP will automatically have the authorized consumer, authorized contributor, and user manager roles. The RP does not need to register for any additional roles.
2. The RP may delegate PRD access to an authorized user or proxy.
3. The Director of Operations, Chief Pilot or Director of Maintenance can fill this position.

1.8.4.4.2.2 User Manager

An individual who is employed by the operator that has been assigned access rights to the PRD on behalf of that employer by the RP. The following points describe the responsibilities and authority of each user manager.

1. A user manager may delegate, remove, and otherwise manage users within the PRD on behalf of a



RP.

2. A user manager may not access or enter PRD records. This does not prohibit a user manager from also being given other roles by the RP.

The Chief Pilot may fill this position in addition to being a RP.

1.8.4.4.2.3 Authorized Contributor

An individual who is employed by the operator who has been assigned access rights to the PRD on behalf of that employer by the RP. The following points describe the responsibilities and authority of each authorized contributor.

1. An authorized contributor can upload airman records to the PRD and can also edit records in the PRD which are currently under the control of the same reporting entity.
2. An authorized contributor may not delegate access to the PRD to any other person.
3. The RP can generate additional internal requirements for authorized contributors prior to granting PRD access.

Company Training Officers and Check Airman will fill this role.

1.8.4.4.2.4 Authorized Consumer

An individual who is employed by the operator that has been assigned access rights to the PRD on behalf of his/her employer by the RP. The following points describe the responsibilities and authority of each authorized consumer.

1. An authorized consumer can only view or download pilot records via the PRD.
2. An authorized consumer may not delegate access to the PRD to any other person.
3. The RP can generate additional internal requirements for authorized consumer prior to granting PRD access.

Company Human Resources or Admin will fill this position.

1.8.4.4.2.5 Pilot

The PRD populates with pilots who hold a:

1. Commercial pilot certificate and FAA medical,
2. Airline transport pilot (ATP) certificate and FAA medical, or
3. Remote pilot certificate and FAA medical.

The individual pilot will have to grant consent to requesting RP's to access their individual pilot records for pre-employment purposes.

1.8.4.4.3 PRD Implementation

The RP is overall responsible for the implementation of the PRD at the operation. They are responsible to



ensure that all records are uploaded appropriately and within regulatory times.

1. For Pre-employment purposes, the RP and Authorized Consumer will review the PRD to determine if the candidate applicant meets hiring requirements. Records for the applicant will be requested for the previous 5 years.
2. Authorized Contributors will ensure that appropriate records are updated within 30 days of completion of the training, or within 30 days of the finalized disciplinary action.

1.8.4.4.4 Pilot Consent and Right of Review

No reviewing entity may retrieve records in the PRD pertaining to any pilot prior to receiving that pilot's written consent authorizing the release of that pilot's information maintained in the PRD.

Any pilot who submits written consent to a reviewing entity may request a copy of any State motor vehicle driving records the reviewing entity obtained regarding that pilot. The reviewing entity must provide to the pilot all copies of State motor vehicle driving records obtained within 30 days of receiving the request from that pilot.

1.8.4.4.5 PRD Account Registration

Registering as an RP. RP registrations are approved by the PRD administrator. After a registration request is received, the PRD administrator will ensure that the person registering as the RP meets the requirements of § 111.15 by comparing the submitted information with information the FAA has stored about each operator, such as in the SAS or in the OpSpecs.

Registering as an Authorized Consumer, Contributor, or User Manager. These roles are approved, edited, or rejected by the operator's RP. After a registration request is submitted for one of these roles, the RP, or a user manager, approves or rejects the request.

Registering as a Pilot. Pilot registrations are an automated process completed by the PRD. After a pilot logs in via MyAccess, the pilot will arrive at the PRD pilot registration screen unless they have already created a pilot role in the PRD. The pilot will be prompted to enter various information from their pilot certificate, which will be verified by the PRD.

For specific instructions on how to submit a registration application, review the Resources section found at : https://www.faa.gov/regulations_policies/pilot_records_database

1.8.4.4.6 Required Training Records uploaded into PRD



6/28/21

AC 120-68J
Appendix A**Table A-4. Part [135](#) Training Record Options**

Data Element	Regulation	Allow Evaluator's Comments	Task or Maneuver Which was Unsatisfactory or Incomplete	Description
Initial Knowledge Test; Written/Oral	135.293(a)	Yes	Required if unsatisfactory or incomplete	Knowledge testing.
Recurrent Knowledge Test; Written/Oral	135.293(a)	Yes	Required if unsatisfactory or incomplete	Knowledge testing.
Initial Competency Check	135.293(b)	Yes	Required if unsatisfactory or incomplete	Flight test to meet the currency requirements of § 135.293.
Recurrent Competency Check	135.293(b)	Yes	Required if unsatisfactory or incomplete	Flight test to meet the currency requirements of § 135.293.
Instrument Proficiency Check	135.297	Yes	Required if unsatisfactory or incomplete	Instrument proficiency check (IPC).
Initial Line Check	135.299	Yes	Required if unsatisfactory or incomplete	A line check for the PIC.
Recurrent Line Check	135.299	Yes	Required if unsatisfactory or incomplete	A line check for the PIC.
Initial Training	135.345 , 135.347	N/A	N/A	The training required for crewmembers who have not qualified and served in the same capacity on an aircraft.
Transition Training	135.345, 135.347	N/A	N/A	The training required for crewmembers who have qualified and served in the same capacity on another aircraft.
Upgrade Training	135.345, 135.347	N/A	N/A	The training required for crewmembers who have qualified and served as SIC on a particular aircraft type, before they serve as PIC on that aircraft.



Data Element	Regulation	Allow Evaluator's Comments	Task or Maneuver Which was Unsatisfactory or Incomplete	Description
Differences Training	135.345, 135.347	N/A	N/A	The training required for crewmembers who have qualified and served on a particular type aircraft, when the Administrator finds differences training is necessary before a crewmember serves in the same capacity on a particular variation of that aircraft.
Recurrent Training	135.351(b) , 135.351(c)	N/A	N/A	The training required for crewmembers to remain adequately trained and currently proficient for each aircraft, crewmember position, and type of operation in which the crewmember serves.
Requalification Training	135.321(a)(8)	N/A	N/A	The training required for crewmembers previously trained and qualified, but who have become unqualified due to not having met within the required period.
Initial Crew Resource Management Training	135.330	N/A	N/A	Crew resource management training.
Recurrent Crew Resource Management Training	135.330	N/A	N/A	Crew resource management training.
Emergency Training	135.331	N/A	N/A	Each training program must provide emergency training under this section for each aircraft type, model, and configuration, each crewmember, and each kind of operation conducted, as appropriate for each crewmember and the certificate holder.
Other	Other	Optional	Optional	A pilot training or performance record kept in accordance with § 135.329(e) .

1.8.4.4.7 Required Drug Test Records



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Appendix B**Table B-2. Report Type Options**

(Record Type) Report Type	Type of Test Options	Description
(Drug Test) Verified Positive Drug Test Result	Random	Record of a random drug test conducted in accordance with 14 CFR § 120.109(b) that was verified positive by the employer's Medical Review Officer.
(Drug Test) Verified Positive Drug Test Result	Post-Accident	Record of a post-accident drug test conducted in accordance with 14 CFR § 120.109(c) that was verified positive by the employer's Medical Review Officer.
(Drug Test) Verified Positive Drug Test Result	Reasonable Cause	Record of a reasonable cause drug test conducted in accordance with 14 CFR § 120.109(d) that was verified positive by the employer's Medical Review Officer.
(Drug Test) Verified Positive Drug Test Result	Return-to-Duty	Record of a return-to-duty drug test conducted in accordance with 14 CFR § 120.109(e) that was verified positive by the employer's Medical Review Officer.
(Drug Test) Verified Positive Drug Test Result	Follow-Up	Record of a follow-up drug test conducted in accordance with 14 CFR § 120.109(f) that was verified positive by the employer's Medical Review Officer.
(Drug Test) Verified Positive Drug Test Result	Pre-employment*	Record of a pre-employment drug test conducted in accordance with 14 CFR § 120.109(a) that was verified positive by the employer's Medical Review Officer.
(Drug Test) Verified Positive Drug Test Result	Other DOT Agency Test (Non-FAA modes)*	Record of a pilot that submitted to a DOT drug test conducted by an employer under another DOT agency regulation that is verified positive by the employer's Medical Review Officer.
(Drug Test) Refusal to Submit to Drug Test	Random	Record of a refusal to submit to a random drug test required under 14 CFR § 120.109(b), including any conduct listed under 49 CFR § 40.191 .
(Drug Test) Refusal to Submit to Drug Test	Post-Accident	Record of a refusal to submit to a post-accident drug test required under 14 CFR § 120.109(c), including any conduct listed under 49 CFR § 40.191.



1.8.4.4.8 Required Disciplinary Actions

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Appendix C**APPENDIX C. FINAL DISCIPLINARY ACTION RECORDS****Table C-1. Data Fields (all required)**

Reporting entity designator:	Designator (or PRD equivalent) of the air carrier or operator entering the record (automatically populates with assigned credential).
Reporting entity name:	Name of the certificate holder or operator as it appears on the certificate or is otherwise recorded in the FAA records (automatically populates with assigned credential).
Pilot certificate number:	Certificate number of the pilot receiving the action.
Pilot last name:	Name of the pilot as it appears on their certificate.
Date of action:	YYYYMMDD
Action type:	The only disciplinary actions permitted to be entered in accordance with § 111.230 are: a written warning, suspension, or termination related to pilot performance. Other types of disciplinary actions are not to be entered. <u>Valid values:</u> Written Warning Suspension Termination
Relevant documents:	These are documents which are relevant to this record and may be requested by the reviewing entity or the pilot in accordance with § 111.230. <u>Valid values:</u> Yes No
Removal from aircraft operations:	Did the disciplinary action result in permanent or temporary removal of the pilot from aircraft operations? <u>Valid values:</u> Yes No

1.8.4.4.9 Employment History Records



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Appendix D**APPENDIX D. EMPLOYMENT HISTORY RECORDS****Table D-1. Data Fields**

Reporting entity designator:	Designator (or PRD equivalent) of the air carrier or operator entering the record (automatically populates with assigned credential).
Reporting entity name:	Name of the certificate holder or operator as it appears on the certificate or is otherwise recorded in the FAA records (automatically populates with assigned credential).
Pilot certificate number:	Certificate number of the pilot receiving the action.
Pilot last name:	Name of the pilot as it appears on their certificate.
PRD date of hire:	MMDDYYYY The earliest date on which an individual begins any form of required training in preparation for the individual's service as a pilot on behalf of an operator or entity subject to the applicability of this part; or performs any duty as a pilot for an operator or entity subject to the applicability of part 111 .
Assignment to duty history: (optional)	Enter the starting dates and ending dates of assignments to a flight position and aircraft. See Table D-2 .
Separation of employment date:	Last-in-time record of any action ending the employment relationship between a pilot and an air carrier or other operator which is not subject to any pending formal or informal dispute initiated by the pilot. No separation from employment may be considered final until 30 days after the action. YYYYMMDD <i>Valid values:</i> TBD
Separation type:	<i>Valid values:</i> Termination-pilot performance Termination-professional disqualification Other-employer-initiated separation not due to pilot performance Other-physical/medical disqualification Other-resignation Other-retirement Other [free form text box] details TBD
Relevant documents:	These are documents which are relevant to this record and may be requested by the reviewing entity or the pilot in accordance with § 111.235 . <i>Valid values:</i> Yes No



PRD records complete date:	This is the date when the air carrier or operator completes the process of entering all applicable records into the PRD for this particular airman. Additional records may be added as they become available but this date indicates a hiring employer no longer needs to request records under PRIA in addition to checking the PRD records as the records are current to this date.
No additional records to report to PRD date:	This is the date when the previous employer attests that all records related to this pilot have been provided to the PRD and no additional records are available or expected to be generated. YYYYMMDD <u>Valid values:</u> Must be equal to or after separation of employment date and separation of employment date and type must have been entered.

Table D-2. Assignment to Duty (entry of this record is optional)

Start date:	YYYYMMDD (required)
End date:	YYYYMMDD (optional; can't be before start date)
Aircraft:	Make and Model (Use make/model lookup table)
Position:	The position in the aircraft that the pilot was assigned during the event. <u>Valid values:</u> PIC (Pilot in command) SIC (Second in command)

1.8.4.4.10 Verification of Motor Vehicle Driving Record Search and Evaluation



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Appendix E

**APPENDIX E. VERIFICATION OF MOTOR VEHICLE DRIVING RECORD SEARCH
AND EVALUATION****Table E-1. Data Fields**

Reporting entity designator:	Designator (or PRD equivalent) of the air carrier or operator entering the record (automatically populates with assigned credential).
Reporting entity name:	Name of the certificate holder or operator as it appears on the certificate or is otherwise recorded in the FAA records (automatically populates with assigned credential).
Pilot certificate number:	Certificate number of the pilot.
Pilot last name:	Name of the pilot as it appears on their certificate.
Date when NDR search and evaluation was completed:	The date the NDR record was received and evaluated: YYYYMMDD

1.9 Eligibility Periods

1.9.1 Eligibility Periods – Pilots

[14 CFR § 135.301(a) and 135.232(b)]

1. Pilot testing will be conducted IAW 14 CFR Part 135.301(a):
 - a. If a crewmember who is required to take a test or a flight check under this part, completes the test or flight check in the calendar month before or after the calendar month in which it is required, that crewmember is considered to have completed the test or check in the calendar month in which it is required.
2. Pilot recurrent will be conducted IAW 14 CFR Part 135.323(b):
 - a. Whenever a crewmember who is required to take recurrent training under this subpart completes the training in the calendar month before, or the calendar month after, the month in which that training is required, the crewmember is considered to have completed it in the calendar month in which it was required.

1.9.2 Eligibility Periods – Instructors/Check Pilots

[14 CFR § 135.339(b), 135.340(b), 135.339(a)(2), and 135.340(a)(2)]

Instructor/Check Pilot observations will be conducted IAW 14 CFR § 135.339(b) and 135.340(b):

1. 14 CFR § 135.339(a)(2) is considered to be completed in the due month if it is completed in the month before or after in regards to Check Pilot observations.



2. 14 CFR § 135.340(a)(2) is considered to be completed in the due month if it is completed in the month before or after in regards to Instructor observations.

For Reference Only



2 Evaluation / Checking

2.1 Procedures for Training and Checking

2.1.1 Preflight Planning

1. Prior to a training flight, the instructor shall always do the following:
 - a. Reference the applicable module.
 - b. Study the content.
 - c. Review past student performance to identify areas needing focus.
 - d. Review required proficiency standards, and;
 - e. Develop a plan for instructing the module.
2. Prior to an evaluation, the check airman shall do the following:
 - a. Prepare and follow a plan of action.
 - b. Reference the evaluation module.
 - c. Study the required items in which proficiency must be displayed.
 - d. Review past student performance to understand areas of potential weakness.
 - e. Develop or utilize a previously made plan of action to ensure the check covers all required items.

2.1.2 Preflight Briefing

1. The pre-flight briefing sets the tone for the whole flight. Pre-flight briefings conducted with company pilots should be kept to approximately an hour.
2. Testing questions should be kept to the minimum required during the pre-flight brief.
3. Tell the candidate to expect an element of training during the check:
 - a. During line checks this may take the form of advice or suggested alternative methods of achieving a task.
 - b. It might involve the clarification of a SOP; the interpretation of a manual or point of the FAA regulations.
 - c. During initial / recurrent checkrides the training is more likely to involve handling exercises to improve skills.
4. Training given during a check flight must not be promoted as an indication of failure.

2.1.3 The Flight

Training captains should attempt to make the flight appear routine. This is easily achieved during line checks but during initial / recurrent checks it can be more difficult. Training captains should be pleasant at all times, try not to appear formal or unduly serious and yet remain professionally focussed throughout the flight.

Assessing during the flight:



1. While it is always desirable to achieve excellence during a check flight, overemphasis on setting high standards can be counter productive.
2. Setting too high a standard for an individual's experience and ability will always make the candidate nervous and ill at ease. He will not therefore demonstrate his true potential.
3. If doubt exists about a candidate's performance, extra training should be given. When this becomes necessary try to promote it as advice; or as a suggested better method of executing a maneuver.
4. Avoid criticism and contentious issues. Avoid stating opinions as facts.
5. If the desired performance standard cannot be met even after extra training the matter should be referred to the Chief Pilot.

2.1.4 Postflight Briefing

At the conclusion of any training or evaluation, the candidate being checked should be debriefed expeditiously. If there is a possibility that a delay may occur, tell the candidate the outcome of the check immediately on vacating the aircraft. The debrief should be pleasant and informal.

Follow the following sequence:

1. Ask the candidate for an assessment of his performance. This has the following advantages:
 - a. The answer will indicate the degree of self-criticism applied by the candidate.
 - b. It might also give a clear indication of under, or over confidence.
2. Comment on the answer given and move on to cover other good points.
3. Use testing questions in an informal manner to clarify points or confirm knowledge.
4. Move on to cover the weak points and emphasize anything that should have been learned from any rectification skills training given. Use testing questions again to confirm understanding.
5. Progress from this point to show how good CRM might have improved the overall result.
6. Debrief the CRM points.
7. Re-cap the good points to finish on a high point.
8. Give a clear indication of overall assessment. Show the candidate the assessment given in the training form according to the 9 core competencies.
9. Lastly, invite comments from the candidate on the overall conduct of the test. Ask for a critique of your performance as an examiner. Find out if the candidate feels that he has been assessed fairly and that the flight was conducted sympathetically.

If an evaluation is failed for any reason, follow the debriefing guidelines below:

1. Don't belabor minor points but emphasize the main weakness.
2. Make sure that the candidate is fully aware **that the check has been failed and that he cannot exercise the privileges of his license** until he has successfully passed a requalification.
3. Progress from this point to show how good CRM might have improved the overall result.
4. Debrief any general CRM points.
5. Give clear guidance to show **what** was wrong, **why** it was wrong and **what** is necessary for him to



do to achieve a satisfactory standard.

6. Tell him exactly what sequence of events will occur next, covering in particular any extra training that may be given and state the level of performance that will be required for a pass.

2.1.5 Assessment Criteria

The following rating scale will be used by flight instructor/examiner to evaluate performance of each individual maneuver or procedure attempting during training.

2.1.5.1 Grading system

Grade	Explanation
5	Excellent – The candidate performed the operation, maneuver, or assigned task to near zero tolerances quickly and efficiently with an excellent degree of: planning, coordination, smoothness, and airmanship, which left little room for improvement. The candidate always demonstrated all observable behaviors when required, which significantly enhanced safety, effectiveness and efficiency.
4	Above Average – The candidate performed the operation, maneuver, or assigned task well within accepted standards; demonstrating a well developed sense of aircraft control, coordination, and knowledge. The candidate regularly demonstrated all the observable behaviors when required, which enhanced safety.
3	Required Average – The candidate easily performed the operation, maneuver, or assigned task within HeliService USA standards. Minor errors, whether attributed to extenuating circumstances or not, were readily recognized and corrected in a proper manner. The candidate regularly demonstrating most of the observable behaviors when required, which resulted in a safe operation.
2	Below Average – The candidate was able to perform the operation, maneuver, or assigned task within only the minimum or to a minimum level of performance that is normally accepted for a maneuver. Repeated attempts to satisfy these minimum standards were required. The pilot applied procedures at the minimum acceptable level, by only occasionally demonstrating some of the observable behaviors when required, but which overall did not yet result in an unsafe situation.
1	Below Acceptable Standards – The candidate lacked sufficient knowledge, skill, or ability to perform the operation, maneuver, or assigned task without assistance. Examples are: failure to achieve the stated objective of the maneuver for whatever reason; obvious or gross lack of control, or any loss of control to the extent which required the instructor to take over the controls in order to maintain safety of flight. The candidate did not apply procedures correctly, by rarely demonstrating any of the behavioral indicators when required, which resulted in an unsafe situation.

2.1.5.2 Core Competencies

The following tables guides an instructor on how to assess different aspects of the training flight according to the 9 Core Competencies with their desired observable behaviors.

Every HeliService USA instructor must understand these guidelines before a check or training flight and



grade the candidate accordingly. The aim of this guide is to make grading objective and transparent whilst finding the root cause for any sub-standard performance.

Competency Description	Observable Behaviors
Application of procedures and compliance with regulations Identifies and applies appropriate procedures in accordance with published operating instructions and applicable regulations.	OB 1.1 Identifies where to find procedures and regulations. OB 1.2 Applies relevant operating instructions, procedures and techniques in a timely manner. OB 1.3 Follows SOP's unless a higher degree of safety dictates an appropriate deviation. OB 1.4 Operates aeroplane systems and associated equipment correctly. OB 1.5 Monitors aircraft systems status. OB 1.6 Complies with applicable regulations. OB 1.7 Applies relevant procedural knowledge.
Competency Description	Observable Behaviors



Communication Communicates through appropriate means in the operational environment, in both normal and non normal situations.	OB 2.1 Determines that the recipient is ready and able to receive information. OB 2.2 Selects appropriately what, when, how and with whom to communicate. OB 2.3 Conveys messages clearly, accurately and concisely. OB 2.4 Confirms that the recipient demonstrates understanding of important information. OB 2.5 Listens actively and demonstrates understanding when receiving information. OB 2.6 Asks relevant and effective questions. OB 2.7 Uses appropriate escalation in communication to resolve identified deviations. OB 2.8 Uses and interprets non-verbal communication in a manner appropriate to the organizational and social culture. OB 2.9 Adheres to standard radiotelephone phraseology and procedures.
Competency Description	Observable Behaviors
Flight Path Management - Automation Controls the flight path through automation	OB 3.1 Uses appropriate flight management, guidance systems and automation, as installed and applicable to the conditions. OB 3.2 Monitors and detects deviations from the intended flight path and takes appropriate action. OB 3.3 Manages the flight path safely to achieve optimum operational performance. OB 3.4 Maintains the intended flight path during flight using automation while managing other tasks and distractions. OB 3.5 Selects appropriate level and mode of automation in a timely manner considering phase of flight and workload. OB 3.6 Effectively monitors automation, including engagement confirmation and automatic mode transitions.



Competency Description	Observable Behaviors
Flight Path Management - Manual Control Controls the flight path through manual control	OB 4.1 Controls the aircraft manually with accuracy and smoothness as appropriate to the situation. OB 4.2 Monitors and detects deviations from the intended flight path and takes appropriate action. OB 4.3 Manually controls the helicopter using the relationship between attitude, speed and power settings and navigation signals or visual information. OB 4.4 Manages the flight path safely to achieve optimum operational performance. OB 4.5 Maintains the intended flight path during manual flight while managing other tasks and distractions. OB 4.6 Uses appropriate flight management and guidance systems, as installed and applicable to the conditions.

Competency Description	Observable behaviours
Leadership and Teamwork Influences others to contribute to a shared purpose. Collaborates to accomplish the goals of the team.	OB 5.1 Encourages team participation and open communication. OB 5.2 Demonstrates initiative and provides direction when required. OB 5.3 Engages others in planning. OB 5.4 Considers inputs from others. OB 5.5 Gives and receives feedback constructively. OB 5.6 Addresses and resolves conflicts and disagreements in a constructive manner. OB 5.7 Exercises decisive leadership when required. OB 5.8 Accepts responsibility for decisions and actions. OB 5.9 Carries out instructions when directed. OB 5.10 Applies effective intervention strategies to resolve identified deviations. OB 5.11 Manages cultural and language challenges, as applicable.



Competency Description	Observable Behaviors
Problem Solving and Decision Making Identifies precursors, mitigates problems; and makes decisions.	OB 6.1 Identifies, assesses and manages threats and errors in a timely manner. OB 6.2 Seeks accurate and adequate information from appropriate sources. OB 6.3 Identifies and verifies what and why things have gone wrong, if appropriate. OB 6.4 Perseveres in working through problems while prioritizing safety. OB 6.5 Identifies and considers appropriate options. OB 6.6 Applies appropriate and timely decision-making techniques. OB 6.7 Monitors, reviews and adapts decisions as required. OB 6.8 Adapts when faced with situations where no guidance or procedure exists. OB 6.9 Demonstrates resilience when encountering an unexpected event.

Competency Description	Observable Behaviors
Situational awareness Perceives, comprehends and manages information and anticipates its effect on the operation.	OB 7.1 Monitors and assesses the state of the aeroplane and its systems. OB 7.2 Monitors and assesses the aeroplane's energy state, and its anticipated flight path. OB 7.3 Monitors and assesses the general environment as it may affect the operation. OB 7.4 Validates the accuracy of information and checks for gross errors. OB 7.5 Maintains awareness of the people involved in or affected by the operation and their capacity to perform as expected. OB 7.6 Develops effective contingency plans based upon potential risks associated with threats and errors. OB 7.7 Responds to indications of reduced situation awareness.



Competency Description	Observable Behaviors
Workload Management Maintain available workload capacity by prioritizing and distributing tasks using appropriate resources.	OB 8.1 Exercises self-control in all situations. OB 8.2 Plans, prioritizes and schedules appropriate tasks effectively. OB 8.3 Manages time efficiently when carrying out tasks. OB 8.4 Offers and gives assistance. OB 8.5 Delegates tasks. OB 8.6 Seeks and accepts assistance when appropriate. OB 8.7 Monitors, reviews and cross-checks actions conscientiously. OB 8.8 Verifies that tasks are completed to the expected outcome. OB 8.9 Manages and recovers from interruptions, distractions, variations and failures effectively while performing tasks.
Competency Description	Observable Behaviors
Knowledge Demonstrates knowledge and understanding of relevant information, operating instructions, aircraft systems and the operating environment.	OB 9.1 Demonstrates practical and applicable knowledge of limitations and systems and their interaction. OB 9.2 Demonstrates required knowledge of published operating instructions. OB 9.3 Demonstrates knowledge of the physical environment, the air traffic environment including routings, weather, airports and the operational infrastructure. OB 9.4 Demonstrates appropriate knowledge of applicable legislation. OB 9.5 Knows where to source required information. OB 9.6 Demonstrates a positive interest in acquiring knowledge. OB 9.7 Is able to apply knowledge effectively.

2.1.6 Evaluation Standards

The standard required for basic checks is at least that required for obtaining the certificate which must be held to act as a PIC. The minimum HeliService USA Part 135 operation standard required of HSUSA Pilots is to meet "commercial certificate with an instrument rating" standards of proficiency.



HeliService USA pilots will always be able to fly the helicopter in a manner that will demonstrate they are obviously in command of the aircraft with the successful outcome of each maneuver never in doubt. Additionally maneuvers will be completed within the standards set by the Practical Test Standards (PTS) or to the standards in the maneuvers guide listed in Appendix A, whichever standard is more restrictive.

All written or CBT examinations must be completed with a minimum score of 80%. Subject areas found deficient will be reviewed and the individual's knowledge corrected to 100%. The Check Airman will determine the applicable standards prior to the start of any evaluation session. The required standards will be discussed with the pilot being tested.

2.1.7 Checking - Multi-pilot in a Single-pilot Aircraft

HeliService USA operates certain aircraft that do not require an SIC, but chooses to operate with two pilots (dual PIC's) as an extra measure of safety and to satisfy customer requirements. While not regulatory in nature, HeliService USA will make every effort to ensure checkrides are conducted with two pilots to most closely mimic it's daily operations.

CREW COMPLEMENT:

Checking is generally conducted as a crew; however, a pilot may complete the evaluation alone (single PIC) or with another pilot acting as second crewmember. The pilot acting as second crewmember must be:

1. A qualified line pilot, or
2. A qualified flight instructor and/or check airmen, or
3. A pilot being evaluated at the conclusion of the same course

SINGLE-PILOT STANDARDS:

1. All maneuvers will be evaluated IAW the standards and maneuvers described in this manual.
2. All emergency procedures will be handled IAW with the procedures listed in Appendix A of this manual and the OM-A
3. The pilot will not be expected to make the call-outs listed in Appendix A of this manual and the OM-A.

DUAL-PILOT STANDARDS:

1. The pilot being evaluated shall be the acting PIC and must demonstrate the maneuvers required for the appropriate checking module. That PIC may still choose to hand off the controls at times or delegate tasks to the other pilots as needed utilizing CRM to most effectively complete the maneuver or procedure.
2. All maneuvers will be evaluated IAW the standards described in this manual.
3. All procedures and emergency procedures will be handled IAW with the procedures listed in Appendix A of this manual and the OM-A.
4. All appropriate call-outs by the PF/PM will be made IAW Appendix A of this manual and the OM-A.



2.2 14 CFR § 135.293 (Initial/Recurrent)

2.2.1 14 CFR § 135.293(a)

[14 CFR § 135.293(a)]

1. The appropriate provisions of 14 CFR Parts 61, 91, and 135, the operations specifications and manual of the certificate holder.
2. For each type of aircraft to be flown by the pilot, the aircraft powerplant, major components and systems, major appliances, performance and operating limitations, standard and emergency operating procedures, and the contents of the approved Rotorcraft Flight Manual, or equivalent, as applicable.
3. For each type of aircraft to be flown by the pilot, the method of determining compliance with weight and balance limitations for takeoff, landing and enroute operations.
4. Navigation and use of air navigation aids appropriate to the operation or pilot authorization, including, when applicable, instrument approach facilities and procedures.
5. Air Traffic Control procedures, including IFR procedures when applicable.
6. Meteorology in general, including the principles of frontal systems, icing, fog, thunderstorms, and windshear, and, if appropriate for the operations of the certificate holder, high altitude weather.
7. Procedures for:
 - a. Recognizing and avoiding severe weather situations.
 - b. Escaping from severe weather situations, in case of inadvertent encounters, including low altitude windshear (except that rotorcraft pilots are not required to be tested on escaping from low altitude windshear).
 - c. Operating in or near thunderstorms (including best penetrating altitudes), turbulent air (including clear air turbulence), icing, hail, and other potentially hazardous meteorological conditions.
8. New equipment, procedures, or techniques, as appropriate.
9. For rotorcraft pilots, procedures for aircraft handling in flat light, whiteout, and brownout conditions, including methods for recognizing and avoiding those conditions.

2.2.2 14 CFR § 135.293(b)

[14 CFR § 135.293(b)]

1. Competency check (simulator).
2. Competency check (aircraft).

Note: While not required, the preferred method of conducting this type of check is the simulator. This enables the Check Airmen to induce a greater number of different emergency procedures and



consistently simulate variable weather conditions.

2.3 14 CFR § 135.299 (Line check)

[14 CFR § 135.299(a)(1)(2)(3)(b)(c)]

The line check is considered a particularly important factor in the development, maintenance, and refinement of high operating standards. It can provide a valuable indication of the effectiveness of company training policy and methods.

In addition to the regulatory requirement, the primary purpose of a HeliService USA line check is to assess a pilot's ability to perform a complete line operation from start to finish. This includes pre-flight / post-flight procedures, the use of the equipment provided, and shall be considered an overall assessment of the ability to perform the duties required. The route chosen shall be chosen to accurately represent the scope of a pilot's normal operation. The line check is not intended to determine competence on any particular route. While only required for pilots who will fly as PIC (not HSUSA designated SIC's), every pilot shall undergo a line check on the helicopter to demonstrate their competence in carrying out normal line operations as described in the Operations Manual.

1. The flight check shall:
 - a. Be given by an approved check pilot or by the Administrator.
 - b. Consist of at least one flight over one route segment.
 - c. Include takeoffs and landings at one or more representative airports. In addition to the requirements of this paragraph, for a pilot authorized to conduct IFR operations, at least one flight shall be flown over a civil airway, an approved off-airway route, or a portion of either of them.
2. The pilot who conducts the check shall determine whether the pilot being checked satisfactorily performs the duties and responsibilities of a pilot in command in operations under this part, and shall so certify in the pilot training record.
3. Each certificate holder shall establish in the manual required by 14 CFR § 135.21 a procedure which will ensure that each pilot who has not flown over a route and into an airport within the preceding 90 days will, before beginning the flight, become familiar with all available information required for the safe operation of that flight.

2.3.1 Line Experience Module (LEM)

2.3.1.1 Purpose

The Line Experience Module (LEM) is a training component designed to reinforce HeliService USA's standard operating procedures, CRM principles, TEM (Threat and Error Management) skills, integrate HFDM trends, industry trends and safety culture into HSUSA's training program. LEMs simulate real-world operational environments to meet the requirements of 135.337(d)(2) and 135.338(d)(2) for a check pilot or instructor who is only designated for the flight simulator. Using an approved flight simulator and



structured Line-Oriented Flight Training (LOFT) methodology.

2.3.1.2 Objectives

1. Reinforce operational decision-making under real-time pressures.
2. Assess crew performance in normal and abnormal operational conditions.
3. Provide a no-jeopardy environment for evaluating and coaching CRM and TEM skills.
4. Integrate safety observations, HFDM findings and industry trends into the company's training program.

2.3.1.3 Scheduling and Applicability

1. HSUSA has developed and will continue to develop LEM scenarios to ensure flight instructors and check pilots do not repeat the same scenario within four training cycles.
2. If the ground and flight training segments occur more than 30 days apart, then the HSUSA will maintain a separate tracking record for each segment.
 - a. In this case each segment will then have its own 12-month currency cycle.

2.3.1.4 Ground Training Elements

1. Clearances - Duties of a check pilot or flight instructor with regards to the use of ATC clearances in a training environment.
 - a. Format of clearances to accurately simulate ATC.
 - b. Ability to provide timeliness of responses typical of controllers.
 - c. Using clearances accurately to create a more realistic scenario in the training environment.
 - d. Accurately providing realistic clearances to enhance instructor credibility and professionalism.
 - e. Using clearances accurately to simulate correct ATC spacing and controller knowledge of aircraft system performance parameters.
2. Terminology
 - a. Common ATC terminology variations.
 - b. International language and terminology considerations (if applicable).
 - c. Realistic use in the training environment.
3. Vectoring - Appropriate use and training benefits/pitfalls.
 - a. Commonly misused vectoring practices.
 - b. Training benefits and the importance of correct vectoring to preserve realism.
 - c. Negative training and the associated impact of inaccurate or unrealistic vectors/practices.
4. Runway/Taxiway Signage, Surface Movement Guidance and Control System (SMGCS), etc.
 - a. Recent changes in signage and markings.
 - b. Large airports/small airport variances.
 - c. Runway markings.
 - d. Taxiway markings.



- e. Ground and approach lighting systems.
- f. SMGCS: who must comply, under what weather conditions it applies, and how to comply with clearances and procedures (if applicable).

2.3.1.5 Flight Training Elements

Flight Training - Training must be conducted in an FSTD and will be designed as an LEM event on the required training/checking form. The LEM event is intended to provide replication of operations within the National Airspace System (NAS) with emphasis on the crew's interaction and functions. It must be designed to realistically simulate (within the limitations of FSTD technology) a view of the ATC environment that flightcrews may encounter when operating in the NAS. Flight training must include any combination of the following events, as applicable, that are tailored to reflect the operational environments of HeliService USA's operations:

1. Normal operations;
2. ATC normal, abnormal, and emergency operations;
3. Low visibility taxi and takeoff (use of enhanced taxiway markings and ATC interactions);
4. Short approach request;
5. Tailwind landing request <10 knots;
6. Speed restrictions and expedite requests;
7. Expedite to an altitude (may be up or down);
8. Unexpected clearance to a new fix (flight management system (FMS) exercise);
9. Visual approach;
10. Special approaches (e.g., Required Navigation Performance (RNP) subject to a special requirement, localizer performance with vertical guidance (LPV), and precision runway monitor (PRM));
11. Traffic alert;
12. Traffic Alert and Collision Avoidance System (TCAS) alert;
13. Uncontrolled field approach;
14. Special airport operation;
15. Special route authorizations;
16. RNP approaches and departures;
17. Equipment failures affecting navigation reliability;
18. Offshore IFR approach (HEDA/OSAP)
19. Offshore IFR missed approach
20. Offshore IFR holding
21. Crew Change flights in MVFR condition
22. Night approach to Wind Turbine, Vessel, Rig, Hospital
23. NVG failure (single pilot/dual pilot)
24. Low visibility with NVGs (recognition and recovery)
25. Single engine failure during Part 133 operations
26. Low visibility approach to a vessel (day/night)



27. HAA mission to hospital (single pilot/dual pilot)
28. HAA mission in Local Area (LFA) (single pilot/dual pilot)
29. HAA mission out side the Local Area (single pilot/dual pilot)
30. PinS approach Day/Night (LPV/LNAV mins)

2.3.1.6 Complexity

Scenarios must incorporate environmental variability, regulatory challenges, communication complexity, and operational decision points. Scenario development shall follow the structure outlined in AC 120-35D and this section.

2.3.1.7 LEM Structure

1. Pre-Flight Briefing
 - a. Review of scenario objectives, weather, NOTAMs, aircraft MEL status.
 - b. CRM emphasis and procedural expectations.
2. Operational Segment
 - a. Scenario execution in the aircraft or an approved FSTD.
 - b. Instructor-controlled event triggers simulating abnormalities.
3. Debriefing
 - a. Facilitated review of performance using TF 03 Evaluation Form.
 - b. Grading based on performance, communication, and CRM application.

2.3.1.8 Scenario Development

Each LEM scenario shall be developed to reflect realistic HeliService USA missions. All scenarios will incorporate the required ground and flight elements as described in sections 2.3.1.5 & 2.3.1.6 and any other items the instructor/check pilot deem necessary. With an emphasis on CRM and Situational Awareness.

2.3.1.8.1 Examples of LEM Scenarios

2.3.1.8.1.1 Scenario Template A – Offshore Hoist Operation

1. Mission: Standard hoisting mission to/from offshore wind turbine.
2. Weather: Marginal VFR, rising winds.
3. Events: Radio failure, task specialist illness, power fluctuation during hoist.
4. Objectives: CRM, ADM, TEM evaluation, procedural adherence, abnormal management.

2.3.1.8.1.2 Scenario Template B – Class D Load to Vessel

1. Mission: External sling of turbine component.
2. Weather: Gusting winds, comms interference.



3. Events: Hook snags, load oscillation, airspace conflict with nearby traffic.
4. Objectives: Decision-making, load coordination, airspace management, TEM evaluation, CRM, ADM.

2.3.1.8.1.3 Scenario Template C – Night SAR Mission (NVG)

1. Mission: Locate and retrieve injured worker from support vessel.
2. Weather: Clear but dark night, confined LZ.
3. Events: NVG failure, conflicting coordinates, medical emergency.
4. Objectives: Night operations proficiency, risk assessment, crew communication TEM evaluation, CRM, ADM.

2.3.1.8.1.4 Scenario Template D – Offshore IFR Day/Night Crew Change

1. Mission: Crew transfer to/from offshore vessel.
2. Weather: IFR conditions with low visibility
3. Events: Comm issues with Center/Approach, Conflicting coordinates, ill passenger
4. Objective: Complete mission safely, Offshore IFR proficiency, CRM, ADM, TEM evaluation.

2.3.1.9 Evaluation and Feedback

Performance assessments will be documented and retained in the pilot's training file and reviewed by the Director of Training. Findings relevant to safety or SOP compliance will be reported into SMS.

2.3.1.10 Pilot Training Recordkeeping

LEM records shall be maintained in accordance with HSUSA's Training program. If LEM is supplementing Part 135.299 Line Check records will be kept IAW Part 135.63.

2.4 14 CFR § 135.297 (Instrument)

[14 CFR § 135.297(c)(1)(2)(g)(1)(2)(3)]

The instrument proficiency check consists of an oral or written equipment test and a flight check under simulated or actual IFR conditions. The equipment test includes questions on emergency procedures, engine operation, fuel and lubrication systems, and hydraulic, mechanical, and electrical systems, as appropriate. The flight check includes navigation by instruments, recovery from simulated emergencies, and standard instrument approaches involving navigational facilities which that pilot is to be authorized to use. Each pilot taking the instrument proficiency check must show that standard of competence required by 14 CFR § 135.293(e).

The instrument proficiency check must:

1. For a pilot in command of an airplane or helicopter under 14 CFR § 135.243(c), include the



procedures and maneuvers for a commercial pilot certificate with an instrument rating and, if required, for the appropriate type rating.

2. The instrument proficiency check must be given by an authorized check airman or by the Administrator.

If the pilot in command is authorized to use an autopilot system in place of a second in command, that pilot must show, during the required instrument proficiency check, that the pilot is able (without a second in command) both with and without using the autopilot to:

1. Conduct instrument operations competently; and
2. Properly conduct air-ground communications and comply with complex air traffic control instructions.
3. Each pilot taking the autopilot check must show that, while using the autopilot, the airplane can be operated as proficiently as it would be if a second in command were present to handle air-ground communications and air traffic control instructions. The autopilot check need only be demonstrated once every twelve (12) calendar.

Note: While not required, the preferred method of conducting this type of check is the simulator. This enables the Check Airmen to induce a greater number of different emergency procedures and consistently simulate variable weather conditions.

2.5 Training to Proficiency

When a HeliService USA Check Pilot determines that an event is unsatisfactory, the Check Pilot may conduct training and repeat the testing of that event. This provision is made in the interest of fairness and to avoid undue hardship and expense for pilots and operators. Training may not be conducted, however, without recording the failure of these events. The quality control (QC) of a training program is accomplished, among other means, by identifying those events on checks which HeliService USA Pilots fail. The following is the guidance for conducting training during checking and testing:

1. Training and checking cannot be conducted simultaneously. When training is required, the check must be temporarily suspended, training conducted, and then the check resumed.
2. When training to proficiency is required;
 - a. The Check Pilot must record the events which were initially failed and in which training was given.
 - b. The event(s) will be recorded by entering a comment in the "Remarks" section of the 8410 Airman Competency / Proficiency Check form by reference to the maneuver number(s) on the form.

When training to proficiency is conducted and the check is subsequently completed within the original session, the overall grade for the check may be recorded as satisfactory. When the training required to reach proficiency cannot be completed in the original checking session, the check must be recorded as unsatisfactory and the HSUSA Pilot entered into requalification training.



When training to proficiency is required and it is practical to do so, the remaining events of the flight test phase should be completed before training in the failed event is conducted. If it is more practical, the failed event may be repeated at the end of a logical sequence. For example, training on steep turns might be conducted at altitude after all other air work has been completed, but before returning to the traffic pattern. If, after having received training, the pilot fails an event again, the failure must be recorded, and the pilot must be entered into requalification training.

Note: If for mechanical or other reasons the check cannot be completed after the failure of an event and before training and retesting can be accomplished, the check is considered terminated; however, the pilot may not serve in HeliService USA revenue operations until the check is successfully completed.

Written Test Scores / Grades:

All written or CBT examinations must be completed with a minimum score of 80%. Subject areas found deficient will be reviewed and the individual's knowledge corrected to 100%.

2.6 Training Documentation

All training and evaluation events conducted under HeliService USA must be documented promptly and accurately. In accordance with company policy and regulatory requirements, all associated training forms, checklists, evaluation records, and digital entries must be completed and submitted within **24 hours** of the event. This applies to initial, recurrent, and proficiency checks, as well as any ground or flight instruction. Instructors and Check Airmen are responsible for ensuring that all training documentation is properly signed, uploaded to the designated system (e.g., Digital AirWare, SharePoint, or Complete Flight), and reviewed for completeness. Delays or incomplete records may result in administrative follow-up and may affect crew scheduling or qualification status.



3 Training Curriculum

3.1 Categories of Training

There are six (6) basic categories of training at HeliService USA. The primary factor determining the appropriate category of training is the pilot's previous experience. Each category of training consists of several curriculums which are specific to an aircraft and duty position. Each individual being assigned to a different duty position and/or aircraft type must complete either Initial, Initial Equipment, Transition, Upgrade, Recurrent, or Requalification Training (depending upon the aircraft type and duty position for which they were previously qualified). All initial new hires will complete a curriculum that includes: Basic Indoctrination, Emergency Training and Hazardous Material.

Categories of training:

Courses of training which provide the necessary training and checking or testing for various types of crewmembers who have not previously qualified (or have or will become unqualified) to serve unsupervised in specific duty positions. Each category of training consists of one or more curriculums.

Initial New-Hire:

This training category is for Pilots who have no previous experience with HeliService USA (newly hired). Initial New-Hire Training includes basic indoctrination training and training for a specific duty position and aircraft type. Except for a basic indoctrination curriculum segment, the regulatory requirements for initial new-hire (INH) and initial equipment (IE) training are the same. Since initial new-hire training is usually the employee's first exposure to specific HSUSA methods, systems, and procedures, it is the most comprehensive of the six categories of training. For this reason, initial new-hire training is a distinct, separate category of training and should not be confused with initial equipment training. Initial equipment training is a separate category of training.

Transition Training:

This category of training is for personnel who have been previously trained and qualified for a specific duty position by the operator and are being re-assigned to the same duty position on a different aircraft type.

Recurrent Training:

This category of training is for personnel who have been trained and qualified by the operator and will continue to serve in the same duty position and aircraft type and must receive recurring training or checking within an appropriate eligibility period to maintain currency.

Upgrade Training:

This category of training is for a HeliService USA Pilot who has been previously trained and qualified as a SIC by HSUSA and is being qualified as a PIC.

Re-qualification Training:

This category of training is for personnel who have been trained and qualified by the operator, but have



become unqualified to serve in a particular duty position and/or aircraft, due to not receiving recurrent training and/or, a required flight or competency check, within the appropriate eligibility period.

Initial Equipment Training:

This category of training is for personnel who have been previously trained and qualified in a duty position by the operator in an aircraft and are being reassigned to another duty position on a different aircraft type which they were not previously trained and qualified by the operator for that duty position and aircraft type.

3.2 Initial New Hire PIC

3.2.1 INH-A (AW169)

OBJECTIVE OF TRAINING:

To provide newly hired HeliService USA pilots with the training required to operate HSUSA aircraft while engaged in the business of air commerce.

PREREQUISITES:

Commercial Pilot Certificate, Rotorcraft - Helicopter, Instrument Helicopter or ATP.

TRAINING HOURS, APPLICABLE CURRICULUM SEGMENTS, & MODULES:

Applicable Curriculum Segments	Planned Training Hours	References	Chapter / Modules
Basic Indoctrination	24	FAR 135.329(a)(1)	BI Modules 1-18 (HAA/SAR crews BI 16 & 17) (Offshore IFR crews BI 18)
General Emergency Training	4	FAR 135.331	General Emergency Modules 1-18
Hazardous Materials (Will-Carry)	4	FAR 135.501 through 135.507	HAZMAT Module 1
CRM	4	FAR 135.330	CRM Module 1
AW169 Ground Training	24	FAR 135.345	AW169 General Operational Subjects Modules 1-4 & AW169 Systems Modules 1-28



AW169 Flight Training	16	FAR 135.347	AW169 Sim Module 1
NVG (If Applicable)	4	FAR 61.57	NVG Ground 1-17 and Flight 1-2 Modules.
Qualification	2	FAR 135.293/297/299	Qualification Modules 1-4 (as required)

3.2.2 INH-B (AW169)

OBJECTIVE OF TRAINING:

To provide newly hired HeliService USA pilots the training required to operate HSUSA aircraft while engaged in the business of air commerce. This curriculum is tailored for pilots whom already have experience operating the type of aircraft they will be assigned as a pilot with HSUSA.

PREREQUISITES:

1. Commercial Pilot Certificate, Rotorcraft - Helicopter, Instrument Helicopter or ATP.
2. INH-B is designed for incoming pilots with previous experience in the AW-169 helicopter who meet one of the following specific criteria:
 - a. Must have successfully completed a full manufacturer (Leonardo) or equivalent (Coptersafety) AW169 initial training or type rating course in the past six months.
 - i. Initial Ground School Completion Certificate or equivalent documentation Flight Training.
 - ii. Recurrent Flight Training Completion Certificate or equivalent.
3. Or alternatively;
 - a. Must have logged a minimum of 25 hours of recent specific (past year) airframe flight time.
 - b. Must have completed the manufacturer's Recurrent Ground School Program.
 - c. Must have completed the manufacturer's Recurrent Flight Training Program.
 - d. Must have a record of successfully completing the above listed training within the previous twelve (12) calendar months. Acceptable documentation includes:
 - i. Recurrent Ground School Completion Certificate or equivalent documentation Flight Training.
 - ii. Recurrent Flight Training Completion Certificate or equivalent.

TRAINING HOURS, APPLICABLE CURRICULUM SEGMENTS, & MODULES:



Applicable Curriculum Segments	Planned Training Hours	References	Chapter / Modules
Basic Indoctrination	24	FAR 135.329(a)(1)	BI Modules 1-16 (HAA/SAR crews BI 16 & 17) Offshore IFR crews BI 18
General Emergency Training	4	FAR 135.331	General Emergency Modules 1-18
Hazardous Materials (Will-Carry)	4	FAR 135.501 through 135.507	HAZMAT Module 1
CRM	4	FAR 135.330	CRM Module 1
AW169 Ground Training	8	FAR 135.345	AW169 General Operational Subjects Modules 1-4, AW169 Ground Modules 1, 8, 9, 10, 12, 13, 14, 17, 19, 20, 23, 24, 27, or All AW169 Systems CTS
NVG (If Applicable)	4	FAR 61.57	NVG Ground 1-17 and Flight 1-2 modules.
AW169 Flight Training	4	FAR 135.347	AW169 Sim Recurrent Modules 1-2
Qualification	2	FAR 135.293/297/299	Qualification Modules 1-4 (as required)

3.3 Transition PIC

OBJECTIVE OF TRAINING:

To provide pilots with the training required to operate HeliService USA aircraft while engaged in the business of air commerce.

PREREQUISITES:

Pilots entering this category of training must be a HeliService USA PIC who is already fully qualified as a 14 CFR Part 135 PIC in an assigned HeliService USA helicopter.

TRAINING HOURS, APPLICABLE CURRICULUM SEGMENTS, & MODULES:



Applicable Curriculum Segments	Planned Training Hours	References	Chapter / Modules
General Emergency Training	4	FAR 135.331	General Emergency Modules 1-18
AW169 Ground Training	24	FAR 135.345	AW169 General Operational Subjects Modules 1-4 & AW169 Systems Modules 1-28
AW169 Flight Training	16	FAR 135.347	AW169 Sim Module 1
Qualification	2	FAR 135.293/297/299	Qualification Modules 1-4 (as required)

3.4 Recurrent PIC

OBJECTIVE OF TRAINING:

This category of training is for pilot who has been trained and qualified by HeliService USA, who will continue to serve in the same duty position and helicopter type, and who must receive recurring training and/or checking within an appropriate eligibility period. It is intended to refresh the pilot in the same ground and flight subjects as INH training and assure their continued proficiency.

PREREQUISITES:

Currently qualified as a HeliService USA PIC on the aircraft type recurrent training is required.

TRAINING HOURS, APPLICABLE CURRICULUM SEGMENTS, & MODULES:

Applicable Curriculum Segments	Planned Training Hours	Chapter / Modules
General Subjects	8	BI Modules 1-5 and CTS A/B/C rotation.
General Emergency	1	Review General Emergency Modules 1-18



Hazardous Materials (Will-Carry)	4	HAZMAT Module 1
CRM	2	CRM Module 1
AW169 Systems	8	All AW169 Systems CTS and AW169 Recurrent A/B/C rotation of classroom.
AW169 Flight Training	4	AW169 Sim Recurrent Modules 1-2
NVG	2	NVG Ground 1-4 and Flight 1-2 modules.
Qualification	2	Qualification Modules 1-4 (as required)

Note 1: A/B/C rotation enables all subjects and emergency procedures to be covered in a three year period. Charts laying out the yearly requirement are listed in Chapter 4.

Note 2: Satisfactory completion of HeliService USA's 14 CFR § 135.293 may be substituted for the recurrent flight curriculum segments.

3.5 Requalification PIC

OBJECTIVE OF TRAINING:

To return a pilot to a qualified status in the aircraft and re-establish qualification for crewmembers that have become unqualified in their assigned crew position.

PREREQUISITES:

Pilot previously qualified as a HeliService USA PIC that has become unqualified due to not having completed any required recurrent ground or flight training or a required competency/proficiency check within the appropriate eligibility period.

COMPLETION STANDARDS:

Refer to Qualification SEGMENT in table below.

TRAINING HOURS, APPLICABLE CURRICULUM SEGMENTS, & MODULES:

Months Past Due	Recurrent Ground Training SEGMENT	Recurrent Flight Training SEGMENT	Qualification SEGMENT
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Up to twelve (12) calendar months	The portion of Recurrent Ground Training not accomplished when due.	The elements not accomplished when due.	The modules not accomplished in the eligibility period: Competency Check, Line Check, or special.
Twelve (12) to 35 months	50% of hours for initial new hire	50% of hours for initial new hire	All Qualification Modules of the Transition Curriculum.
More than 35 months	SAME AS INITIAL NEW HIRE		

3.6 Flight Instructor

OBJECTIVE OF TRAINING:

To provide pilots with the necessary knowledge to qualify for duty as an instructor pilot in the simulator and/or aircraft.

PREREQUISITES:

Commercial Pilot Certificate, Rotorcraft - Helicopter, Instrument Helicopter or ATP.

TRAINING HOURS, APPLICABLE CURRICULUM SEGMENTS, & MODULES:

Instructor Ground / Flight Hours				
Instructor / Check Airmen Curriculum Segments	Initial	Transition	Recurrent	Requalification
Flight Instructor (Aircraft)	4/4	2/2	2/2	See specialty curriculum requalification.
Flight Instructor (Simulator)	4 4 IAW Simulator IOS	2/2	2/2	

3.7 Check Airman

OBJECTIVE OF TRAINING:

To provide pilots with the necessary knowledge to qualify for duty as a Check Airman in the simulator and/or aircraft.

PREREQUISITES:

Commercial Pilot Certificate, Rotorcraft - Helicopter, Instrument Helicopter or ATP.



TRAINING HOURS, APPLICABLE CURRICULUM SEGMENTS, & MODULES:

Check Airman Ground / Flight Hours				
Instructor / Check Airmen Curriculum Segments	Initial	Transition	Recurrent	Requalification
Check Airman (Aircraft)	2/2	1/2	2/2	See specialty curriculum requalification.
Check Airman (Simulator)	2/2	1/2	2/2	

3.8 Additional Training

In addition to initial, transition, upgrade and recurrent training, HeliService USA will provide ground and flight training, instruction, and practice necessary to ensure that each crewmember:

1. Remains adequately trained and currently proficient for each aircraft, crewmember position, and type of operation in which the crewmember serves; and
2. Qualifies in new equipment, facilities, procedures, and techniques, including modifications to aircraft.

This may include, but is not limited to, additional ground training, computer training, time in the aircraft, simulator, line training, or any client requirements. These are not considered regulatory in nature. Listed below are two primary types of additional training HeliService USA may conduct:

1. Line training.
2. HUET.
3. Offshore multi-crew CRM.

Note: Completion of either of the above shall be documented for client requirements.

3.8.1 Line Training

Line training is conducted to give new hires additional opportunity to familiarize themselves with offshore operations and multi-pilot operations while flying with a Check Airmen, Instructor, or senior PIC. The Chief Pilot or Director of Operations may waive this requirement due to extenuating circumstances like the start of a new base of operations.

HeliService USA designated SIC:

1. Line Flying with a Check Airmen/Instructor/Senior PIC (5 Legs).



2. Total deck landings day (5 landings).
3. Total deck landings at night (5 landings optional).

HeliService USA PIC:

1. Line flying with a Check Airmen/Instructor/Senior PIC (10 legs).
2. Line flying with a Check Airmen/Instructor/Senior PIC at night (1 leg optional).
3. Total supervision time (10h).
4. Total deck landings day (10 each).
5. Total deck landings night (5 each optional).

Note 1: In an effort avoid any confusion or potential regulatory non-compliance, prior to conducting any line training or revenue operations all pilots will be fully current and qualified PIC's under Part 135.

Note 2: Senior PIC's will be scheduled to conduct line training by the DO or CP based off maturity, good judgement, excellent adherence to regulations/SOP's, and time with HSUSA.

Note 3: Completion of line training will be documented in the pilots records and provided to customers at their request.

3.8.2 HUET (Helicopter Underwater Egress Training)

1. Flight crew who fly extended legs over water shall complete a HUET course to a recognized standard (e.g., OPITO) or greater, that includes the use of a Modular Egress Training Simulator (METS) at least every four years, unless local regulation requires greater frequency.
2. In HUET devices the emergency exit types and sizes should be representative of the aircraft flown in offshore operations.
3. All HUET training will be documented and recorded IAW with HSUSA training program requirements.

Note 1: Military initial and recurrent HUET training with EBS shall be considered sufficient for aircrew as long as documentation is provided to the training department.

Note 2: If the HUET training facility is not OPITO certified, HeliService USA will conduct an audit of the facility to ensure it meets or exceeds OPITO standards. HeliService USA has audited the following and assessed them to meet or exceed OPITO standards:

1. Survival Systems USA, Groton, CT

3.8.3 Differences Training Single-Pilot vs. Dual-Pilot Operations

Objective:

To ensure pilots transitioning between single-pilot and dual-pilot operations are trained, evaluated, and proficient in the specific operational, procedural, and CRM-related differences.

Applicability:



All pilots transitioning from single-pilot to dual-pilot operations or vice versa under HeliService USA Part 135 operations.

3.8.3.1 Module 1 – Regulatory Overview and Flight Crew Roles

- Review of 14 CFR Part 135 crew qualification regulations
- OM-A and OM-D sections pertaining to crew configuration
- Responsibilities and authority of PIC vs. SIC in dual-pilot ops
- Decision-making authority in single-pilot scenarios

3.8.3.2 Module 2 – Crew Resource Management (CRM) and Communication

- Effective communication protocols in dual-pilot ops
- Transfer of controls, sterile cockpit procedures
- CRM challenges and techniques in single-pilot operations

3.8.3.3 Module 3 – Normal Procedures Comparison

- Checklist flow differences between single- and dual-pilot operations
- Task-sharing during all phases of flight
- FMS, radio, and ATC workload management in single-pilot ops

3.8.3.4 Module 4 – Abnormal and Emergency Procedures

- Division of responsibilities during emergencies in dual-pilot operations
- Prioritization and workload management under emergency in single-pilot ops
- Case studies and scenario-based training

3.8.3.5 Module 5 – Aircraft Systems and Equipment Considerations

- Review of any equipment that may differ in operation or use between crew configurations (e.g., autopilot, dual controls, radios)
- Use of autopilot as SIC substitute (if authorized) in single-pilot ops



3.8.3.6 Module 6 – Flight Scenarios

- Conducted in aircraft or simulator (if available)
- One scenario in each configuration (single-pilot and dual-pilot) focusing on:
 - CRM/SPCRM (Crew Recourse Management / Single Pilot Crew Recourse Management)
 - TEM (Threat and Error Management)
 - Situational Awareness
 - Departure
 - En-route
 - Arrival and approach
 - Emergency simulation

3.8.3.7 Module 7 – Evaluation

- Oral examination on regulatory and procedural differences
- Flight check including CRM demonstration in the alternate configuration
- Completion of 14 CFR §135.299-line check if transitioning to PIC role

3.8.4 Multi-Crew CRM

OBJECTIVE OF TRAINING:

Multi-crew CRM training is done at operator discretion in excess of regulatory guidance to meet customer specific requirements and promote more effective interaction between HeliService USA flight crews, enhancing safety. Emphasis is placed on the importance of effective coordination and communication between all crew members in various situations and environments. This includes management of stress, workload, utilization of SOP's, and other available resources.

CRM INSTRUCTOR REQUIREMENTS:

1. Pilot on AW139 or/and AW169.
2. Operational knowledge of HeliService USA SOP's.
3. CRMT Course at an approved course by the Training Manager.
4. Demonstrated adequate knowledge and facilitation skills regarding to CRM instruction to the Training Manager.

For recency CRM trainer shall:

1. Conduct at least two CRM training events in any 12-month period.



2. Be assessed within the last twelve months of the 3-year validity period by the operator.
3. Complete CRM trainer refresher training within the 3-year validity period.

HeliService USA provides recurrent, operator conversion, and command CRM training. The training is given in-house and is tailored to the specific HeliService USA operational requirements.

3.8.4.1 Initial CRM Training

The CRM initial course is designed to provide initial CRM training for new pilots, crew members, or communications specialists for HeliService USA. This also offers an additional introduction into HeliService USA safety culture, SOP's, communication, and company specific related CRM elements.

Initial CRM training is a one-time training requirement. Each successive year (12 months after completion of initial) pilots, crew members, and dispatchers will complete the regular recurrent CRM training described and outlined in section 3.8.4.3.

Note: Every HeliService USA pilot shall have completed initial CRM before commencing unsupervised line flying. Line training however is still permitted prior to completion. All required subjects of CRM initial training course are included in the following syllabus.

All the required subjects of the Initial CRM Training are included in the following syllabus:

Initial CRM Training	Duration
Human factors & Human performance and limitations	0:45
Threat and Error Management	0:45
Automation	0:45
Specific type related differences*	0:45
Situational Awareness & Workload Management in Windfarms	1:15
Operator Safety Culture, SOP's	1:30
Effective Communication	1:00
Case study, incorporated in any of the above	1:00
Total Time:	7:45 hours



3.8.4.2 Command CRM Training

As part of the progress to upgrade to an HSUSA designated PIC for offshore operations, the Command Course CRM training program must be completed. The Command Course CRM training program consists of one full day of training (8 hours) utilizing the syllabus specified below.

Command CRM training need only be completed once at HeliService USA and has no expiration date.

All the required subjects of the Command CRM Training are included in the following syllabus:

Command Course CRM	Duration
Human Factors & Threat and Error Management	1:00
Personality awareness, professionalism, stress & fatigue	1:30
Situational Awareness & workload management	1:30
Monitoring & Intervention	1:00
Effective communication & Leadership	0:45
Resilience development, Surprise & Startle Effect	0:45
Operator Safety Culture	1:00
Case study, incorporated in any of the above	-
Total Time:	7:30 hours

3.8.4.3 Recurrent CRM Training

The recurrent CRM Training programme will alternate each year between program A, B and C detailed below; starting with Programme A in 2024.



Each programme consists of one full day of classroom training (8 hours) and is valid for 12 months with a renewal period of 3 months prior to expiration.

3.8.4.3.1 Recurrent CRM Training - A

Training Syllabus - Programme A	Duration
CRM principles and objectives	0:30
Stress and stress management	1:30
Fatigue and vigilance	1:30
Workload management	1:00
Effective communication and coordination inside and outside the flight crew compartment	0:45
Leadership, synergy, delegation and professionalism	0:45
Effective communication and coordination with other operational personnel and ground services	1:00
Case study, incorporated in any of the above.	N/A
Total time:	7:00 hours

3.8.4.3.2 Recurrent CRM Training - B

Training Syllabus - Programme B - MC/SP	Duration
Human factors in aviation	0:30
Human performance and limitations	0:45
Personality awareness, human error and reliability, attitudes and behaviours, self-assessment, self-critique and defence mechanisms	1:00
Cultural differences	0:45
Automation and philosophy on the use of automation	1:30
Specific type related differences for pilots flying multiple helicopters	0:45
Resilience development	1:00

Surprise and startle effect	0:45
Case study, incorporated in any of the above.	1:00
Total time:	8:00 hours

3.8.4.3.3 Recurrent CRM Training - C

Training Syllabus - Programme C - MC/SP	Duration
Threat and error management	0:45
Assertiveness, situational awareness, information acquisition and processing in multi-crew environments	1:30
Monitoring and intervention	0:45
Operators safety culture and company culture, SOPs and factors linked to the type of operation	1:00
Decision making and the FORDEC system	1:30
FORDEC system applied to simulated situation	1:00
Case study, incorporated in any of the above	1:00
Total time:	7:30 hours

3.9 Communications Specialist - Initial Hire

OBJECTIVE OF TRAINING:

To provide communications specialists with the necessary knowledge and preparation to understand HeliService USA policies, procedures, and means of compliance with the Federal Aviation Regulations, while providing flight following services and processing customer requests.

PREREQUISITES:

Each pilot must meet appropriate HeliService USA employment requirements.

COMPLETION STANDARDS:

The crewmember must attend the course and demonstrate appropriate knowledge of basic indoc by successfully completing an oral or written examination.

Applicable Segments	Curriculum	Planned Hours	Training	Chapter / Modules



General Subjects	8	BI Modules 1-5 & 7.
CRM	8	Muli-Crew CRM (OM-D Section 3.8)

3.10 HAA/SAR Personnel

3.10.1 Initial

OBJECTIVE OF TRAINING:

To provide each HAA/SAR crewmember with the necessary knowledge and preparation to understand HeliService USA policies, procedures, and means of compliance with the Federal Aviation Regulations, while providing flight following services and processing customer requests.

PREREQUISITES:

Each HAA/SAR crewmember must meet appropriate HeliService USA employment requirements.

COMPLETION STANDARDS:

The crewmember must attend the course and demonstrate appropriate knowledge of basic indoc by successfully completing an oral or written examination.

Training documentation will be maintained in the individual training records for a period of 24 months following the completion of training utilizing the TF-03 form. Additionally, every 24 months or less, a recurrent training must be accomplished.

Applicable Segments	Curriculum	Planned Training Hours	Chapter/Modules
General Subjects		4	BI Modules 1-3,15 and 17
General Emergency		4	Module 1-18
CRM		8	Muli-Crew CRM (OM-D Section 3.8)
NVG		4	NVG Ground 1-17 and Flight 1-2 Modules

3.10.2 Recurrent

OBJECTIVE OF TRAINING:

This category of training is for medical/SAR crewmembers who have been trained and qualified by HeliService USA. It is intended to refresh the crewmember in the same subjects covered during initial



training and to ensure their continued proficiency. Recurrent training must be accomplished every 24 months.

PREREQUISITES:

Currently qualified as a Medical/SAR crewmember on the aircraft type recurrent training is required.

Applicable Curriculum Segments	Planned Training Hours	Chapter/Modules
General Subjects	1	BI Module 17
General Emergency	3	Module 1-18
CRM	8	Muli-Crew CRM (OM-D Section 3.8)
NVG	2	NVG Ground 1-4 and Flight 1-2 Modules

3.11 Offshore IFR Operating Procedures

3.11.1 Objective of Training

To provide pilots with the necessary knowledge and preparation to conduct safe IFR offshore operations.

3.11.2 Planned Training Hours

Initial	Recurrent
Ground- 2 hours	Ground- 1 hour
Flight- 2 Hours	Flight- 1 hour

3.11.3 Curriculum Prerequisites

Each pilot must meet the following prerequisites:

1. Pilot must possess a valid Pilot Certificate and instrument rating

3.11.4 Completion Standards [14 CFR § 135.293(a)(1) and 135.297(c)(1)(ii)]

The pilot must attend the course and demonstrate adequate knowledge and preparation by successfully completing a written or oral examination. Prior to conducting HEDA/OSAP operations pilots will perform



at a minimum 5 HEDAs and 5 OSAPs from each pilot position.

3.11.5 Curriculum Segment Outline: IFR Offshore Operating Procedures Modules

For Reference Only



3.11.5.1 Initial Ground Training Module

1. Weather requirements:
 - a. Offshore IFR dispatch weather
 - b. Offshore IFR arrival weather
2. HeliService authorized heliports/Areas
3. IFR departures offshore
4. Enroute procedures:
 - a. Letters of agreement
 - b. Offshore routes map
 - c. Route coordinates
5. Offshore flights:
 - a. Offshore/Onshore alternates
 - b. Offshore departure
 - c. Onshore destination
6. Authorized offshore IFR alternates.
7. Offshore weather deviation:
 - a. One station
 - b. Two stations
 - c. Offshore missed approach lost communications procedures.
8. OSAP procedures
 - a. Equipment
 - b. Weather reporting
 - c. Crew qualifications
 - d. Operating limitations
 - e. Landing minimums
 - f. Intermediate segment
 - g. Final approach
 - h. OSAP charts
 - i. OSAP locations
 - j. OSAP update list.
9. HEDA procedures:
 - a. Weather reporting
 - b. Definitions
 - c. Intermediate segment
 - d. Final descent
 - e. Special limitations
 - f. Missed HEDA procedure.
 - g. HEDA chart
 - h. HEDA locations
10. Weather stations



3.11.5.2 Initial Flight Training Module

(May be accomplished in either an approved simulator or the Aircraft).

1. Preflight procedures:
 - a. Weather observations
 - b. Obtaining IFR clearance
 - c. Cockpit management
 - d. CRM briefing
 - e. Sterile cockpit procedures
2. Takeoff and departure phase:
 - a. Area departure
3. Inflight maneuvers:
 - a. Holding
 - b. Navigation
 - c. Communication
 - d. Radar control/ADS-B
 - e. HEDA both seats x 5
 - f. Unusual attitude recovery
4. Landing and approach to landing:
 - a. Approach briefing
 - b. Area arrival
 - c. Stabilized approach procedures
 - d. OSAP/Circling both seats x 5
5. Normal and abnormal systems
6. Postflight procedures:
7. Debriefing

3.11.5.3 Recurrent Ground

OBJECTIVE OF TRAINING:

This category of training is for crewmembers who have been trained and qualified by HeliService USA to utilize OSAP and HEDA approaches. It is intended to refresh the crewmember in the same subjects covered during initial training and to ensure their continued proficiency. Recurrent training must be accomplished every 12 months.

3.11.5.3.1 Module

1. Weather requirements:
 - a. Offshore IFR dispatch weather
 - b. Offshore IFR arrival weather
2. HeliService USA authorized Heliports/Areas
3. IFR departures offshore
4. Enroute procedures:



- a. Letters of agreement
- b. Offshore routes map
- c. Route coordinates
- 5. Offshore flights:
 - a. Offshore/Onshore alternates
 - b. Offshore departure
 - c. Onshore destination
- 6. Authorized offshore IFR alternates.
- 7. Offshore weather deviation:
 - a. One station
 - b. Two stations
 - c. Offshore missed approach lost communications procedures.
- 8. OSAP procedures
 - a. Equipment
 - b. Weather reporting
 - c. Crew qualifications
 - d. Operating limitations
 - e. Landing minimums
 - f. Intermediate segment
 - g. Final approach
 - h. OSAP charts
 - i. OSAP locations
 - j. OSAP update list.
- 9. HEDA procedures:
 - a. Weather reporting
 - b. Definitions
 - c. Intermediate segment
 - d. Final descent
 - e. Special limitations
 - f. Missed HEDA procedure.
 - g. HEDA chart
 - h. HEDA locations
- 10. Weather stations

3.11.5.4 Recurrent Flight

1. OBJECTIVE OF TRAINING:

This category of training is for crewmembers who have been trained and qualified by HeliService USA to utilize OSAP and HEDA approaches. It is intended to refresh the crewmember in the same subjects covered during initial training and to ensure their continued proficiency. This training will serve as the pilot's annual proficiency check for the continued use of OSAP's and HEDA's for HeliService USA. Recurrent training must be accomplished every 12 months.



3.11.5.4.1 Module

1. Preflight procedures:
 - a. Weather observations
 - b. Obtaining IFR clearance
 - c. Cockpit management
 - d. CRM briefing
 - e. Sterile cockpit procedures
2. Takeoff and departure phase:
 - a. Area departure
3. Inflight maneuvers:
 - a. Holding (Either Onshore/Offshore)
 - b. Navigation
 - c. Communication
 - d. Radar control/ADS-B
 - e. a minimum of 1 HEDA and 1 OSAP either seat
 - f. Unusual attitude recovery
4. Landing and approach to landing:
 - a. Approach briefing
 - b. Area arrival
 - c. Stabilized approach procedures
 - d. Landing offshore facility
5. Normal and abnormal systems
6. Postflight procedures:
7. Debriefing



4 Training Segments & Modules

4.1 Basic Indoc

OBJECTIVE OF TRAINING:

To provide pilots with the necessary knowledge and preparation to understand HeliService USA policies, procedures, and means of compliance with the Federal Aviation Regulations, while engaged in the business of air commerce. Additionally, successful completion of this curriculum segment ensures the pilot is fully prepared to enter Aircraft Ground and Flight Training Curriculum Segments.

PREREQUISITES:

Each pilot must meet appropriate HSUSA 14 CFR Part 135 employment requirements.

CURRICULUM SEGMENTS:

The Basic Indoctrination Training curriculum segment is comprised of the following subject areas: Operator Specific and Airman Specific.

COMPLETION STANDARDS:

The crewmember must attend the course and demonstrate appropriate knowledge of basic indoc by successfully completing an oral or written examination.

4.1.1 Operator Specific

4.1.1.1 BI - Module 1

Duties and Responsibilities of HeliService USA Flight Crew

1. Company history, organization, and management structure.
2. Operational concepts, policies, and kind of operation.
3. Company forms, records, and administrative procedures.
4. Employee standards and rules of conduct.
5. Employee compensation, benefits, and contracts.
6. Authority and responsibilities of duty position.
7. Company-required equipment.
8. Company manual organization, revisions, and employee responsibilities concerning manuals.

4.1.1.2 BI - Module 2

Appropriate provisions of the Code of Federal Regulations

1. Flight crewmember's certification, training, and qualification requirements.
2. Medical certificates, physical examination, and fitness for duty requirements.
3. Operational control requirements (dispatch, flight release, or flight locating).
4. Flight duty and rest requirements.



5. Record keeping requirements.
6. Operational Rules in 14 CFR Part 91 and Part 135 (as appropriate) and any other applicable regulations.
7. Appropriate portions of HeliService USA operating manuals.
8. Other appropriate regulations such as flight crew emergency authority, interference with crewmembers, and reporting requirements.
9. Applicable portions of transportation security regulations (Paragraph 1542 - Airport Security, and Paragraph 1544 - Aircraft Operator Security).
10. National Transportation Safety Board (NTSB 830 - Notification and Reporting of Aircraft Accidents or Incidents and Overdue Aircraft, and Preservation of Aircraft Wreckage, Mail, Cargo, and Records).
11. HeliService USA drug and alcohol testing program.
 - a. Drug and alcohol video.
 - b. Drug and alcohol handbook.
 - c. Receipt form signed.
12. Airworthiness (MEL).

4.1.1.3 BI - Module 3

Contents of Certificate and Operations Specifications

1. Regulatory Basis in 14 CFR Part 135 and Title 49 of the United States Code (49 U.S.C.) (formerly the Federal Aviation Act (FA Act) of 1958).
2. Definitions, description, and organization of Operations Specifications.
3. Limitations and authorizations of Operations Specifications.
4. Description of Certificate.
5. Description of FAA Certificate Holding District Office and responsibilities of FAA Principal Operations Inspector.

4.1.2 Airmen Specific

4.1.2.1 BI - Module 4

Operational Control

1. Dispatch, flight release, or flight locating systems and procedures (as applicable).
2. Organization, duties, and responsibilities.
3. Weather and NOTAM information.
4. HeliService USA communications.
5. Aircraft tracking software use and procedures.

4.1.2.2 BI - Module 5

Weight and Balance



1. Definitions (such as zero fuel weight, moments, and inches of datum).
2. General loading procedures and center of gravity computations.
3. Effects of fuel burn and load shifts in flight.
4. Weight and balance forms, load manifests, fuel slips, and other applicable documents.

4.1.2.3 BI - Module 6

Aircraft Performance and Airport Analysis

1. Definitions (such as balanced field, visual meteorological conditions (VMC), obstruction planes, and maximum endurance).
2. Effects of temperature and pressure altitude.
3. General Terminal Instrument Procedures (TERPS) Criteria (obstacle clearance standards).
4. Airport analysis system as appropriate to the type of operation and family or families of aircraft.
5. Effects of contaminated runways.
6. Land and Hold Short Operations (LAHSO) procedures.
7. Runway limitations.

4.1.2.4 BI - Module 7

Meteorology

1. Basic weather definitions (such as forecasts, reports, and symbols).
2. Temperature, pressure, and winds.
3. Atmosphere moisture, fog, and clouds.
4. Air masses and fronts.
5. Thunderstorms, icing, and wind shear.

4.1.2.5 BI - Module 8

Navigation

1. Definitions (such as Class I, Class II navigation).
2. Basic navigational instruments.
3. Dead reckoning (DR) and pilotage concepts and procedures.
4. Navigational Aids (NAVAID).
5. Very high frequency (VHF), very low frequency (VLF), and self-contained systems (as applicable).

4.1.2.6 BI - Module 9

Airspace and ATC Procedures

1. Definitions (such as precision approaches, airways, and automated terminal information service (ATIS)).
2. Description of airspace.
3. Navigation performance and separation standards.



4. Controller and pilot responsibilities.
5. ATC communications.
6. Air traffic flow control.
7. Wake turbulence recognition and avoidance.

4.1.2.7 BI - Module 10

En Route and Terminal Area Charting and Flight Planning

1. Terminology of charting services (such as Jeppesen or FAA National Aeronautical Charting Office (NACO)).
2. Takeoff minimums, landing minimums, and alternate requirements.
3. General company flight-planning procedures.
4. Flight service and international procedures (as applicable).
5. Airport diagrams.
6. Airport ground operational safety (AC 120-74 and AC 91-73).

4.1.2.8 BI - Module 11

Concepts of Instrument Procedures

1. Definitions (such as minimum descent altitude (MDA), height above airport (HAA), height above touchdown (HAT), decision height (DH), Category II (CAT II) instrument landing system (ILS), and no procedure turn required (NOPT)).
2. Holding patterns and procedure turns.
3. Precision approaches.
4. Nonprecision approaches.
5. Circling, visual, and contact approaches (as applicable).
6. Landing minimums (basic turbojet vs standard).
7. Visual cues before and during descent below DH or MDA.

4.1.2.9 BI - Module 12

Flight Hazards

1. Bird strike.
2. CFIT.
3. Inadvertent IMC.
4. Collision avoidance.
5. Brown/white out/flat light.
6. Dynamic rollover.
7. Ground resonance.
8. Settling with power.
9. Loss of Tail Rotor Effectiveness (LTE).



10. Other hazards as appropriate.

4.1.2.10 BI - Module 13

All Weather Surface Operations

1. Surface Movement:
 - a. Day.
 - b. Night.
 - c. Low visibility/reduced visibility.
 - d. Surface Movement Guidance and Control Systems (SMGCS).
 - e. Airport Surface Detection Equipment (ASDE-3).
 - f. Airport Movement Area Safety System (AMASS).
2. Airport markings.
3. Airport signs.
4. Airport lighting.
5. Heliport markings.
6. Heliport signs.
7. Heliport lighting.

4.1.2.11 BI - Module 14

Airport Ground Operational Safety

1. Standard Operating Procedures (AC 120-74 and AC 91-73, as amended).

4.1.2.12 BI - Module 15

Flatlight, Whiteout, Brownout Recognition and Procedures

1. Definitions:
 - a. Brownout.
 - b. Whiteout.
 - c. Flat light.
2. Methods for recognition of brownout, whiteout and flat light conditions:
 - a. High recon.
 - b. Low recon.
3. Methods for avoiding brownout, whiteout and flat light conditions.
4. General Operations Manual:
 - a. Grades of brownout and whiteout.
 - b. Mitigation of brownout and whiteout.
5. Procedures for aircraft handling in whiteout, brownout, and flat light conditions.

4.1.2.13 BI- Module 16



HAA/SAR Pilot Training

1. Risk Analysis- Review HeliService Risk Assessment
2. Local Flying Area (LFA)
 - a. LFA borders;
 - b. Locations of terrain features and obstructions that may pose a hazard
 - c. Available (emergency) and operator-approved airports, heliports, and off-airport landing sites/areas along with appropriate ingress/egress routes and limitations or guidance;
 - d. Preferred low-altitude flyways and minimum cruising altitudes throughout the LFA ;
 - e. Air traffic control (ATC) frequencies, instrument approaches, and the terms and conditions of any local Memorandums of Understanding (MOU) established between the ATC facilities and the operator;
 - f. Weather resources available within the LFA ;
 - g. Local communications procedures between the aircraft and any communications centers, specialists, or Operations Control Centers (OCC);
 - h. Specific local area adverse weather cues; and
3. Flight planning and weather minimums (refer to § 135.615).
4. Flightcrew functions and responsibilities, including CRM as it pertains to interface between medical personnel and flightcrew members.
5. Obstacle recognition and avoidance.
6. Handling, loading, and unloading of special medical equipment such as stretchers, and ventilators.
7. Appropriate restraint of infants, pediatric patients, and passengers who may pose a threat to the safety of the aircraft and crew, to include patients who are hysterical or combative.
8. Hospital heliport operations and procedures.
9. Day and night unimproved landing area (scene) operations.
10. International operations (if appropriate).
11. BBP, biohazard, and infection control, including prevention and control of infectious diseases.
12. Refueling procedures and methods to ensure fuel quality.

4.1.2.14 BI - Module 17

Medical/SAR Crewmember 135.621 Training

All medical/SAR crewmembers will be trained on the following:

1. Passenger briefing requirements in § 135.117(a)
2. Physiological aspects of flight;
3. Patient loading and unloading;
4. Safety in and around the helicopter;
5. In-flight emergency procedures;



6. Emergency landing procedures;
7. Emergency evacuation procedures;
8. Efficient and safe communications with the pilot; and
9. Operational differences between day and night operations, if appropriate.
10. External power unit (EPU) door and cart
11. Medical equipment – loading and unloading/securing;
12. Oxygen system and outlets;
13. Audio panel and headsets;
14. Lights and vents;
15. Cabin cleaning;
16. Emergency locator transmitter (ELT);
17. Emergency fuel shutoff;
18. Radios –VHF, FM, 800 megahertz (MHz).

**** Briefing requirements in 135.117(a)**

Before each takeoff each pilot in command of an aircraft carrying passengers shall ensure that all passengers have been orally briefed on—

1. Smoking.
2. The use of safety belts
3. The placement of seat backs in an upright position before takeoff and landing;
4. Location and means for opening the passenger entry door and emergency exits;
5. Location of survival equipment;
6. If the flight involves extended overwater operation, ditching procedures and the use of required flotation equipment;
7. If the flight involves operations above 12,000 feet MSL, the normal and emergency use of oxygen; and
8. Location and operation of fire extinguishers.
9. If beyond autorotational glide distance from shore, use of life preservers, ditching procedures and emergency exit from the rotorcraft in the event of a ditching; and the location and use of life rafts and other life preserver devices.

Training record documentation must be maintained in the individual training records for 24 months following the completion of training.

4.1.2.15 BI - Module 18

Offshore IFR Pilot training

1. Weather requirements:
 - a. Offshore IFR dispatch weather
 - b. Offshore IFR arrival weather
2. HeliService authorized heliports/Areas



3. IFR departures offshore
4. Enroute procedures:
 - a. Letters of agreement
 - b. Offshore routes map
 - c. Route coordinates
5. Offshore flights:
 - a. Offshore/Onshore alternates
 - b. Offshore departure
 - c. Onshore destination
6. Authorized offshore IFR alternates.
7. Offshore weather deviation:
 - a. One station
 - b. Two stations
 - c. Offshore missed approach lost communications procedures.
8. OSAP procedures
 - a. Equipment
 - b. Weather reporting
 - c. Crew qualifications
 - d. Operating limitations
 - e. Landing minimums
 - f. Intermediate segment
 - g. Final approach
 - h. OSAP charts
 - i. OSAP locations
 - j. OSAP update list.
9. HEDA procedures:
 - a. Weather reporting
 - b. Definitions
 - c. Intermediate segment
 - d. Final descent
 - e. Special limitations
 - f. Missed HEDA procedure.
 - g. HEDA chart
 - h. HEDA locations
10. Weather stations

4.2 General Emergency

INTRODUCTION:

There are two types of emergency training provided by HSUSA to meet the requirements of 14 CFR § 135.331 Emergency Training.



1. General Emergency Training:

This training is required for all crewmembers on each item specified in 14 CFR § 135.331. The training is divided into two distinct parts, Emergency Drill and Emergency Situation.

- a. Emergency Drill: Instruction and practice in the actual use of certain items of emergency equipment such as fire extinguishers, life vests, and first aid equipment. This training is specific to the type aircraft being trained.
- b. Emergency Situation: Instruction on the factors involved and the procedures to be followed when an emergency occurs. This includes ditching, rapid decompression, aircraft fires and persons needing first aid.

Note 1: All items identified in 14 CFR § 135.331 must be conducted every twelve (12) months (14 CFR § 135.351).

Note 2: The emergency drill training modules, which require the crewmember to actually operate the items of emergency equipment, must be conducted at least every 24 months. The emergency drill training may be accomplished by pictorial presentation or demonstration.

Note 3: Initial and Recurrent General Emergency Training outlined in this segment meets all the regulatory requirements of 14 CFR Part 135. Because of logistic and other requirements particular to general emergency training, modules are presented concurrently with other curriculum segments such as Basic Indoctrination, Aircraft Ground, and Security Training. This permits greater training continuity and flexibility.

2. Aircraft Specific Emergency Training:

This training includes instruction and practice in emergency and abnormal procedures associated with aircraft systems, structural design, and operational characteristics. It provides flight crewmembers with the skills required to perform the procedures outlined in the Rotorcraft Flight Manual. It also includes training on the location of specific items of emergency equipment included in both the Aircraft Ground and Flight Curriculum Segments.

OBJECTIVE OF TRAINING:

To provide crewmembers with the necessary knowledge and preparation to be proficient in the subjects required by 14 CFR § 135.331.

PREREQUISITES:

Each pilot must meet appropriate HSUSA 14 CFR Part 135 employment requirements.

CURRICULUM SEGMENTS:

The General Emergency Training curriculum segment is comprised of the following subject areas: Emergency Drill and Emergency Situation.

COMPLETION STANDARDS:

The crewmember must attend the course and demonstrate appropriate knowledge of the principles associated with General Emergency Training by successfully completing an oral or written examination.



4.2.1 Emergency Drill

4.2.1.1 General Emergency - Module 1

Ditching (If Applicable)

1. Ditching procedures.
2. Survival equipment.

4.2.1.2 General Emergency - Module 2

Emergency Evacuation

1. Evacuation.
2. Actual operation and use of slide/slide raft (if applicable) (this requirement needs to be accomplished only once during initial new hire or initial equipment training).

4.2.1.3 General Emergency - Module 3

Fire Extinguisher and Smoke Control

1. Inspection tags, seals, dates, and proper charge levels.
2. Removal and storage of hand held extinguishers.
3. Actual discharge of each type of hand held extinguisher.
4. Inspection and use of Protective Breathing Equipment (PBE), oxygen masks, and smoke goggles.

4.2.1.4 General Emergency - Module 4

Operation and Use of Emergency Exits

1. Actual operation (open/close) of each type normal exit.
2. Actual operation of each type emergency exit.
3. Procedures for egress.

4.2.1.5 General Emergency - Module 5

Crew and Passenger Oxygen – If Applicable

1. Type.
2. Location.
3. Purpose.
4. Inspection tags, seals, dates, and pressures.
5. Removal and stowage.
6. Procedures for use.
7. Actual operation/demonstration of each type bottle and mask.

4.2.1.6 General Emergency - Module 6



Removal of Life Rafts from the Aircraft, Inflation of the Life Rafts, Use of Life Lines, and Boarding of Passengers and Crew – If Applicable

1. Type.
2. Location.
3. Procedures for use.
4. Actual operation/demonstration.

4.2.1.7 General Emergency - Module 7

Donning and Inflation of Life Vests and the Use of Other Individual Flotation Devices – If Applicable

1. Type.
2. Location.
3. Procedures for use.
4. Actual operation/demonstration.

4.2.2 Emergency Situation

4.2.2.1 General Emergency - Module 8

Crewmember Duties and Responsibilities

1. Emergency assignments.
2. PIC's emergency authority.
3. Reporting incidents and accidents.

4.2.2.2 General Emergency - Module 9

Crew Coordination

1. PIC/SIC.
2. Cabin crew notification procedures.
3. Ground agency notification procedures (FAA, airport authority).
4. HeliService USA's communication procedures.

4.2.2.3 General Emergency - Module 10

Aircraft Fires

1. Fire in flight, or on the surface.
2. Smoke control procedures.
 - a. Electrical equipment and related circuit breakers found in cabin areas.
3. Principles of combustion and classes of fires.
4. Toxic fumes and chemical irritants.
5. Use of appropriate hand held extinguishers.



6. Lavatory fires.
7. Protective Breathing Equipment (PBE), oxygen masks, smoke goggles.
8. Personal electronics fires.

4.2.2.4 General Emergency - Module 11

First Aid Equipment

1. Contents of first aid kit.
2. Requirements for first aid kit integrity.
3. Use of individual items.

4.2.2.5 General Emergency - Module 12

Illness, Injury and Basic First Aid

1. Principles of CPR.
2. Ear and sinus block.
3. Seeking medical assistance.
4. Treatment of shock.
5. Heart attack.
6. Pregnancy situations.

4.2.2.6 General Emergency - Module 13

Ground Evacuation

1. Aircraft configuration.
2. Directing passenger flow.
3. Blocked or jammed exit.
4. Fuel spills and other ground hazards.
5. Handicapped/disabled person.

4.2.2.7 General Emergency - Module 14

Ditching

1. Cockpit and cabin preparation.
2. Passenger briefing.
3. Crew coordination.

4.2.2.8 General Emergency - Module 15

Previous Aircraft Accidents/Incidents

1. HeliService USA accidents/incidents.
2. NTSB accident report reviews.



3. Human factors/considerations.
4. NASA reporting system.

4.2.2.9 General Emergency - Module 16

Crewmember Incapacitation

1. HeliService USA procedures
2. Reporting requirements (NTSB)
3. Interference with crewmembers

4.2.2.10 General Emergency - Module 17

Basic Survival – As Required

1. Area of operation
2. Desert
3. Over water
4. Jungle
5. Arctic

4.2.2.11 General Emergency - Module 18

Hijacking/Bomb Threat and Other Unusual Situations

1. Hijack procedures
2. Bomb threat procedures
3. Security coordinator responsibilities
4. In flight intercept signals and procedures

4.3 CRM

OBJECTIVE OF TRAINING:

To provide crewmembers with the necessary Operator Specific knowledge to effectively manage all available resources and to develop skills in recognizing and responding to the conditions that lead to aircrew error. The intended effect of this training is to reduce the frequency and severity of crew errors and thereby reduce the frequency of accidents and incidents.

PREREQUISITES:

Each pilot must meet appropriate HSUSA 14 CFR Part 135 employment requirements.

COMPLETION STANDARDS:

The crewmember must attend the course and demonstrate appropriate knowledge of the principles associated with CRM by successfully completing an oral or written examination.

4.3.1 CRM - Module 1



1. Authority of the PIC.
2. Communication processes including:
 - a. Decisions and coordination.
 - b. Communication with Air Traffic Control, personnel performing flight locating and other operational functions, and passengers.
3. Building and Maintaining of a Flight Team.
4. Workload and Time Management.
5. Situational Awareness.
6. Effects of fatigue on performance, avoidance strategies and countermeasures Effects of stress and stress reduction strategies.
7. Aeronautical Decision-Making (ADM) and judgement training tailored to the operator's (HeliService USA's) flight operations and aviation environment.
8. Risk Management.
9. Operational Pitfalls and Human Factors.

4.4 HAZMAT

OBJECTIVE OF TRAINING:

To provide crewmembers and persons performing or directly supervising any interaction with Hazardous Material with the necessary knowledge and preparation to accomplish their specific job functions, as required by 14 CFR § 135.503. This training is laid out in detail within the HAZMAT training manual.

PREREQUISITES:

Each pilot must meet appropriate HSUSA 14 CFR Part 135 employment requirements.

COMPLETION STANDARDS:

The crewmember must attend the course and demonstrate appropriate knowledge of the principles associated with HAZMAT by successfully completing an oral or written examination.

GENERAL:

[14 CFR § 135.507(a)]

1. The training program for the Hazardous Material (HM) shall apply to all crewmembers and persons performing or directly supervising the acceptance, rejection, handling, storage incident to transport, packing of company material, or loading of any Hazardous Material.
2. The scope of training should cover all functions listed in the "Hazardous Materials Training Requirements Table" according to the individuals specific job related functions.
3. Training records will be maintained for all personnel requiring and undergoing training for Hazardous Materials and will show all information relative to training.
4. The schedule of training shall be coordinated with the Federal Aviation Administration (FAA) Certificate Holding District Office and the nearest office of FAA, Civil Aviation Security.
5. The person assigned responsibility for HM training will contact the nearest office of FAA, Civil



Aviation Security, at least annually and coordinate HM procedures and training requirements to ensure that procedures and training remain up to date. The person assigned responsibility of HM training may also accomplish this requirement by attending a Hazardous Material Transportation training course offered by an external vendor.

Note 1: Please refer to the HAZMAT training manual for all modules and training requirements.

4.5 HNVGO

General

This curriculum specifies the training program that HeliService USA deploys for helicopter Night Vision Goggle Operations (HNVGO). This section applies to PICs, SICs, Check Pilot, Instructors and non-pilot crew who are operating this equipment. The flight and ground training prescribed herein is to be used by instructors in conducting a course of training to ensure that HeliService USA pilots employing NVGs have and maintain the necessary skills, knowledge, proficiency and judgment to perform all assigned flight duties employing night vision goggles. Further, so that HeliService USA pilots can perform their duties in a safe and efficient manner in compliance with all applicable regulations.

Prerequisites

Pilots must have completed:

1. Basic Indoctrination Ground Training.
2. Helicopter Specific Ground Training.
3. Complete all the CBT NVG Modules.

Training Aids/Equipment/Courseware

1. NVGs (Type i.e. brand/model).
2. Computer overhead projectors
3. PowerPoint Presentations
4. Computer Workstation
5. HeliService USA Helicopters
6. NVG User's Manual (Type i.e. brand)/model).
7. RFM NVG supplement
8. Minimum Equipment List.
9. 14 CFR 61.57(g) 91.207(h)
10. OpsSpec A050
11. CBT Modules

Completion Standards

Upon completion of this curriculum the HSUSA pilot will demonstrate proficiency and orally show



knowledge of the course material.

Note 1: Hours may be reduced based on pilot proficiency provided all training elements are conducted to standard. Enough training and practice must be conducted from both the right and left seat (If applicable) to ensure the individuals competency in conducting the required normal, abnormal and emergency training and checking.

Note 2: NVG training may be conducted concurrently with the Helicopter specific flight training

Testing and Checking Method

1. Computer Examination (80% minimum score corrected to 100%)
2. Written Test (80% minimum score corrected to 100%)
3. Oral Evaluation
4. Practical Exercise Evaluation

Authorized Instructors

The following HSUSA personnel are authorized to teach modules of this curriculum segment as appropriate:

1. Director of Operations
2. Chief Pilot
3. HSUSA Check Pilot
4. HSUSA Instructors

4.5.1 NVG Initial Training

Planned Training Hours

Ground	Flight
6	5

4.5.1.1 NVG-Initial Pilot Specific Ground Modules

4.5.1.1.1 NVG-I-Ground Module 1 - NVG Human Factors

1. Introduction to NVGs-General
2. Anatomy and physiology of the Eye
3. Common Visual Deficiencies
4. Limitations of Night Vision



5. Types of Vision
6. Proper Night Vision Viewing Technique
7. Methods Used to Protect Night Vision
8. Self-Imposed Stress and Night Vision
9. Factors which Effect Dark Adaptation
10. Visual Illusion

4.5.1.1.2 NVG-I-Ground Module 2 - Introduction to Night Vision Goggle Equipment and Operation

1. Description of equipment (i.e. Make/Model of NVG)
2. Additional Equipment
3. Components and Operation Sequence
4. Nomenclature, Functions, and Pre-Operational Inspection
5. Operator Checks
6. Mounting to Helmet
7. Flipping up and down
8. Power
9. Adjusting and focusing
10. Operational Defects
11. Cosmetic Blemishes
12. Dismounting the NVG from the Helmet
13. Care and Cleaning, for Adverse Environmental Conditions

4.5.1.1.3 NVG-I-Ground Module 3 - Environmental Considerations

1. Light Sources
 - a. Natural light sources
 - b. Visual and Near IR heat energy
2. Inadvertent IMC while using NVG's
 - a. Signs of IMC
 - b. Disorientation of Conditions
 - c. Ambient Light
 - d. Reflectivity
 - e. Altitude and Angle of Moon and Sun
3. Cautions and Hazards for NVG Use
 - a. Precipitation
 - b. Environmental Hazards Affecting
 - c. Visibility
 - d. Terrain Features Hindering or
 - e. Aiding Visibility Conditions
 - f. Towers and Antennas



4.5.1.1.4 NVG-I-Ground Module 4 - Planning and Emergency Operations

1. Preflight Planning
 - a. Features Near Route
2. Limited Instrument Flight
3. Spatial Disorientation
4. Inadvertent IMC
5. Personal Minimums
6. White Out, Brown Out Conditions and Flat Light
7. Use of External Lights
8. Failure of NVG's

4.5.1.1.5 NVG-I-Ground Module 5 - Practical Exercise NVG Mounting/Focus/Operational

1. Techniques Elements
2. Assemble NVG equipment
3. Perform initial focus (using indoor eye lane or manufacturer's testing/focusing device)
4. Practice night viewing outdoors

4.5.1.2 NVG-Initial-Ground Operator Specific Modules**4.5.1.2.1 NVG-I-Ground Module 6 - Authorized Types of Operations**

1. Landing and Takeoff
2. Enroute Procedures

4.5.1.2.2 NVG-I-Ground Module 7 - Forms, Records, and Administrative Procedures

1. Night Vision Goggle Log
2. Pilot Flight and Duty Time Summary

4.5.1.2.3 NVG-I-Ground Module 8 - Authority and Responsibility of Duty Position

1. Required Aircraft Documents for NVG Operations
2. NVG STC RFM Supplement
3. Aircraft Logbooks
4. Risk Assessment
5. Preflight Actions at the start of duty for NVG operations
 - a. NVG preflight inspection, operational checkout, and discrepancy log
 - b. Aircraft cockpit instrument
6. Equipment and unauthorized lighting
7. Before Takeoff Briefing
8. Post Flight Actions



4.5.1.2.4 NVG-I-Ground Module 9 - Overview of FAR's Applicable to NVG Operations

1. FAR 61.57
2. FAR 91: 91.13, 91.209
3. FAR 135: 135.159, 135.179, 135.203, 135.205, 135.207, 135.229

4.5.1.2.5 NVG-I-Ground Module 10 - Air Carrier Certificate and Operations Specifications

1. Elements
2. Regulatory Basis, Authorization, and Limitations
3. Definitions and Description
4. A050 OpSpecs

4.5.1.2.6 NVG-I-Ground Module 11 – HeliService USA OMA NVG

1. General Information
2. NVG Operation Requirements
3. Recording of NVG Flight Time
4. Operational Compliance
5. Minimum Safe Altitudes for NVG Operations
6. NVG Weather Minimums
7. Aircraft Modification Required for NVG Flight
8. Aircraft External Lighting
9. Use of EFB
10. Additional Lighting Sources Authorized
11. Authorized Areas of Operations for NVG's
12. NVG Maintenance and Inspections
13. Reporting of NVG Irregularities and Discrepancies
14. Pilot in Command Qualification for NVG Operations
15. Inadvertent IMC Procedures
16. Flight Operations
17. Emergency Egress

4.5.1.3 NVG Initial Ground Aircraft Specific Modules

The subjects in this section are specific to HeliService USA's Helicopter type that the Pilot/Crewmember is going to operate using night vision goggles. This segment is designed to be done in and around the helicopter, at night or in a dark hangar, using external power.

4.5.1.3.1 NVG-I-Ground Module 12 - Lighting Systems

1. Internal
 - a. Cockpit Lighting
 - b. Instrument Lighting



- c. Avionics Lighting
- 2. External
 - a. Anticollision Lights
 - b. Position Lights
 - c. Landing and Search Lights

4.5.1.3.2 NVG-I-Ground Module 13 - Caution/Warning System

Interpretation of System while wearing NVGs

4.5.1.3.3 NVG-I-Ground Module 14 - Cockpit Familiarization

- 1. Familiarization of aircraft in night environment
- 2. NVG equipment assembly
- 3. Familiarization of aircraft internal and external lighting systems
- 4. Instruments and controls visibility

4.5.1.3.4 NVG-I- Ground Module 15 - Other System Familiarization

- 1. Radar Altimeter
- 2. GPS

4.5.1.3.5 NVG-I Ground Module 16 - NVG CRM

- 1. NVG CRM general considerations
- 2. NVG CRM aspects in scene operations
- 3. NVG CRM aspects in confined area operations
- 4. Safety risk assessment and hazard identification
- 5. Air Medical Resource Management (AMRM)
- 6. Hazard recognition and mitigation techniques.
- 7. LZ Command
- 8. CFIT Avoidance Tools
- 9. Moving Map
- 10. HTAWS
- 11. GPS
- 12. Radar Altimeters
- 13. Autopilots

4.5.1.3.6 NVG Initial Ground Testing Module

- 1. **Testing Element**-A CBT electronic test or a written exam can be administered. A minimum score of 80% corrected to 100% is required for both types of tests for satisfactory completion of this segment.
- 2. **Record Requirements**-The HSUSA instructor pilot will certify and document satisfactory completion



of Initial NVG Ground Training for each pilot. The instructor pilot certification of satisfactory completion will be placed in the pilot training record.

Note:

The month the pilot completes the training will be designated as the Base Month (Eligibility Period) for recurrent training.

4.5.1.4 NVG Initial Flight Training Modules

Prerequisite:

Pilots must have completed the Initial NVG Initial Ground Training curriculum prior entering this training.

NVG Initial Flight Module Lessons

HSUSA Pilots will be trained at night or simulated night if a simulator is to be used. Using night vision goggles with access to the flight controls. The order and number of events to be performed will be at the discretion of the instructor to assure maximum training benefit and efficiency.

4.5.1.4.1 NVG-I-Flight Module 1 (2 Hours)

1. Preflight planning
2. Preflight Inspection of NVG
3. Checklist usage
4. Aircraft systems
5. Internal lighting
6. External lighting techniques
7. Engine start procedures
8. Lifting to a hover
9. Hovering maneuvers
10. Hover taxi
11. Air taxi
12. Quick stops
13. Land from a hover
14. Slope operations
15. OEI prior to TDP (If applicable)
16. Normal takeoff
17. CAT A Profiles (If applicable)
18. Climbs (aided/unaided)
19. Turns (aided/unaided)
20. Climbing/descending turns (aided/unaided)
21. Pinnacle operations
22. Confined area operations



23. Pattern entry/exit (aided/unaided)
24. Scanning techniques
25. Ground hazards recognition
26. Normal approach to landing
27. Steep approach to landing
28. Cat A Profiles (If Applicable)
29. OEI after to LDP (If Applicable)

4.5.1.4.2 NVG-I-Flight Module 2 (2 Hours)

1. IFR Takeoff (If applicable)
2. OEI after TDP
3. Climb (Aided/Unaided)
4. NVG cross country to another airport
5. IFR procedures (If applicable)
6. System Emergencies
7. Unusual attitude recovery
8. NVG inadvertent IMC
9. Forced landing
10. Emergency Decent
11. NVG failure (with recovery)
12. Flat light operations (oral if applicable)
13. Scanning techniques
14. External lighting techniques
15. Pattern entry
16. Normal approach
17. OEI prior to LDP
18. Running landing

4.5.1.4.3 Qualification Module Checking Initial NVG Flight Training

The required maneuvers (events) for the Initial NVG Checking Module will consist of an oral evaluation and checking of the maneuvers (elements) listed below.

Written or Oral Test

1. 135.293 oral
2. Tail rotor failure (oral/simulator only)
3. Dynamic rollover (oral or simulator only)
4. Low rotor rpm (oral or simulator only)
5. Anti-torque system failure (oral or simulator only)

Ground Operations



1. Preflight inspections
2. Start procedures
3. Taxi and ground hover
4. Pre-takeoff checks

Takeoff and departure

1. Normal takeoff
2. With powerplant failure
3. Rapid deceleration

In flight maneuvers

1. Settling with power (Simulator or may be waived)
2. Unusual attitude recovery

Landing and Approaches to Landing

1. Normal landing
2. Landing with engine out (ME Only)

Non-Normal and Emergency Procedures

1. System malfunction
2. Recovery from IIMC
3. Power failure and autorotation to a power recovery (SE Only)
4. Hovering autorotation (SE only)
5. Confined Area
6. Pinnacle operations
7. Slope operations
8. Ground hazard recognition
9. Brownout/Whiteout/Flat light operations (Oral only)
10. Use of external lighting

Pilot Initial NVG Flight Qualification will be documented on the TF-03 (8410) form and posted in the Pilot Training Record Folder. The month the NVG Qualification is completed will be designated as the Pilot's NVG Base Month (Eligibility Period).

4.5.2 NVG Recurrent

Planned Training Hours

Ground	Flight
2	2

4.5.2.1 NVG-Recurrent Ground

Pilots will complete the following CBT-CTS NVG Training Modules for Recurrent Ground Training. Electronic tests will be administered and corrected to 100%. Authorized company personnel will certify



that the required training and checking was completed.

4.5.2.1.1 NVG-R-Ground Module 1 - NVG Human Factors

1. Introduction to NVGs-General
2. Anatomy and physiology of the Eye
3. Common Visual Deficiencies
4. Limitations of Night Vision
5. Types of Vision
6. Proper Night Vision Viewing Technique
7. Methods Used to Protect Night Vision
8. Self-Imposed Stress and Night Vision
9. Factors which Effect Dark Adaptation
10. Visual Illusion

4.5.2.1.2 NVG-R-Ground Module 2 - Introduction to Night Vision Goggle Equipment and Operation

1. Description of equipment (i.e. Make/Model of NVG)
2. Additional Equipment
3. Components and Operation Sequence
4. Nomenclature, Functions, and Pre-Operational Inspection
5. Operator Checks
6. Mounting to Helmet
7. Flipping up and down
8. Power
9. Adjusting and focusing
10. Operational Defects
11. Cosmetic Blemishes
12. Dismounting the NVG from the Helmet
13. Care and Cleaning, for Adverse Environmental Conditions

4.5.2.1.3 NVG-R-Ground Module 3 - Environmental Considerations

1. Light Sources
 - a. Natural light sources
 - b. Visual and Near IR heat energy
2. Inadvertent IMC while using NVG's
 - a. Signs of IMC
 - b. Disorientation of Conditions
 - c. Ambient Light
 - d. Reflectivity
 - e. Altitude and Angle of Moon and Sun



3. Cautions and Hazards for NVG Use
 - a. Precipitation
 - b. Environmental Hazards Affecting
 - c. Visibility
 - d. Terrain Features Hindering or
 - e. Aiding Visibility Conditions
 - f. Towers and Antennas

4.5.2.1.4 NVG-R-Ground Module 4 - Planning and Emergency Operations

1. Preflight Planning
2. Limited Instrument Flight
3. Spatial Disorientation
4. Inadvertent IMC
5. Personal Minimums
6. White Out, Brown Out Conditions and Flat Light
7. Use of External Lights
8. Failure of NVG's

4.5.2.2 NVG Recurrent Flight Training

Pilots will complete the following NVG Training Modules for Recurrent flight training

4.5.2.2.1 NVG-R-Flight Training-Module 1-Ground Operations and Normal Maneuvers

1. Preflight Inspection
2. Start Procedures
3. Checklist usage
4. Taxi and Ground Hover
5. Normal Takeoff
6. Rapid deceleration
7. Confined Area Operations
8. Pinnacle Operations
9. Slope Operations
10. Use of external Lighting
11. Normal Approach to Landing x3

4.5.2.2.2 NVG-R-Flight Training-Module 2-Emergency/Malfunctions Procedures

1. Takeoff with powerplant failure (If Applicable)
2. Landing with powerplant failure (If Applicable)
3. Settling with Power**
4. Unusual attitude recovery
5. Systems malfunctions



6. Recovery from IIMC
7. Ground Hazard Recognition
8. Brown Out/White Out/Flat Light Operations (Oral Only)

4.5.3 Medical Personnel (MP) NVG Curriculum Segment

Purpose

This curriculum specifies the training program for ensuring MP comply with the regulatory requirements to conduct helicopter flight employing night vision goggles (NVGs). The ground and demonstration training prescribed herein is to be used by HSUSA AIs in conducting a course of training to ensure that all MP employing NVGs have and maintain the necessary skills, knowledge, proficiency and judgment to perform all assigned duties employing night vision goggles. Further, that MP can perform those duties in a safe and efficient manner and in compliance with FARs.

Objective

The primary objective of NVG training is to provide MP with the necessary knowledge for understanding the basic function and operation of NVG systems, including the integration with aircraft systems and operational procedures.

Planned Training Hours

This training includes 5 hours of ground training and will include 1 hour of NVG demonstration and use conducted at night in the helicopter.

Initial	Demonstration	Recurrent CBT
4	1	4

Primary References

HSUSA OMA
HSUSA OMD

Prerequisites

MP will complete all CBT- Modules listed in the MP NVG Initial Ground Training section as the initial entry into the MP NVG Curriculum segment. The CBT- Modules are part of the Planned Training Hours for this curriculum segment.

Training Aids/Equipment/Courseware



1. NVGs (Make/Model)
2. Computer overhead projectors
3. Computer Workstation-CBT
4. White Board
5. HSUSA Helicopters
6. ITT User's Manual TM F4949-1
7. RFM NVG supplement
8. Minimum Equipment List

Instructional Delivery Methods

1. CBT
2. Lecture-PowerPoint
3. Cockpit Instruction (EPU power systems training)
4. Practical Exercises (PE)

Completion Standards

Upon completion of this curriculum the MP will demonstrate proficiency and knowledge of the course material.

Testing and Checking Methods

1. CBT- Computer Examination (80% minimum score corrected to 100%) Written Test (80% minimum score corrected to 100%)
2. Oral Evaluation
3. Practical Exercise Evaluation
4. Qualification Curriculum Segment requirements are listed paragraphs 20.11.13 and 20.11.14.2

Authorized Instructors (AI)

The following HSUSA personnel are authorized to teach modules of this Curriculum Segment as appropriate:

1. Director of Operations
2. Chief Pilot
3. HSUSA Check Pilot
4. HSUSA Instructor



4.5.3.1 NVG Initial MP Ground Training

4.5.3.1.1 NVG Initial MP Specific Training Modules

4.5.3.1.1.1 NVG-I-MP Ground Module 1 - Night Vision Physiology (CBT)

NVG-I-MP Ground Module 1 - Night Vision Physiology (CBT)

1. Introduction
2. Loss of visual cue
3. Target acquisition and tracking
4. The effects of ground lighting
5. Eye Physiology
6. Anterior portion of the eye
7. Receptor cells, rods and cones
8. Scotopic vision
9. Photopic vision
10. Visual Anomalies
11. Central and Peripheral visual processing
12. Spatial disorientation of the Labyrinthine and Proprioceptive sensory systems Autokinesis
13. Techniques to reduce visual illusions
14. The Black Hole Illusion
15. The Distance Illusion Phenomenon
16. False Horizon and Understanding false perceptions
17. Flicker Vertigo

4.5.3.1.1.2 NVG-I-MP Ground Module 2 - Night Operations Training (CBT)

1. Factors that affect night flight vision
2. Night Operations
3. Supplemental lighting colors
4. Night vision recovery and Rhodopsin
5. Night vision and the blue-green spectrum
6. Lightning and strobe lights
7. The night-blind spot of the Fovea
8. Protecting Night Vision
9. Oxygenation
10. The ill-effects of smoking and night flight
11. Rhodopsin and vitamin A

4.5.3.1.1.3 NVG-I-MP Ground Module 3- Night Goggles Operations (CBT)

1. What are NVG?
2. Why are NVG needed and what are the risks in using NVG?



3. How do NVG work?
4. Four generations of NVG.
5. Regulations: NVG Applicability and Definitions (FAR Part 61.1).
 - a. FAR Part 61.31
 - i. 61.31(k) Additional training required for night vision goggle operations.
 - b. FAR Part 61.51
 - i. 61.51(k) Logging night vision goggle time.
 - ii. 61.51(b)(3) Conditions of flight.
 - c. FAR Part 61.57 61.57(f) Night vision goggle operating experience.
 - i. 61.57(g) Night vision Goggles proficiency check.
 - d. FAR Part 61.195
 - i. 61.195(k) Training for Night Vision Goggles operations.
 - e. FAR Part 91.205
 - i. 91.205(h) Night Vision goggles operations.
 - f. Components, Operation, and Characteristics of NVGs
 - i. Night Vision Goggle System Overview (Make/Model)
 - ii. Features and Benefits
 - iii. Specifications
 - iv. Performance Parameters
 - g. Standard Accessories
 - i. The Binocular Assembly
 - ii. The Mount Assembly
 - h. Preflight Adjustment
 - i. Assessment Procedures
 - ii. Alignment procedures
 - iii. Focusing procedures
 - i. Assessment of function
 - i. Operational Defects
 - ii. Shading
 - iii. Edge Glow
 - iv. Flashing, Flickering, or Intermittent Operation
 - v. Cosmetic Blemishes
 - vi. Bright spots
 - vii. Dark spots
 - viii. Honeycomb
 - ix. Distortion
 - x. Cleaning Steps
 - j. NVG Limitations
 - i. Field of View (FOV)
 - ii. Monochromatic Image
 - iii. Monochromatic Adaptation



- iv. Required Light (Ambient or Artificial)
- v. Hight Illumination and Sky Glow
- vi. Sky Glow
- vii. Moon Illumination
- viii. Dark Adaptation
- ix. Depth Perception and Object Recognition
- x. Low Contrast Terrain and Contrast Reversal
- k. Environmental Issues - Fog and Rain
 - i. Winter Operations
 - ii. Snow
 - iii. Ice
 - iv. Airframe Ice
 - v. Airborne Obscurants
 - vi. Haze, Dust, Sand, Smoke
 - vii. Rotor wash Affects
 - viii. Brownouts
- l. Spatial Disorientation
- m. Fatigue and Stress
- n. Complacency and Overconfidence
- o. External Lighting Precautions
 - i. Red LED lighting systems - FAA SAFO 09007
 - ii. Laser Pointers
- p. NVG Emergency Procedures
 - i. Single Tube NVG Failure
 - ii. Dual Tube NVG Failure
- q. Night Planning and Risk Assessment
 - i. NVG Preflight Check
 - 1. Preflight NVG adjustment
 - 2. Exterior Lighting Check

- NVG Flight Evaluation

- 1. Risk Assessment

4.5.3.1.1.4 NVG-I-MP Ground Module 4 - Practical Exercise NVG

- 1. Mounting/Focus/Operations Techniques
 - a. Assemble NVG equipment-Helmet Mounting
 - b. Perform initial focus
 - c. Perform infinity focus outdoors at night
 - d. Practice night viewing outdoors

4.5.3.1.2 NVG Initial MP Operator Specific Training Segment



4.5.3.1.2.1 NVG-I-MP Ground Module 5 - Authorized Types of Operation

1. Landing and Takeoff
2. Enroute Procedures

4.5.3.1.2.2 NVG-MP-I-Ground Module 6 - Forms, Records, and Administrative

1. Night Vision Goggle Log
2. Recording Deficiencies

4.5.3.1.2.3 NVG-I-MP Ground Module 7 - MP Duty Position Requirements

1. HSUSA Risk Assessment-NVG Aspect for PICs
2. Preflight Actions at the start of duty for NVG operations
3. NVG preflight inspection, operational checkout, and discrepancy log
4. Equipment and unauthorized lighting
5. Before Takeoff Briefing
6. Post Flight Actions

4.5.3.1.2.4 NVG-I-MP Ground Module 8-Regulatory Guidance Applicable to MP NVGs

1. HSUSA Helicopter STCs
2. Applicable Operations Specifications

4.5.3.1.2.5 NVG-I-MP Ground Module 9 - HSUSA Operations Manual (OMA)

1. General Information
2. NVG Operation Requirements
3. Minimum Safe Altitudes for NVG Operations
4. NVG Weather Minimums
5. Aircraft External Lighting Requirements
6. Additional Lighting Sources Authorized
7. Inadvertent IMC Procedures-MP Duties

4.5.3.1.2.6 NVG-I-MP Ground Module 10 - NVG CRM/AMRM

1. NVG CRM/AMRM general considerations
2. NVG CRM aspects in scene operations
3. NVG CRM aspects in confined area operations
4. LZ Command
5. CFIT Avoidance Tools

4.5.3.1.3 NVG Initial MP Demonstration-Use Training

The subjects in this section are specific to HSUSA Helicopters in which the MP are going to use the NVGs. This training is designed to be instructed using a combination of the classroom and HSUSA helicopters.



The demonstration / use training in the helicopter will be at night, augmented by external power.

Note: MPs are required to complete one hour of NVG demonstration and use, which must be accomplished at night and may be accomplished in flight or on the ground. HSUSA will conduct this training on the ground.

4.5.3.1.3.1 NVG-I-MP Demo Module 1—Lighting Systems

1. Internal
2. Cockpit Lighting
3. Instrument Lighting
4. Avionics Lighting
5. Cabin Lighting
6. External
7. Anticollision Lights
8. Position Lights
9. Landing and Search Lights

4.5.3.1.3.2 NVG-I-MP Demo Module 2—Cockpit Familiarization

1. Familiarization of aircraft in night environment
2. Instruments and NVG compatibility

4.5.3.1.4 Qualification Module—NVG Initial Ground Training/Demonstration Use

4.5.3.1.4.1 Testing Element NVG Initial Ground Training

A CBT- electronic test will be administered for the CBT modules. A written exam will be administered for the non-CBT classroom modules. A minimum score of 80% corrected to 100% is required for both type tests for satisfactory completion of this segment.

4.5.3.1.4.2 Record Requirement

Authorized Instructors will certify satisfactory completion of this curriculum on the appropriate HSUSA form. The completed form will be filed in the MP Training Record Folder.

Note: The month the MP completes the NVG initial training will be designated as the Base Month (Eligibility Period) for recurrent training.

4.5.3.1.5 MP NVG Recurrent Training



MP will complete the following CBT- NVG Training Modules for Recurrent Ground Training in their Base Month (Eligibility Period).

4.5.3.1.5.1 NVG-R Recurrent Modules (CBT)

4.5.3.1.5.2 NVG-R-MP Ground Module 1 - Night Vision Physiology (CBT)

4.5.3.1.5.3 NVG-R-MP Ground Module 2 - Night Operations Training (CBT)

4.5.3.1.5.4 NVG-R-MP Ground Module 3 - Night Vision Goggles Operations (CBT)

4.5.3.1.5.5 Qualification Module-NVG Recurrent Training (CBT)

Testing Element

After the completion of the required material a CBT- electronic test will be administered. A minimum score of 80% corrected to 100% is required for satisfactory completion of this segment.

4.5.3.1.6 Record Requirements

Authorized Instructors will certify the satisfactory completion of the NVG training modules by printing or downloading the CBT certificate and adding it to the MP training file. All module records will be recorded on the appropriate HSUSA training form by an authorized instructor and be placed in the MP training folder.

4.6 Instructor Pilots

OBJECTIVE OF TRAINING:

To provide pilots with the necessary knowledge to qualify for duty as a Flight Instructor (Aircraft).

Note: Training is specific to the aircraft for which the instructor is being trained.

REQUIREMENTS:

1. HeliService USA Flight Instructors shall meet the requirements of 14 CFR § 135.338 and 135.340.
2. Contract Flight Instructors shall meet the appropriate requirements of 14 CFR § 135.338, 135.340, and 14 CFR Part 142.
3. Flight Instructor (Aircraft) Ground may be given credit for Flight Instructor (Simulator) Ground.
4. For Instructor Pilots of NVGs,
 - a. A Flight Instructor may not conduct HSUSA Pilot or MP training for Night Vision Goggle Operations unless that Flight Instructor or Check Airman:
 - i. Is current and qualified as a Part 135 PIC IAW with the HSUSA Training Program and this chapter.
 - ii. Has completed HSUSA's Initial HNVGO Ground and Flight curriculum.



- iii. Has a minimum of 50 hours NVG flight time. Or has been approved by the Director of Operations.
- iv. Has logged at least 100 HNVGO's as the sole manipulator of the controls while operating helicopters. Or has been approved by the Director of Operations.
- v. Completes the HSUSA Training Program requirements for Flight Instructors or Check Airman.

PREREQUISITES:

1. Holds the airman certificates and ratings required to serve as a PIC in operations under 14 CFR Part 135.
2. Has satisfactorily completed the appropriate training phases for the aircraft, including recurrent training, that are required to serve as a PIC in operations under 14 CFR Part 135.
3. Has satisfactorily completed the appropriate proficiency or competency checks that are required to serve as a PIC in operations under 14 CFR Part 135.
4. Has satisfactorily completed the applicable training requirements of 14 CFR § 135.340.
5. Holds at least a Class III medical certificate unless serving as a required crewmember, in which case holds a Class I or Class II medical certificate as appropriate.
6. Has satisfied the recency of experience requirements of 14 CFR § 135.247.

Note 1: Flight Instructors who do not hold a medical certificate may function as Flight Instructors in an aircraft if functioning as Non-required crewmembers, but may not serve as flight crewmembers in operations under 14 CFR Part 135.

Note 2: Flight instructors must conduct instruction within the preceding 24 calendar months while under the observation of an FAA inspector, an HSUSA check airman, or an aircrew-designated examiner employed by the certificate holder.

4.6.1 Simulator**4.6.1.1 Initial****4.6.1.1.1 Instructor Pilot Simulator Initial - Module 1****Ground Training Module**

Initial ground training for flight instructor:

1. Flight Instructor duties, functions, and responsibilities.
2. The applicable Code of Federal Regulations and HeliService USA policies and procedures.
3. The applicable methods, procedures, and techniques for conducting flight instruction.
4. Proper evaluation of student performance including the detection of:
 - a. Improper and insufficient training.
 - b. Personal characteristics of an applicant that could adversely affect safety.



5. The corrective action in the case of unsatisfactory training progress.
6. The approved methods, procedures, and limitations for performing the required normal, abnormal, and emergency procedures in the aircraft.
7. Except for holders of a flight instructor certificate:
 - a. The fundamental principles of the teaching learning process.
 - b. Teaching methods and procedures.
 - c. The instructor student relationship.

4.6.1.1.2 Instructor Pilot Simulator - Module 2

Simulator Instructor Panel Operation Module

Proper operation of flight simulator and flight training device controls and systems:

1. Simulator initialization:
 - a. Aircraft configuration.
 - b. Environmental conditions.
 - c. Airport selection.
2. Aircraft positioning/repositioning.
3. Windshear selections and activation.
4. System malfunction selection, indications, and clearing procedures.
5. Area and approach map tracking and clearing.
6. Initialization and termination of control loading and motion base.

Proper operation of environmental and fault panels:

1. Controls.
2. Simulator safety and emergency procedures:
 - a. Crash override.
 - b. Automatic safety devices.
 - c. Emergency shutoff switches.
 - d. Fire procedures.
 - e. Emergency evacuation capabilities.
 - f. Maintenance interphone/P.A. system.

Limitations of simulation:

1. General elements.

Minimum equipment requirements for each curriculum:

1. General elements.

4.6.1.1.3 Instructor Pilot Simulator Initial - Module 3

Flight Training Module



Initial flight training for flight instructors (simulator):

1. Training and practice in the required normal, abnormal, and emergency procedures to ensure competence to conduct the flight instruction required by 14 CFR Part 135. These maneuvers and procedures must be accomplished in full or in part in a flight simulator or in a flight training device.

4.6.1.2 Transition

4.6.1.2.1 Instructor Pilot Simulator Transition - Module 1

Ground Training Module

Transition ground training for flight instructor:

1. The approved methods, procedures, and limitations for performing the required normal, abnormal, and emergency procedures applicable to the type, class, or category aircraft to which the flight instructor is in transition.

4.6.1.2.2 Instructor Pilot Simulator Transition - Module 2

Simulator Instructor Panel Operation Module

Proper operation of flight simulator and flight training device controls and systems:

1. Simulator initialization:
 - a. Aircraft configuration.
 - b. Environmental conditions.
 - c. Airport selection.
2. Aircraft positioning/repositioning.
3. Windshear selections and activation.
4. System malfunction selection, indications, and clearing procedures.
5. Area and approach map tracking and clearing.
6. Initialization and termination of control loading and motion base.

Proper operation of environmental and fault panels:

1. Controls.
2. Simulator safety and emergency procedures:
 - a. Crash override.
 - b. Automatic safety devices.
 - c. Emergency shutoff switches.
 - d. Fire procedures.
 - e. Emergency evacuation capabilities.
 - f. Maintenance interphone/P.A. system.

Limitations of simulation:



1. General elements.

Minimum equipment requirements for each curriculum:

1. General elements.

4.6.1.2.3 Instructor Pilot Simulator Transition - Module 3

Flight Training Module

Transition flight training for flight instructor (simulator):

1. Training and practice in the required normal, abnormal, and emergency procedures to ensure competence to conduct the flight instruction required by 14 CFR Part 135. These maneuvers and procedures must be accomplished in full or in part in a flight simulator or in a flight training device.

4.6.1.3 Recurrent

4.6.1.3.1 Instructor Pilot Simulator Recurrent - Module 1

Ground Training Module

Recurrent ground training for flight instructor:

1. Duties, functions, and responsibilities.
2. Fundamentals of instructions (not required for individuals who hold a current CFI/CFII).
3. HeliService USA policies and procedures.
4. HeliService USA Pilot Training Manual.
5. Record keeping.
6. Training/testing media.
7. Applicable Code of Federal Regulations (to include 14 CFR Part 135 subpart H training).
8. Human factors.
9. Proper evaluation of student's performance (grading).
10. The approved methods, procedures, briefings, and limitations for performing the required normal, abnormal, and emergency procedures.
11. Instructor facilitated training.
12. Instructor Operating Station (IOS) SIM/FTD.

4.6.1.3.2 Instructor Pilot Simulator Recurrent - Module 2

Flight Training Module

Recurrent flight training for flight instructors:

1. Complete requirements of Line Observation Training (LOT) special Curriculum (14 CFR § 135.338).
2. Complete all required emergency procedures IAW the emergencies outlined in the Recurrent



Lesson Plan.

3. Conducting safety measures for emergency situations that are likely to occur during instruction.

4.6.2 Aircraft

4.6.2.1 Initial

4.6.2.1.1 Instructor Pilot Aircraft - Module 1

Ground Training Module

Initial ground training for flight instructor:

1. Flight Instructor duties, functions, and responsibilities.
2. The applicable Code of Federal Regulations and HeliService USA policies and procedures.
3. The applicable methods, procedures, and techniques for conducting flight instruction.
4. Proper evaluation of student performance including the detection of:
 - a. Improper and insufficient training.
 - b. Personal characteristics of an applicant that could adversely affect safety.
5. The corrective action in the case of unsatisfactory training progress.
6. The approved methods, procedures, and limitations for performing the required normal, abnormal, and emergency procedures in the aircraft.
7. Except for holders of a flight instructor certificate:
 - a. The fundamental principles of the teaching learning process.
 - b. Teaching methods and procedures.
 - c. The instructor student relationship.

4.6.2.1.2 Instructor Pilot Aircraft - Module 2

Flight Training Module

Initial flight training for flight instructor (aircraft):

1. The safety measures for emergency situations that are likely to occur during instruction.
2. The potential results of improper, untimely, or non-execution of safety measures during instruction.
3. Training and practice from the left and right pilot seats in the required normal, abnormal and emergency procedures to ensure competence to conduct the flight instruction required by 14 CFR Part 135.
4. The safety measures to be taken from either pilot seat for emergency situations that are likely to develop during instruction.

4.6.2.2 Transition



4.6.2.2.1 Instructor Pilot Transition Aircraft - Module 1

Ground Training Module

Transition ground training for flight instructor:

1. The approved methods, procedures, and limitations for performing the required normal, abnormal, and emergency procedures applicable to the type, class, or category aircraft to which the flight instructor is in transition.

4.6.2.2.2 Instructor Pilot Transition Aircraft - Module 2

Flight Training Module

Transition flight training for flight instructor (aircraft):

1. The safety measures for emergency situations that are likely to occur during instruction.
2. The potential results of improper, untimely, or non-execution of safety measures during instruction.
3. Training and practice from the left and right pilot seats in the required normal, abnormal and emergency procedures to ensure competence to conduct the flight instruction required by 14 CFR Part 135.
4. The safety measures to be taken from either pilot seat for emergency situations that are likely to develop during instruction.

4.6.2.3 Recurrent

4.6.2.3.1 Instructor Pilot Recurrent Aircraft - Module 1

Ground Training Module

Recurrent ground training for flight instructor:

1. Duties, functions, and responsibilities.
2. Fundamentals of instructions (not required for individuals who hold a current CFI/CFII).
3. HeliService USA policies and procedures.
4. HeliService USA Pilot Training Manual.
5. Training/testing media.
6. Applicable Code of Federal Regulations (to include 14 CFR Part 135 subpart H training).
7. Human Factors.
8. Proper evaluation of student's performance (Grading).
9. The approved methods, procedures, briefings, and limitations for performing the required normal, abnormal, and emergency procedures.
10. Instructor facilitated training.



11. Instructor operating station (IOS) SIM/FTD.

4.6.2.3.2 Instructor Pilot Recurrent Aircraft - Module 2

Flight Training Module

Recurrent flight training for flight instructors:

1. Complete all required emergency procedures IAW the emergencies outlined in the Recurrent Lesson Plan.
2. Conducting Safety Measures for Emergency Situations that are likely to occur during instruction.

4.7 Check Pilots

OBJECTIVE OF TRAINING:

To provide pilots with the necessary knowledge to qualify for duty as a Check Pilots (Simulator). NOTE: Training is specific to the aircraft for which the check airman is being trained.

REQUIREMENTS:

1. HSUSA Check Pilots shall meet the requirements of 14 CFR § 135.337 and 135.339, and be approved as such by the CHDO.
2. Contract Check Pilots shall meet the appropriate requirements of 14 CFR § 135.337, 135.339, and 14 CFR Part 142; be approved as a training center evaluator by the TCPM and then be designated by the POI as a Contract Check Pilot for HSUSA.
3. Check Pilot (Simulator) Ground may be given credit for Check Pilot (Aircraft) Ground.
4. For Check Pilots of NVGs,
 - a. A Check Pilot may not conduct HSUSA Pilot or MP checking for Night Vision Goggle Operations unless that Flight Instructor or Check Airman:
 - i. Is current and qualified as a Part 135 PIC IAW with the HSUSA Training Program and this chapter.
 - ii. Has completed HSUSA's Initial HNVGO Ground and Flight curriculum.
 - iii. Has a minimum of 50 hours NVG flight time. Or has been approved by the Director of Operations.
 - iv. Has logged at least 100 HNVGO's as the sole manipulator of the controls while operating helicopters. Or has been approved by the Director of Operations.
 - v. Completes the HSUSA Training Program requirements for Check Pilot.

PREREQUISITES:

1. Holds the airman certificates and ratings, except medical certificate, required to serve as a PIC in operations under 14 CFR Part 135.
2. NOTE: Check Pilot who do not hold an appropriate medical certificate may function as Check Pilot (Simulator), but may not serve as flight crewmembers in operations under 14 CFR Part 135.
3. Has satisfactorily completed the appropriate training phases for the aircraft, including recurrent



training, that are required to serve as a PIC in operations under 14 CFR Part 135.

4. Has satisfactorily completed the appropriate proficiency or competency checks that are required to serve as a PIC in operations under 14 CFR Part 135.
5. Has satisfactorily completed the applicable training requirements of 14 CFR § 135.339. Has been approved by the administrator for the Check Pilot duties involved.
6. A Check Pilot (Aircraft) must have satisfied the recency of experience requirements of 14 CFR § 135.247.
7. All HSUSA Check Pilots will be trained and qualified to conduct NVG training and evaluations.
8. A Check Pilot (Simulator) must accomplish the following:
 - a. Fly at least two flight segments as a required crewmember for the type, class, or category aircraft involved within the 12 month period preceding the performance of any flight instructor duty in a flight simulator, or
 - b. Satisfactorily complete an approved line observation program within the period prescribed by that program and that must precede the performance of any check airman duty in a flight simulator.

Note 1: The required flight segments or line observation program are considered completed in the month required if completed in the calendar month before, or in the calendar month after, the month in which they are due.

Note 2: Check Pilot must have an observed ride by and FAA inspector within the preceding 24 calendar months.

4.7.1 Simulator

4.7.1.1 Initial

4.7.1.1.1 Check Airman Simulator Initial - Module 1

Ground Training Module

Initial ground training for Check Airman:

1. Check Airman duties, functions, and responsibilities.
2. The applicable Code of Federal Regulations and the certificate holder's policies and procedures.
3. The applicable methods, procedures, and techniques for conducting the required checks.
4. Proper evaluation of student performance including the detection of:
 - a. Improper and insufficient training.
 - b. Personal characteristics of an applicant that could adversely affect safety.
5. The corrective action in the case of unsatisfactory checks.
6. The approved methods, procedures, and limitations for performing the required normal, abnormal, and emergency procedures in the aircraft.
7. Conducting checks IAW the HeliService USA Testing Standards.



8. Record keeping.
9. Testing media.
10. Common mistakes and emergency situations
11. NVG teaching techniques
12. Teaching visual cues
13. Teaching the mounting and operation of the NVGs
14. Required NVG maneuvers for FAR 135 evaluations.
15. General review of NVG Ground Modules (Refer to Chapter 4.5)

Note: Item 6. above is normally accomplished during aircraft specific ground training.

4.7.1.1.2 Check Airman Simulator Initial - Module 2

Simulator Instructor Panel Operation Module

Proper operation of flight simulator and flight training device controls and systems:

1. Simulator initialization:
 - a. Aircraft configuration.
 - b. Environmental conditions.
 - c. Airport selection.
2. Aircraft positioning/repositioning.
3. Windshear selections and activation.
4. System malfunction selection, indications, and clearing procedures.
5. Area and approach map tracking and clearing.
6. Initialization and termination of control loading and motion base.

Proper operation of environmental and fault panels:

1. Controls.
2. Simulator safety and emergency procedures:
 - a. Crash override.
 - b. Automatic safety devices.
 - c. Emergency shutoff switches.
 - d. Fire procedures.
 - e. Emergency evacuation capabilities.
 - i. Maintenance interphone/P.A. system.

Limitations of simulation:

1. General elements.

Minimum equipment requirements for each curriculum:

1. General elements.



4.7.1.1.3 Check Airman Simulator Initial - Module 3

Flight Training Module

Initial flight training for Check Airman (simulator):

1. Training and practice in conducting flight checks in the required normal, abnormal, and emergency procedures to ensure competence to conduct the flight checks required by 14 CFR Part 135. This training and practice must be accomplished in a flight simulator, or in a flight training device.

4.7.1.2 Transition

4.7.1.2.1 Check Airman Simulator Transition - Module 1

Ground Training Module

Transition ground training for Check Airman:

1. The approved methods, procedures, and limitations for performing the required normal, abnormal, and emergency procedures applicable to the aircraft to which the check airman is in transition.

Note: The above is normally accomplished during aircraft specific ground training.

4.7.1.2.2 Check Airman Simulator Transition - Module 2

Simulator Instructor Panel Operation Module

Proper operation of flight simulator and flight training device controls and systems:

1. Simulator initialization:
 - a. Aircraft configuration.
 - b. Environmental conditions.
 - c. Airport selection.
2. Aircraft positioning/repositioning.
3. Windshear selections and activation.
4. System malfunction selection, indications, and clearing procedures.
5. Area and approach map tracking and clearing.
6. Initialization and termination of control loading and motion base.

Proper operation of environmental and fault panels:

1. Controls.
2. Simulator safety and emergency procedures:
 - a. Crash override.
 - b. Automatic safety devices.
 - c. Emergency shutoff switches.



- d. Fire procedures.
- e. Emergency evacuation capabilities.
- f. Maintenance interphone/P.A. system.

Limitations of simulation:

- 1. General elements.

Minimum equipment requirements for each curriculum:

- 1. General elements.

4.7.1.2.3 Check Airman Simulator Transition - Module 3

Flight Training Module

Transition flight training for Check Airman (simulator):

- 1. Training and practice in conducting flight checks in the required normal, abnormal, and emergency procedures to ensure competence to conduct the flight checks required by 14 CFR Part 135. This training and practice must be accomplished in a flight simulator, or in a flight training device.

4.7.1.3 Recurrent

4.7.1.3.1 Check Airman Simulator Recurrent - Module 1

Ground Training Module

Recurrent ground training for Check Airman:

- 1. Duties, functions, and responsibilities.
- 2. HeliService USA policies and procedures.
- 3. HeliService USA Pilot Training Manual.
- 4. Record keeping.
- 5. Testing media.
- 6. Applicable Code of Federal Regulations (to include 14 CFR Part 135 subpart H training).
- 7. Proper evaluation of student's performance IAW the HeliService USA Testing Standards.
- 8. The approved methods, procedures, briefings, and limitations for performing the required normal, abnormal, and emergency procedures.
- 9. Instructor Operating Station (IOS) SIM/FTD.
- 10. Common mistakes and emergency situations
- 11. NVG teaching techniques
- 12. Teaching visual cues
- 13. Teaching the mounting and operation of the NVGs
- 14. Required NVG maneuvers for FAR 135 evaluations.
- 15. General review of NVG Ground Modules (Refer to Chapter 4.5)



4.7.1.3.2 Check Airman Simulator Recurrent - Module 2**Flight Training Module**

Recurrent flight training for Check Airman:

1. Conducting safety measures for emergency situations that are likely to occur during instruction.

4.7.2 Aircraft**4.7.2.1 Initial****4.7.2.1.1 Check Airman Aircraft Initial - Module 1****Ground Training Module**

Initial ground training for Check Airman:

1. Check Airman duties, functions, and responsibilities.
2. The applicable Code of Federal Regulations and the certificate holder's policies and procedures.
3. The applicable methods, procedures, and techniques for conducting the required checks.
4. Proper evaluation of student performance including the detection of:
 - a. Improper and insufficient training.
 - b. Personal characteristics of an applicant that could adversely affect safety.
5. The corrective action in the case of unsatisfactory checks.
6. The approved methods, procedures, and limitations for performing the required normal, abnormal, and emergency procedures in the aircraft.
7. Conducting checks IAW the HeliService USA Testing Standards.
8. Record keeping.
9. Testing media.
10. Common mistakes and emergency situations
11. NVG teaching techniques
12. Teaching visual cues
13. Teaching the mounting and operation of the NVGs
14. Required NVG maneuvers for FAR 135 evaluations.
15. General review of NVG Ground Modules (Refer to Chapter 4.5)

Note: Item 6. above is normally accomplished during aircraft specific ground training.

4.7.2.1.2 Check Airman Aircraft Initial - Module 2**Flight Training Module**

Initial flight training for Check Airman (Aircraft):

1. The safety measures for emergency situations that are likely to develop during a check.



2. The potential results of improper, untimely, or non-execution of safety measures during a check.
3. Training and practice in conducting flight checks from the left and right pilot seats in the required normal, abnormal and emergency procedures to ensure competence to conduct the pilot flight checks required by 14 CFR Part 135.
4. The safety measures to be taken from either pilot seat for emergency situations that are likely to develop during checking.

Note 1: Flight Training may be accomplished in full or in part in flight, in a flight simulator, or in a flight training device, as appropriate.

4.7.2.2 Transition

4.7.2.2.1 Check Airman Aircraft Transition - Module 1

Ground Training Module

Transition ground training for Check Airman:

1. The approved methods, procedures, and limitations for performing the required normal, abnormal, and emergency procedures applicable to the aircraft to which the check airman is in transition.

Note: The above is normally accomplished during aircraft specific ground training.

4.7.2.2.2 Check Airman Aircraft Transition - Module 2

Flight Training Module

Transition flight training for Check Airman (Aircraft):

1. The safety measures for emergency situations that are likely to develop during a check.
2. The potential results of improper, untimely, or Non-execution of safety measures during a check.
3. Training and practice in conducting flight checks from the left and right pilot seats in the required normal, abnormal and emergency procedures to ensure competence to conduct the pilot flight checks required by 14 CFR Part 135.
4. The safety measures to be taken from either pilot seat for emergency situations that are likely to develop during checking.

Note: Flight Training may be accomplished in full or in part in flight, in a flight simulator, or in a flight training device, as appropriate.

4.7.2.3 Recurrent

4.7.2.3.1 Check Airman Aircraft Recurrent - Module 1



Ground Training Module

Recurrent ground training for Check Airman (Aircraft):

1. Duties, functions, and responsibilities.
2. HeliService USA policies and procedures.
3. HeliService Pilot Training Manual.
4. Record keeping.
5. Testing media.
6. Applicable Code of Federal Regulations (to include 14 CFR Part 135 subpart H training).
7. Proper evaluation of student's performance IAW the HeliService USA testing standards.
8. The Approved Methods, Procedures, Briefings, and Limitations for performing the required Normal, Abnormal, and Emergency Procedures.
9. Instructor Operating Station (IOS) SIM/FTD.
10. Common mistakes and emergency situations
11. NVG teaching techniques
12. Teaching visual cues
13. Teaching the mounting and operation of the NVGs
14. Required NVG maneuvers for FAR 135 evaluations.
15. General review of NVG Ground Modules (Refer to Chapter 4.5)

4.7.2.3.2 Check Airman Aircraft Recurrent - Module 2

Flight Training Module

Recurrent flight training for Check Airman (Aircraft):

1. Conducting safety measures for emergency situations that are likely to occur during instruction.

4.8 Aircraft Ground & Flight

OBJECTIVE OF TRAINING:

To provide newly hired HeliService USA pilots with the training required to operate HSUSA aircraft while engaged in the business of air commerce. This will include demonstration, instruction, practice of maneuvers, and procedures pertinent to any assigned aircraft. At the end of a flight training curriculum, the pilot will be able to safely and efficiently operate the aircraft while performing the appropriate pilot duties and responsibilities.

PREREQUISITES:

Each pilot must meet appropriate HSUSA 14 CFR Part 135 employment requirements.

PROGRAMMED TRAINING HOURS:

See appropriate table in either the training curriculum or aircraft ground & flight modules.

TRAINING AIDS/EQUIPMENT/COURSEWARE:



1. Computer overhead projectors
2. Computer Workstation- CBT
3. White Board Handouts
4. Aircraft
5. Simulator
6. Aircraft Manufacturer's Training Manual
7. Aircraft Rotorcraft Flight Manual (RFM)
8. Helicopters Electronic Flight Bag (EFB)

INSTRUCTIONAL DELIVERY METHODS:

1. CBT
2. Lecture-PowerPoint
3. Cockpit Instruction (EPU power systems training)
4. Practical Exercises (PE)

COMPLETION STANDARDS:

At the end of the Ground and Flight Training Curriculum Segment the pilot shall be able to demonstrate to the Administrator, or Check Airmen, normal, abnormal, and emergency procedures and checklists in a timely and sequentially correct manner, and perform all the maneuvers and procedures in accordance with the qualification modules listed in this chapter.

4.8.1 AW169**OBJECTIVE OF TRAINING:**

To provide newly hired HeliService USA pilots with the training required to operate the Leonardo AW-169 while engaged in the business of air commerce. This will include demonstration, instruction, practice of maneuvers, and procedures pertinent to any assigned aircraft. At the end of a flight training curriculum, the pilot will be able to safely and efficiently operate the aircraft while performing the appropriate pilot duties and responsibilities.

PREREQUISITES:

Each pilot must meet appropriate HSUSA 14 CFR Part 135 employment requirements.

PROGRAMMED TRAINING HOURS:

See appropriate table in either the training curriculum or aircraft ground & flight modules.

COMPLETION STANDARDS:

At the end of the Ground and Flight Training Curriculum Segment the pilot shall be able to demonstrate to the Administrator, or Check Airmen, normal, abnormal, and emergency procedures and checklists in a timely and sequentially correct manner, and perform all the maneuvers and procedures in accordance with the qualification modules listed in this chapter.



4.8.1.1 AW169 Initial

4.8.1.1.1 AW169 Ground Modules

4.8.1.1.1.1 AW169 General Operational Subjects Modules

4.8.1.1.1.1.1 AW169 General Operational Subjects Module - 1 **Dispatch, Forms, Weight and Balance**

1. Communications and flight-locating procedures.
2. W&B calculation methods and documentation.
3. Flight manifest procedures.

4.8.1.1.1.1.2 AW169 General Operational Subjects Module - 2 **Performance & Hazardous Conditions**

1. The use of charts, tables, tabulated data, and other related manual information.
2. Fuel consumption calculations.
3. Normal, abnormal, and emergency performance calculations.
4. Meteorological and weight-limiting performance factors.
5. Inoperative equipment performance limiting factors (MEL/CDL).

4.8.1.1.1.1.3 AW169 General Operational Subjects Module - 3 **Aircraft Specific Adverse Weather Practices:**

1. Icing.
2. Turbulence.
3. Heavy precipitation.
4. Thunderstorms.
5. Low visibility.
6. Recognizing and Avoiding Flat-Light, Whiteout, and Brownout Conditions.

4.8.1.1.1.1.4 AW169 General Operational Subjects Module - 4 **Electronic Flight Book:**

1. Overview.
2. Weight & Balance.
3. ForeFlight - Flight Planning.
4. Synchronization.
5. Digital Documents.
6. Limitations.
7. Emergency procedures.

4.8.1.1.2 AW169 Systems Modules



4.8.1.1.1.2.1 AW169 Systems Module - 1

Helicopter General

1. Helicopter Description.
2. Technical Publications.

4.8.1.1.1.2.2 AW169 Systems Module - 2

Dimensions and Areas

1. General.
2. Principal Dimensions.

4.8.1.1.1.2.3 AW169 Systems Module - 3

Leveling and Weighing

1. General.
2. Mass and C.G. Data.

4.8.1.1.1.2.4 AW169 Systems Module - 4

Servicing

1. General.

4.8.1.1.1.2.5 AW169 Systems Module - 5

Aircrew Information

1. General.
2. Aircraft Release / Operation Limitations.
3. Normal Procedure.
4. Emergency Procedures.
5. Performance Data.

4.8.1.1.1.2.6 AW169 Systems Module - 6

Vibration and Noise Analysis and Attenuation

1. General.

4.8.1.1.1.2.7 AW169 Systems Module - 7

Environmental Control

1. General.
2. Distribution.
3. Heating.
4. Cooling.
5. Temperature Control.



6. Integrated Environmental Control System.

4.8.1.1.1.2.8 AW169 Systems Module - 8

Auto Flight

1. General.
2. Autopilot.
3. Speed-Attitude Correction.

4.8.1.1.1.2.9 AW169 Systems Module - 9

Communications

1. General.
2. Speech Communications.
3. Audio Integrating and Voice Command Systems.

4.8.1.1.1.2.10 AW169 Systems Module - 10

Electrical Power

1. General.
2. DC Generation.
3. External Power.
4. DC Electrical Load Distribution.

4.8.1.1.1.2.11 AW169 Systems Module - 11

Equipment / Furnishings

1. General.
2. Flight Compartment.
3. Passenger / Operating Crew Compartment.
4. Emergency.

4.8.1.1.1.2.12 AW169 Systems Module - 12

Fire Protection

1. General.
2. Detection.
3. Extinguishing.

4.8.1.1.1.2.13 AW169 Systems Module - 13

Fuel

1. General.
2. Storage.



3. Distribution.
4. Indicating.

4.8.1.1.1.2.14 AW169 Systems Module - 14

Hydraulic Power

1. General.
2. Main.
3. Auxiliary.
4. Indicating.

4.8.1.1.1.2.15 AW169 Systems Module - 15

Ice and Rain Protection

1. General.
2. Pitot and Static.
3. Windows, Windshields, Canopies and Doors.

4.8.1.1.1.2.16 AW169 Systems Module - 16

Indicating / Recording System

1. General.
2. Independent Instruments.
3. Recorders.
4. Central Warning System.
5. Central Display System.

4.8.1.1.1.2.17 AW169 Systems Module - 17

Landing Gear

1. General.
2. Main Gear and Doors.
3. Nose / Tail Gear and Doors.
4. Extension and Retraction.
5. Wheels and Brakes.
6. Steering.
7. Position and Warning.

4.8.1.1.1.2.18 AW169 Systems Module - 18

Lights

1. General.
2. Flight Compartment.
3. Passenger Compartments.



4. Cargo and Service Compartments.
5. Exterior.
6. Emergency Lighting.

4.8.1.1.1.2.19 AW169 Systems Module - 19

Navigation

1. General.
2. Flight Environment Data.
3. Attitude and Direction.
4. Landing and Taxiing Aids.
5. Independent Position Determining.
6. Dependent Position Determining.
7. Flight Management Computing.

4.8.1.1.1.2.20 AW169 Systems Module - 20

System Integration and Display

1. General.
2. Processing and Integration.

4.8.1.1.1.2.21 AW169 Systems Module - 21

Fuselage

1. General.
2. Airframe Structure.

4.8.1.1.1.2.22 AW169 Systems Module - 22

Main Rotors

1. General.
2. Rotor Blades.
3. Rotor Heads.
4. Rotating Controls, Rotor Shafts / Swash Plate Assemblies.
5. Indicating.

4.8.1.1.1.2.23 AW169 Systems Module - 23

Main Rotor Drives

1. General
2. Engine / Gearbox Couplings.
3. Gearboxes.
4. Mounts and Attachments.
5. Indicating.



4.8.1.1.1.2.24 AW169 Systems Module - 24

Tail Rotor

1. General.
2. Rotor Blades.
3. Rotor Head.
4. Rotating Controls.

4.8.1.1.1.2.25 AW169 Systems Module - 25

Tail Rotor Drive

1. General.
2. Shafts.
3. Gearboxes.
4. Indicating.

4.8.1.1.1.2.26 AW169 Systems Module - 26

Rotors Flight Control

1. General.
2. Rotor Control.
3. Anti-Torque Rotor Control (Yaw Control).
4. Servo-Control System.

4.8.1.1.1.2.27 AW169 Systems Module - 27

Power Plant

1. General.

4.8.1.1.1.2.28 AW169 Systems Module - 28

Crew Escape and Safety

1. General.
2. Impact Protection and Flotation.

4.8.1.1.2 Simulator Modules

PLANNED TRAINING HOURS:

Each Flight/Simulator/FTD Training Module is scheduled for the total hours indicated in the table below. Both Pilot Flying (PF) and Pilot Monitoring (PM) training times are shown. This module may be broken down into separate sessions at the instructors discretion.

CREW COMPLEMENT:

Flight training is generally conducted as a crew; however, a pilot training may complete the course alone (single PIC) or with another pilot acting as second crewmember. The pilot acting as second crewmember



must be:

1. A qualified line pilot, or
2. A qualified flight instructor and/or check airmen, or
3. A pilot being trained in the same course

If the course is conducted single PIC the total time must be conducted as PF.

Flight Simulator	PF	PM	Total Time
Flight Simulator Module No. 1	8.0	8.0	16.0
HEMS Module No. 1 (Night)	1.0	1.0	2.0

BRIEF/DEBRIEF:

The instructor will ensure an hour prior and half hour after each session is scheduled to allow for a thorough brief and debrief.

4.8.1.1.2.1 AW-169 Simulator/Flight Training Modules

The following flight training modules indicate the training session where the training event is first addressed. If the pilot does not demonstrate proficiency in that session, the event will be carried forward until proficiency is demonstrated. Once proficiency is demonstrated, the event may be considered optional for subsequent training. Based on demonstrated proficiency, events scheduled in a subsequent module may be introduced in an earlier module.

1. Pre-Flight Procedures:

- a. Trip planning
- b. Weight and balance
- c. Aircraft performance
- d. Human factors
- e. Preflight inspection
- f. Emergency equipment
- g. Engine starts - Normal/abnormal
- h. Pre-takeoff checks
- i. Lift to hover
- j. Hovering maneuvers
- k. Taxiing - Ground/hover/air
- l. Slopes
- m. Hot weather - high DA operations (optional)
- n. Cold weather operations (optional)
- o. Helicopter operational hazards (discussed only):



- i. Dynamic rollover
- ii. Ground resonance
- iii. Retreating blade stall
- iv. Loss of tail rotor effectiveness
- v. Spatial Disorientation during Night Unaided Helipad/Deck landing/departures.

2. Takeoff and Departure Phase:

- a. Normal takeoff
- b. Crosswind takeoff
- c. Running/rolling takeoff
- d. Maximum performance takeoff
- e. CAT B takeoff
- f. CAT A clear area takeoff
- g. CAT A G/E helipad takeoff
- h. CAT A Variable TDP takeoff
- i. Offshore helideck takeoff
- j. Rejected takeoff - Quick stop

3. In Flight Maneuvers:

- a. Engine power check
- b. CFIT avoidance - Use of TAWS
- c. CFIT avoidance - Use of WX radar
- d. Navigation and communication
- e. Basic attitude flying
- f. Traffic pattern work
- g. Standard rate turns
- h. Steep turns
- i. Rapid decelerations
- j. Recovery from unusual attitudes
- k. Recovery from settling with power
- l. Use of AP modes
- m. Use of CCD and CCJ
- n. Demonstration of Collective safety function

4. Landings and Approaches to Landing:

- a. Normal approach
- b. Steep approach
- c. Confined areas or pinnacle operations
- d. Shallow approach to a running landing
- e. CAT B landing
- f. CAT A clear area landing
- g. CAT A G/E helipad landing
- h. CAT A Variable TDP landing
- i. Rejected landing - Go around



- j. Ship Deck/WTG orientation Day and Night
- k. Local Helipad orientation Day and Night
- l. Unimproved LZ Day and Night

5. Emergency Procedures (Complete for Applicable Aircraft):

- a. Autorotations
- b. Ditching/emergency equipment
- c. In flight fire and smoke removal
- d. Tail rotor malfunctions
- e. Fuel systems malfunctions
- f. Hydraulic malfunctions
- g. Powertrain (main gearbox and tail rotor drive) malfunctions
- h. Electrical system malfunctions
- i. Powerplant malfunctions
- j. Auxiliary power unit malfunctions
- k. Landing gear malfunctions
- l. Flight controls/auto flight systems malfunctions
- m. Avionics and GPS malfunctions
- n. Engine control malfunctions
- o. Engine failure prior to TDP from ship deck/helipad
- p. Engine failure after TDP from ship deck/helipad
- q. Engine failure prior to LDP from ship deck/helipad
- r. Engine failure after LDP from ship deck/helipad
- s. Pitot-static system malfunctions
- t. AHRS malfunctions
- u. Display/EDCU malfunctions
- v. Exhaust fire after shutdown / dry motor procedure

6. Normal and Abnormal Systems:

- a. Other systems, subsystems, and devices specific to the type of helicopter

7. IFR Procedures:

- a. Straight and level flight
- b. Standard rate turns
- c. Steep turns
- d. Climbing and descend turns
- e. Recovery from IIMC & unusual attitudes
- f. IMC autorotations & Emergency descent
- g. Instrument departures (ODP)
- h. SID/STAR
- i. Holding procedures
- j. Enroute and Airways
- k. Radio navigation (NDB, GPS, & VOR)
- l. ILS Approach



- m. RNAV approach LPV
- n. RNAV approach BARO/VNAV
- o. RNAV approach LNAV
- p. VOR approach
- q. LOC approach
- r. Circle to land
- s. Missed-approach with GA mode
- t. Missed-approach hand flown
- u. OSAP/HEDA Procedures

8. IFR Abnormal Procedures

- a. Auto pilot failures
- b. Avionics failures
- c. Hydraulic failures
- d. Engine failures
- e. Engine failure during coupled precision approach
- f. Engine failure during hand flown precision approach
- g. Engine failure during coupled non-precision approach
- h. Engine failure during hand flown non-precision approach
- i. Pitot Static failures

9. HNVGO (If Applicable see Section 4.5 for Flight Modules)

10. Post Flight Procedures:

- a. After landing procedures
- b. Parking and securing

NOTE: Combining events is permissible.

4.8.1.1.2.2 AW169 HEMS Sim/Flight Training Module

1. Discussion Items:

- a. Local flying area (LFA):
 - i. Terrain
 - ii. Airspace
 - iii. ATC facilities
 - iv. Weather (seasonal fog, icing, and convective weather)
 - v. Available airports, heliports, LZ's (approaches/departures)
- b. Special Conditions (including Flat Light/Brownout/Whiteout Ops and Multi-Aircraft Situations)
- c. Hospital heliport operations and procedures (day/night and multi-aircraft)
- d. Unimproved LZ operations (off airport, day/night multi-aircraft)
- e. Lighting (interior/exterior) to include use of searchlight, forced landing considerations

2. Preflight Procedures:

- a. Med-crew Briefing



- b. Hover/Ground Taxi Operations
- 3. **Takeoff and Departure Phase:**
 - a. Normal/Crosswind
 - b. Sidestep
 - c. Maximum Performance
 - d. CAT-A (If Applicable)
- 4. **Cruise:**
 - a. Navigation
 - b. Communication
 - c. Severe Weather Avoidance
 - d. Maintaining Situational Awareness
 - e. Helicopter Terrain Awareness and Warning System (HTAWS)
- 5. **Approach and Landing:**
 - a. High Reconnaissance
 - b. Low Reconnaissance
 - c. Ground/Hazard Recognition
 - d. Normal/Crosswind
 - e. Sidestep
 - f. Confined Area/Steep Approach
 - g. CAT-A (if applicable)
 - h. Emergency and Abnormal Situations
- 6. **HNVGO** (See Section 4.5 for Required Modules)
- 7. **Post-Flight Procedures:**
 - a. Crew Debriefing
 - b. Post-Flight Inspection
 - c. Cleaning/Decontamination of Aircraft and Equipment (biohazards)
 - d. Servicing O2 Systems (if applicable)

Note: If conducted in an FSTD, this module is to be conducted at night and only if hospital's and surrounding hazards are adequately depicted. If conducted in the aircraft then it will be conducted at night.

4.8.1.2 AW169 Recurrent

OBJECTIVE OF TRAINING:

To update the knowledge and information acquired during initial training.

PREREQUISITES:

Each pilot must meet appropriate HSUSA 14 CFR Part 135 employment requirements.

PROGRAMMED TRAINING HOURS:

See appropriate table in either the training curriculum or aircraft ground & flight modules. There is a syllabus for an A/B/C year allowing all subjects to be covered in detailed on a three year cycle.



COMPLETION STANDARDS:

At the end of the Ground and Flight Training Curriculum Segment the pilot shall be able to demonstrate to the Administrator, or Check Airmen, normal, abnormal, and emergency procedures and checklists in a timely and sequentially correct manner, and perform all the maneuvers and procedures in accordance with the qualification modules listed in this chapter.

4.8.1.2.1 AW169 Recurrent Ground**4.8.1.2.1.1 AW169 Recurrent Ground Program A**

Training Syllabus Recurrent Ground Training Program A - AW169	Duration
Hydraulic systems	1:00
RNAV Approaches and RFM Supplement 43	3:00
CRM decision making, situational awareness, communication	1:00
Flight crew incapacitation	0:15
Standard briefings and flight preparation	1:00
Test: aircraft systems and limitations	1:00
	7:15

4.8.1.2.1.2 AW169 Recurrent Ground Program B

Training Syllabus Recurrent Ground Training Program B - AW169	Duration
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Electrical and avionics malfunction	3:00
CAT A procedures & Performance and TO with reduced visibility	1:00
CRM Leadership, cooperation, communication	1:00
Flight crew incapacitation	0:15
Standard briefings and flight preparation	1:00
Test: aircraft systems and limitations	1:00
	7:15

4.8.1.2.1.3 AW169 Recurrent Ground Program C

Training Syllabus Recurrent Ground Training Program C - AW169	Duration
Fuel and engine malfunctions	3:00
TCAS II	1:00
Safety culture, threat and error management, fatigue, communication	1:00
Flight crew incapacitation	0:15
Standard briefings and flight preparation	1:00
Test: aircraft systems and limitations	1:00
	7:15

4.8.1.2.2 Programmed Training Hours

PROGRAMMED TRAINING HOURS:

Each Flight/Simulator/FTD Training Module is scheduled for the total hours indicated in the tables below.

CREW COMPLEMENT:

Flight training is generally conducted as a crew; however, a pilot training may complete the course alone (single PIC) or with another pilot acting as second crewmember. The pilot acting as second crewmember must be:

1. A qualified line pilot, or
2. A qualified flight instructor and/or check airmen

If the course is conducted single PIC the total time must all be met as PF.



Flight Simulator	PF	PM	Total Time
Recurrent Flight Simulator Module No. 1	1.0	1.0	2.0
Recurrent Flight Simulator Module No. 2	1.0	1.0	2.0

BRIEF/DEBRIEF:

The instructor will ensure an hour prior and half hour after each session is scheduled to allow for a thorough brief and debrief.

4.8.1.2.3 AW169 Recurrent Simulator Modules**4.8.1.2.3.1 AW169 Recurrent Sim Program A****Flight Training - Session 1**

Training Syllabus Recurrent Flight Training Program A - AW169
Standard traffic pattern, helicopter familiarization
CAT A ground and elevated helipad procedure
CAT B procedure
Engine failure before/after TDP, CAT A clear area & G/E Helipad
Engine failure before/after LDP, CAT A clear area & G/E Helipad
Engine failure during hover
Dual engine failure during hover
Straight in autorotation VMC and IMC
Engine fire, fuselage fire



Engine failure, relight in flight
Hydraulics abnormal procedures
Engine failure before/after TDP, offshore CAT A elevated helipad
Engine failure before/after LDP, offshore CAT A elevated helipad

Flight Training IFR - Session 2

Training Syllabus Recurrent Flight Training Program A - AW169
IFR take off with low visibility (1/2sm)
Recovery from unusual attitudes
Pilot incapacitation
TCAS II procedures
EGPWS procedures
Precision approach to minima with OEI landing
Go around on instruments from minima with OEI
LPV approach to minima with downgrade to LNAV/VNAV
Autopilot and flight director malfunctions / Flight in Turbulence
Hydraulic System number 2 failure



Pitot static failure
Steep turns, timed standard rate turns and timed standard rate climbing turns
OSAP/HEDA Procedures

4.8.1.2.3.2 AW169 Recurrent Sim Program B**Flight Training - Session 1**

Training Syllabus Recurrent Flight Training Program B - AW169
Standard traffic pattern, helicopter familiarization
CAT A ground and elevated helipad procedure
CAT B procedure
Engine failure before/after TDP CAT A clear area & G/E Helipad
Engine failure before/after LDP CAT A clear area & G/E Helipad
Engine failure during hover
Dual engine failure during hover
Straight in autorotation VMC and IMC
Engine fire, fuselage fire
Electrical malfunctions
Avionics malfunctions



Tail rotor control failure
Tail rotor drive shaft failure
Engine failure before/after TDP, offshore CAT A elevated helipad
Engine failure before/after LDP, offshore CAT A elevated helipad

Flight Training IFR - Session 2

Training Syllabus Recurrent Flight Training Program B - AW169
IFR take off with low visibility (400m)
Recovery from unusual attitudes
Pilot incapacitation
TCAS II procedures
EGPWS procedures
Precision approach to minima with OEI landing
Go around on instruments from minima with OEI
LPV approach to minima with downgrade to LNAV/VNAV
Autopilot and flight director malfunctions / Flight in Turbulence
Offshore approach to a ship
Nav aids failure during approach
Holding procedures



OSAP/HEDA Procedures

4.8.1.2.3.3 AW169 Recurrent Sim Program C**Flight Training - Session 1****Training Syllabus Recurrent Flight Training Program C - AW169**

Standard traffic pattern, helicopter familiarization

CAT A ground and elevated helipad procedure

CAT B procedure

Engine failure before/after TDP CAT A clear area & G/E Helipad

Engine failure before/after LDP CAT A clear area & G/E Helipad

Engine failure during hover

Dual engine failure during hover

Straight in autorotation VMC and IMC

Engine fire, fuselage fire

Engine malfunctions

Engine failure, relight in flight

Tail rotor control failure

Tail rotor drive shaft failure

Powertrain abnormal procedures

Engine failure before/after TDP, offshore CAT A elevated helipad

Engine failure before/after LDP, offshore CAT A elevated helipad

settling with power and vortex ring

Flight Training IFR - Session 2

Training Syllabus Recurrent Flight Training Program C - AW169
IFR take off with low visibility (1/2sm)
Recovery from unusual attitudes
Pilot incapacitation
TCAS II procedures
EGPWS procedures
Precision approach to minima with OEI landing
Go around on instruments from minima with OEI
LPV approach to minima with downgrade to LNAV/VNAV
Autopilot and flight director malfunctions / Flight in turbulence
IFR Autorotations with cloudbase 500 feet
Fuel system failures
Engine Chip and Fail Fixed
OSAP/HEDA Procedures

4.9 Checking

OBJECTIVE OF TRAINING:

To determine a pilot's knowledge of aircraft systems and procedures, and to determine his/her ability to



accomplish the required maneuvers by:

1. The Appropriate Written & Oral Testing.
2. All Items for Completion of a 14 CFR Part 135.293 and/or 135.297 Check.
3. Completion of a 14 CFR Part 135.299 (Line Check).

PREREQUISITES:

Each pilot must meet appropriate HSUSA 14 CFR Part 135 employment requirements appropriate to the duty assignment.

CURRICULUM SEGMENTS:

Exact requirements for each type of qualification module are detailed in the table below. Regulatory requirements in Section 2 of this manual.

COMPLETION STANDARDS:

The standard required for basic checks is at least that required for obtaining the certificate which must be held to act as a PIC. The minimum HeliService USA Part 135 operation standard required of HSUSA Pilots is to meet "commercial certificate with an instrument rating" standards of proficiency.

HeliService USA will always be able to fly the helicopter in a manner that will demonstrate they are obviously in command of the aircraft with the successful outcome of each maneuver never in doubt. Additionally maneuvers will be completed within the standards set by the Practical Test Standards (PTS) or to the standards set by the maneuvers and procedures guide in Appendix A of this manual. When a conflict exists with the PTS, the more restrictive standard applies.

All written or CBT examinations must be completed with a minimum score of 80%. Subject areas found deficient will be reviewed and the individual's knowledge corrected to 100%.

The Check Airman will determine the applicable standards prior to the start of any evaluation session. The required standards will be discussed with the pilot being tested.

4.9.1 Qualification Module 1 - VFR Only Competency Check (135.293)

Regulatory requirements: Must be completed every 12 months when a pilot is authorized to conduct VFR-only operations. Required for each make/model aircraft pilot is authorized to operate.

Test: Oral or Written.

Maneuvers:

1. **Ground Operations**
 - a. Preflight inspection
 - b. Start Procedures
 - c. Taxiing and Ground Hover • Pre-takeoff Checks
2. **Takeoffs**
 - a. Normal
 - b. Takeoff with power-plant failure



- c. Rapid Deceleration (Rejected T/O)
- 3. **Inflight Maneuvers**
 - a. Settling with power
 - b. Unusual Attitude Recovery
- 4. **Approaches and Landings**
 - a. Normal landings
 - b. Landing with engine-out
- 5. **Abnormal and Emergency Procedures**
 - a. System Malfunction (such as power-plant, fuel system, hydraulic system)
 - b. An instrument approach (representative of one the pilot might have to use in an emergency)
 - c. Recovery from Inadvertent IMC
 - d. Oral Only
 - i. Tail Rotor Failure
 - ii. Dynamic Rollover
 - iii. Low Rotor RPM
 - iv. Anti-Torque System Failure
 - v. White-Out
 - vi. Brown-Out
 - vii. Flat Light
 - e. Confined Area
 - f. Pinnacle Operations
 - g. Slope Operations

Note: For IMC recovery, the event should reflect a realistic course of action the pilot might take to escape from an inadvertent encounter with IFR conditions. When the pilot is authorized to operate an aircraft equipped with IFR instruments and navigation equipment, an ILS approach should be demonstrated.

4.9.2 Qualification Module 2 - IFR Competency Check (135.293)

Frequency: Required to be completed every 12 months when a pilot is authorized to conduct IFR revenue operations.

Test: Oral or Written.

Manuevers:

- 1. **GROUND OPERATIONS**
 - a. Preflight inspection
 - b. Start Procedure
 - c. Taxiing and Ground Hover
 - d. Pre-takeoff checks
- 2. **TAKEOFFS**
 - a. Normal



- b. Instrument
- c. With power-plant failure
- d. Rapid Deceleration (Rejected T/O)
- 3. **IN-FLIGHT MANEUVERS**
 - a. Settling with power
 - b. Unusual Attitude Recovery
- 4. **INSTRUMENT PROCEDURES**
 - a. Normal ILS approach
 - b. Engine-out ILS (may be substituted for the normal ILS)
 - c. Coupled approach (may be combined with ILS)
 - d. Non-precision approach (must complete a GPS or RNAV approach if authorized by OpSpec)
 - e. OSAP and HEDA procedures (Annual Proficiency Check)
- 5. **LANDINGS**
 - a. Normal
 - b. Landing with engine-out
- 6. **ABNORMAL AND EMERGENCY PROCEDURES**
 - a. System Malfunction (such as power-plant, fuel system, hydraulic system)
 - b. Instrument approach (representative of one the pilot might have to use in an emergency)
 - c. Recovery from Inadvertent IMC
 - d. Confined Area / Pinnacle Operations
 - e. Slope Operations
 - f. Oral Only
 - i. Tail Rotor Failure
 - ii. Dynamic Rollover
 - iii. Low Rotor RPM
 - iv. Anti-Torque System Failure (fixed pitch)
 - v. White-Out
 - vi. Brown-Out
 - vii. Flat Light

4.9.3 Qualification Module 3 - IFR Proficiency Check (135.297)

Regulatory requirements: Must be completed every 6 months when a pilot is authorized to conduct IFR operations. Required for each make/model aircraft pilot is authorized to operate.

Test: Oral or Written.

Maneuvers:

- 1. **GROUND OPERATIONS**
 - a. Preflight inspection
 - b. Start Procedure
 - c. Taxiing and Ground Hover



- d. Pre-takeoff checks

2. TAKEOFFS

- a. Normal
- b. Instrument
- c. With power-plant failure
- d. Rapid Deceleration (Rejected T/O)
- e. Area Departure

3. IN-FLIGHT MANEUVERS

- a. Steep turns
- b. Settling with power
- c. Unusual Attitude Recovery

4. INSTRUMENT PROCEDURES

- a. Area arrival
- b. Holding
- c. Normal ILS approach
- d. Engine-out ILS
- e. Coupled approach
- f. Non-precision approach (must complete a GPS or RNAV approach if authorized by OpSpec)
- g. Second non-precision approach
- h. Missed approach from an ILS
- i. Second missed approach
- j. Circling approach

5. LANDINGS

- a. Normal landing
- b. Landing from an ILS
- c. Landing with engine-out
- d. Circling approach

6. ABNORMAL AND EMERGENCY PROCEDURES

- a. System Malfunction (such as power-plant, fuel system, hydraulic system)
- b. Instrument approach (representative of one the pilot might have to use in an emergency)
- c. Recovery from Inadvertent IMC
- d. Oral only:
 - i. Tail Rotor Failure
 - ii. Dynamic Rollover
 - iii. Low Rotor RPM
 - iv. Anti-Torque Sys Fail (fixed pitch)
 - v. White-Out
 - vi. Brown-Out
 - vii. Flat Light
- e. Confined Area /Pinnacle Operations



f. Slope Operations

4.9.4 Qualification Module 4 - PIC Line Check (135.299)

Frequency: Required to be completed every 12 months in one of the make and model aircraft that the pilot is assigned to fly.

Test: N/A

Maneuvers:

1. The line check must be conducted over at least one route segment which includes takeoffs and landings at one or more airports that are representative of the operator's type of operation. A route segment for the purpose of a line check consists of a departure from one terminal area and the flight to and arrival at a different terminal area.
2. The pilot must demonstrate to the satisfaction of the inspector or Check Airman, takeoffs and landings at one or more representative airports or landing areas.
3. For a pilot authorized to conduct IFR Operations:
 - a. Complete the first two steps listed above
 - b. At least one flight shall be flown over a civil airway, an approved off-airway route or a portion of either under IFR with an instrument approach procedure

Note 1: The line check should be conducted during revenue flights. If it is not practical to conduct the line check during revenue flights the Check Airman or inspector may authorize that the line check be conducted during the same flight period that the competency check is conducted. If the line check is conducted in this manner, the line portion of this flight must include the requirements of paragraphs 1 through 3 of this section.

Note 2: The pilot must be determined to have satisfactorily performed the duties and responsibilities of a PIC in operations under FAR Part 135. This certification shall be entered in the pilot's training record.

4.10 CTS - Computer Training

4.10.1 Recurrent Training Rotation

CTS Recurrent A
14 CFR Part 1-97 and 49 Part 830 – RW IFR
14 CFR Part 135 – RW IFR
Aeronautical Information Manual – RW IFR



AeroNav Charts Overview
Airspace Overview
Aviation Weather Theory
Classes of Fire and Portable Fire Extinguishers
Hazmat – Will Carry
CRM-ADM - Rotor Wing
Introduction to Safety Management System
HNVGO (If Applicable)
Aircraft Specific Systems (If Available)
CTS Recurrent B
Controlled Flight into Terrain Avoidance (CFIT, TAWS, and ALAR) - RW
CRM-ADM – Rotor Wing
Flat-light, Whiteout, and Brownout Conditions
GPS (RW IFR-VFR)
Hazmat – Will Carry
Helicopter External Lighting (Night Flying)
Land and Hold Short Operations
Introduction to Safety Management System
HNVGO (If Applicable)



Aircraft Specific Systems (If Available)
CTS Recurrent C
METAR and TAF
Minimum Equipment List (MEL)
Physiology and First Aid (RW)
Runway Incursion
Survival
Aviation Safety Action Program (ASAP) Overview
TCAS II
Winter Operations (RW)
CRM-ADM – Rotor Wing
Hazmat – Will Carry
Introduction to Safety Management System
HNVGO (If Applicable)
Aircraft Specific Systems (If Available)

4.11 Special Training

Reserved

4.11.1 HAA/SAR Local Area Training

Each pilot must demonstrate familiarity with each local flying area within which that pilot is authorized to



utilize the local flying area weather minima. This demonstration must be accomplished by passing an examination of the pilot's familiarity with each authorized local flying area (not necessarily a flight check) given by the certificate holder within the 12 calendar-months prior to using the local flying area. BI Module 16 documents the required syllabus for pilot training.

NOTE: A pre/post due month grace period does not apply to local flying are examinations. If a pilot has not successfully passes a local flying area examination within the past 12 calendar-months, cross-country minima must be applied by that pilot into that local flying area.

For Reference Only



5 Appendix A - Maneuvers Guide

5.1 AW169 Maneuvers Guide

5.1.1 Ground Taxi

Ground Taxi

Simultaneously increase collective pitch and move the cyclic slightly forward. When the aircraft is moving, centre the cyclic and anticipate the requirement for collective to control speed and check the operation of the toe brakes.

For normal taxiing the cyclic position is slightly forward and to the left.

Turning, whilst taxiing, should be carried out with collective at minimum pitch and cyclic close to central or as required to compensate for crosswind.

To turn the aircraft, initiate forward movement, then gradually press the respective yaw pedal. For tight turns use differential braking. When turning is completed, reduce pedal deflection towards neutral and reduce power. For turns at high gross weight, slowly increase collective pitch, to offload the nosewheel before initiating the turn.

Taxiing on slippery surface

When taxiing on slippery surfaces (e.g. ice or snow) it is very difficult to turn or straighten the nose wheel. Keep taxi speed to a minimum and avoid sharp turns. If in any doubt as to the controllability of the aircraft or the ability to manoeuvre into a designated parking position, stop the aircraft into wind and have it towed into position.

After taxiing in snow, the brakes may freeze once the aircraft is stopped. If the aircraft is flown immediately, the brakes may still be frozen on landing. Touch down vertically and check that the wheels are free to rotate before continuing with a ground taxi.

Stopping

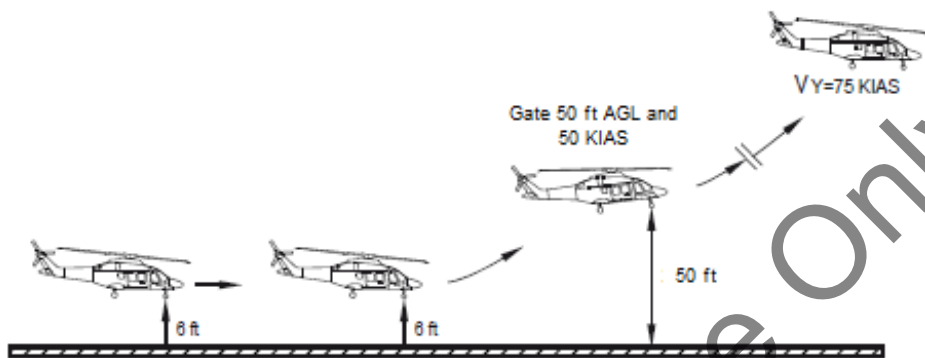
Centre the cyclic and lower the collective to minimum pitch. Smoothly and evenly apply the toe brakes. Do not apply the parking brake until the aircraft has come to a complete stop.

5.1.2 Normal Takeoff

PARAMETERS	
AIRSPEED DEFINITIONS	
Best Rate of Climb Speed (V_Y)	75 KIAS



TAKE-OFF GATE	
Height	50 ft ATS
Airspeed	50 KIAS



Standard procedure and phraseology

Event	PF / PM	<i>"Phraseology"</i> and/or Action	
Before lifting	PF	<i>"Lifting"</i>	
	PM	Checks both AP's engaged	<i>"Two AP's"</i>
Hover	PF	Establishes a 6 ft ATS hover and notes pitch attitude	<i>"Hover checks"</i>
	PM	Performs hover checks by memory	<i>"No warnings, no (further) cautions, T's and P's in the green, PI xx %, target PI xx (hover PI + 5-10) %"</i>
	PF	<i>"Roger, taking-off"</i>	Applies cyclic to rotate 10° nose down with respect to hover attitude and applies collective to increase hover PI by 5 - 10 % to maintain height



	PM	Checks PI If PI < target PI or If PI = target PI	<i>"Power available"</i> <i>or</i> <i>"Power set"</i>
Airspeed indication starts to increase	PM	<i>"Airspeed alive"</i>	
	PF	<i>"Checked"</i>	Reduces pitch down by 5° to start climb and continue acceleration to pass a gate of 50 ft ATS and 50 KIAS
At the gate	PM	<i>"Gate"</i>	
	PF	<i>"Checked"</i>	Continues climb and acceleration to V_Y
Approaching V_Y	PM	<i>"Approaching V_Y"</i>	
	PF	<i>"Checked"</i>	Adjusts pitch attitude to maintain V_Y and continues climb to 1 000 ft ATS
Passing 1 000 ft	PM	<i>"1 000 feet"</i>	
	PF	Checks altitude / height	<i>"Checked, after take-off checks"</i>
	PM	Performs after take-off checks	

5.1.3 Missed Approach

A missed approach may be initiated at anytime during an approach whenever the PF or PNF deems it necessary.

Manual go-around procedure

A manual go-around should be initiated by selecting 95% PI and a pitch attitude to attain V_Y (75 KIAS) as soon as possible. The go-around path should be flown clear of obstacles followed by a wings-level attitude. The climb should at least be continued until reaching the briefed MSA.

Coupled go-around procedure

The Go-Around mode (GA) provides aircraft control for an automatic ascent profile. The function is intended primarily for use during the approach phase of flight to initiate a missed approach.

GA mode is engaged by pressing the GA pushbutton on either collective; any other mode is then



disengaged.

Upon engagement the GA mode commands collective for a 1 000 ft/min climb, roll for wings-level and pitch for an airspeed of 75 KIAS.

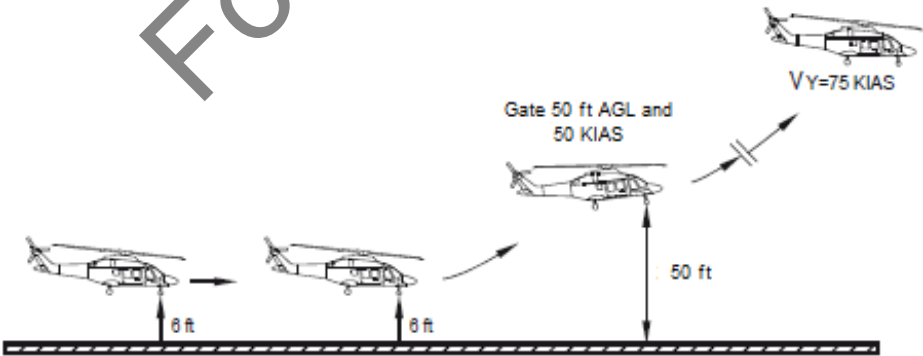
When the aircraft attains 1,000 ft/min ROC and 75 KIAS, the GO AROUND MODE will be replaced by the IAS and VS modes.

5.1.4 CAT-B Takeoff

5.1.4.1 Clear area

The following CAT B Take-Off procedure best ensures the aircraft is within the takeoff portion of the H-V envelope.

PARAMETERS	
AIRSPEED DEFINITIONS	
Best Rate of Climb Speed (V _Y)	75 KIAS
TAKE-OFF GATE	
Height	50 ft ATS
Airspeed	50 KIAS



Standard procedure and phraseology

Event	PF / PM	"Phraseology" and/or Action
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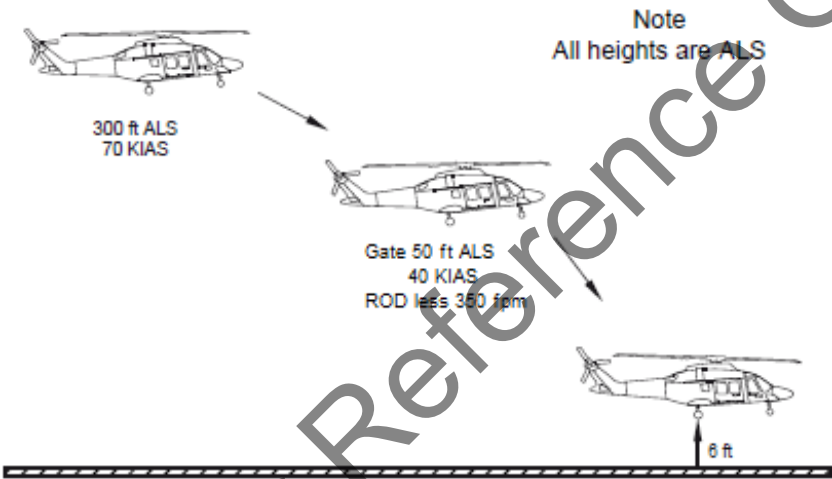
Before lifting	PF	<i>"Lifting"</i>	
	PM	Checks both AP's engaged	<i>"Two AP's"</i>
Hover	PF	Establishes a 6 ft ATS hover and notes pitch attitude	<i>"Hover checks"</i>
	PM	Performs hover checks by memory	<i>"No warnings, no (further) cautions, T's and P's in the green, PI xx %, target PI xx (hover PI + 5-10) %"</i>
	PF	<i>"Roger, taking-off"</i>	Applies cyclic to rotate 10° nose down with respect to hover attitude and applies collective to increase hover PI by 5 - 10 % to maintain height
	PM	Checks PI If PI < target PI or If PI = target PI	<i>"Power available"</i> <i>or</i> <i>"Power set"</i>
Airspeed indication starts to increase	PM	<i>"Airspeed alive"</i>	
	PF	<i>"Checked"</i>	Reduces pitch down by 5° to start climb and continue acceleration to pass a gate of 50 ft ATS and 50 KIAS
At the gate	PM	<i>"Gate"</i>	
	PF	<i>"Checked"</i>	Continues climb and acceleration to V _Y
Approaching V _Y	PM	<i>"Approaching V_Y"</i>	
	PF	<i>"Checked"</i>	Adjusts pitch attitude to maintain V _Y and continues climb to 1 000 ft ATS
Passing 1 000 ft	PM	<i>"1 000 feet"</i>	
	PF	Checks altitude / height	<i>"Checked, after take-off checks"</i>
	PM	Performs after take-off checks	



5.1.5 CAT-B Landing

5.1.5.1 Clear Area

PARAMETERS	
AIRSPEED DEFINITIONS	
Best Rate of Climb Speed (V _y)	75 KIAS
LANDING GATE	
Height	50 ft ALS
Airspeed	40 KIAS
Rate of descent	less than 350 ft/min



Standard procedure and phraseology

Event	PF / PM	<i>"Phraseology"</i> and/or Action	
Initial point	PF	Stabilizes the aircraft on final approach course at 300 ft ALS / 70 KIAS and initiates a descent and deceleration to pass a gate of 50 ft ALS and 40 KIAS with a rate of descent of less than 350 ft/min	<i>"Upper gate"</i>
At the gate	PM	<i>"Gate"</i>	
	PF	<i>"Landing"</i>	Initiates a flare rotating nose up to achieve a hover at 6 ft ALS or to perform a running landing.
Passing 20 ft ALS	PM	Checks pitch attitude for a maximum of 15° nose up at touchdown.	<i>"Pitch below 15"</i> or <i>"Check pitch"</i>
	PF	Checks and adjusts pitch attitude if required and lands	<i>"Checked, landing on"</i>
After touchdown	PF	Centralizes controls, reduces collective to MPOG and applies wheel brakes as required	<i>"After landing checks"</i>
	PM	Performs after landing checks	<i>"After landing checks completed"</i>



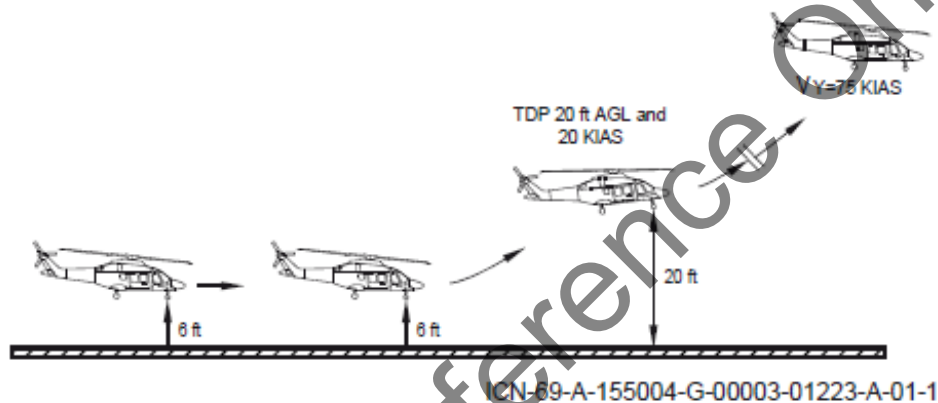
5.1.6 CAT-A Takeoff

DEFINITIONS	
Take Off Decision Point (TDP)	The first point. in the Take-Off path from which a CTO capability is assured and the last point from which a RTO is assured, within the rejected take off distance.
Continued Take Off distance	The horizontal distance from the start of the take off procedure to a point at least 35 ft ATS where V_{TOSS} and a positive rate of climb are attained following an engine failure at or after TDP.
Rejected Take Off distance	The horizontal distance from the start of the take off procedure to a point where the helicopter lands and stops safely following an engine failure prior to TDP.
Take-Off path	The path from the point of commencement of the take off procedure to the point at which the helicopter is 1 000 ft ATS.
PATH 1	The segment between the end of the CTO distance to a height of 200 ft ATS during an OEI 2.5 min power climb at V_{TOSS} and a minimum ROC of 100 ft/min.
PATH 2	The segment between 200 ft ATS and 1000 ft ATS during a Continuous OEI power climb at V_Y and a minimum ROC of 150 ft/min.
Take Off Safety Speed (V_{TOSS})	The airspeed at which the scheduled climb gradient OEI can be achieved.
Elevated Heliport	A heliport which is at least 10 ft above the surrounding surface.
Helideck	A heliport located on a floating or fixed Offshore structure.
Heliport	A defined area of land or structure intended to be used for the arrival and departure of helicopters.



5.1.6.1 Clear area

PARAMETERS	
AIRSPEED DEFINITIONS	
Take-Off Safety Speed (V_{TOSS})	45 KIAS
Best Rate of Climb Speed (V_Y)	75 KIAS
TAKE-OFF DECISION POINT (TDP)	
Height	20 ft ATS
Airspeed	20 KIAS



Standard procedure and phraseology

Event	PF / PM	<i>"Phraseology"</i> and/or Action	
Before lifting	PF	<i>"Lifting"</i>	
	PM	Checks both AP's engaged	<i>"Two AP's"</i>
Hover	PF	Establishes a 6 ft ATS hover and notes pitch attitude	<i>"Hover checks"</i>
	PM	Performs hover checks by memory	<i>"No warnings, no (further) cautions, T's and P's in the green, PI xx %, target PI xx (hover PI + 5-10) %"</i>

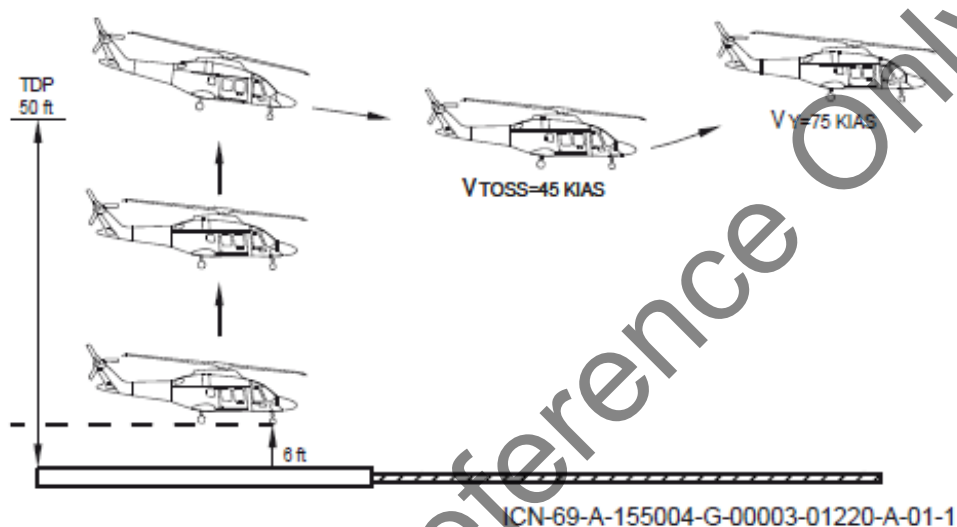


	PF	<i>"Roger, taking-off"</i>	Applies cyclic to rotate 10° nose down with respect to hover attitude and applies collective to increase hover PI by 5 - 10 % to maintain height
	PM	Checks PI If PI < target PI or If PI = target PI	<i>"Power available"</i> <i>or</i> <i>"Power set"</i>
Airspeed indication starts to increase	PM	<i>"Airspeed alive"</i>	
	PF	<i>"Checked"</i>	Reduces pitch down by 5° to start climb to TDP
At TDP	PM	<i>"TDP"</i>	
	PF	<i>"Continuing"</i>	Continues climb and acceleration to V_Y
At V_{TOSS} and positiv ROC	PM	<i>"V_{TOSS} climbing"</i>	
	PF	<i>"Checked"</i>	
Approaching V_Y	PM	<i>"Approaching V_Y"</i>	
	PF	<i>"Checked"</i>	Adjusts pitch attitude to maintain V_Y and continues climb to 1 000 ft ATS
Passing 1 000 ft	PM	<i>"1 000 feet"</i>	
	PF	Checks altitude / height	<i>"Checked, after take-off checks"</i>
	PM	Performs after take-off checks	



5.1.6.2 Ground and elevated heliport/helideck vertical

PARAMETERS	
AIRSPEED DEFINITIONS	
Take-Off Safety Speed (V_{TOSS})	45 KIAS
Best Rate of Climb Speed (V_Y)	75 KIAS
TAKE-OFF DECISION POINT (TDP)	
Height	50 ft ATS



Standard procedure and phraseology

Event	PF / PM	<i>"Phraseology"</i> and/or Action	
Before lifting	PF	<i>"Lifting"</i>	
	PM	Checks both AP's engaged	<i>"Two AP's"</i>
Hover	PF	Establishes a 6 ft ATS hover and notes pitch attitude	<i>"Hover checks"</i>
	PM	Performs hover checks by memory	<i>"No warnings, no (further) cautions, T's and P's in the green, PI xx %, target PI xx (hover PI + 20-25) %"</i>
	PF	<i>"Roger, taking-off"</i>	Increases PI by 20 - 25 % within 2 sec. to climb vertically to TDP (50 ft ATS) maintaining horizontal position

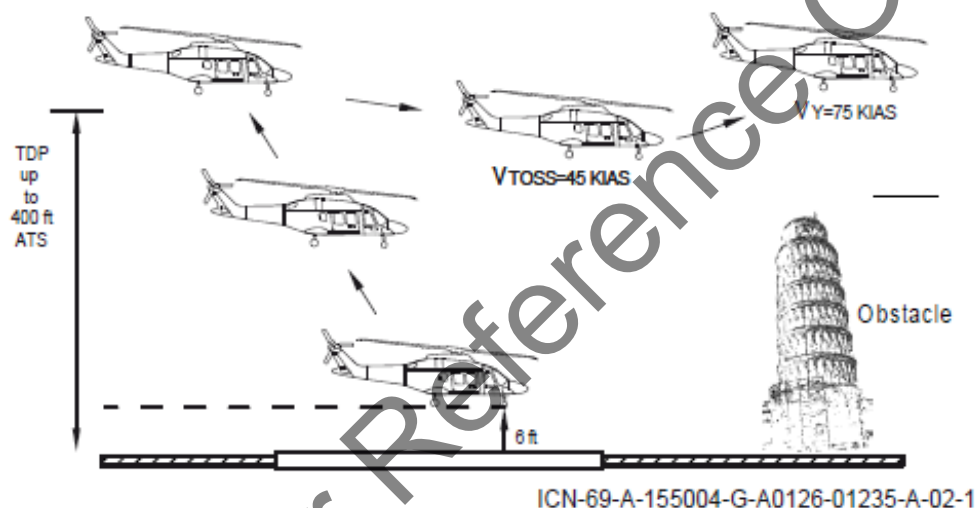


	PM	Checks PI As long as PI < target PI then If PI = target PI	<i>"Power, power, power, ..."</i> <i>then</i> <i>"Power set"</i>
At TDP (50 ft ATS)	PM	<i>"TDP"</i>	
	PF	<i>"Continuing"</i>	Rotates 15° nose down with respect to hover attitude within 2 sec.
Airspeed indication starts to increase	PM	<i>"Airspeed alive"</i>	
	PF	<i>"Checked"</i>	Reduces pitch down by 5° and continues acceleration to V_Y
At V_{TOSS} and positiv ROC	PM	<i>"V_{TOSS} climbing"</i>	
	PF	<i>"Checked"</i>	
Approaching V_Y	PM	<i>"Approaching V_Y"</i>	
	PF	<i>"Checked"</i>	Adjusts pitch attitude to maintain V_Y and continues climb to 1 000 ft ATS
Passing 1 000 ft	PM	<i>"1 000 feet"</i>	
	PF	Checks altitude / height	<i>"Checked, after take-off checks"</i>
	PM	Performs after take-off checks	



5.1.6.3 Ground and elevated heliport/helideck variable TDP

PARAMETERS	
AIRSPEED DEFINITIONS	
Take-Off Safety Speed (V_{TOSS})	45 KIAS
Best Rate of Climb Speed (V_Y)	75 KIAS
TAKE-OFF DECISION POINT (TDP)	
Minimum TDP height, in absence of obstacles	95 ft (80 ft drop down + 15 ft clearance)
Minimum TDP height, with obstacles within CTOD	Obstacle height + 115 ft (80 ft drop down + 35 ft clearance)
Maximum TDP height	400 ft ATS



Standard procedure and phraseology

Event	PF / PM	<i>"Phraseology"</i> and/or Action	
Before lifting	PF	<i>"Lifting"</i>	
	PM	Checks both AP's engaged	<i>"Two AP's"</i>
Hover	PF	Establishes a 6 ft ATS hover and notes pitch attitude	<i>"Hover checks"</i>
	PM	Performs hover checks by memory	<i>"No warnings, no (further) cautions, T's and P's in the green, PI xx %, target rate of climb max 300 ft/min"</i>

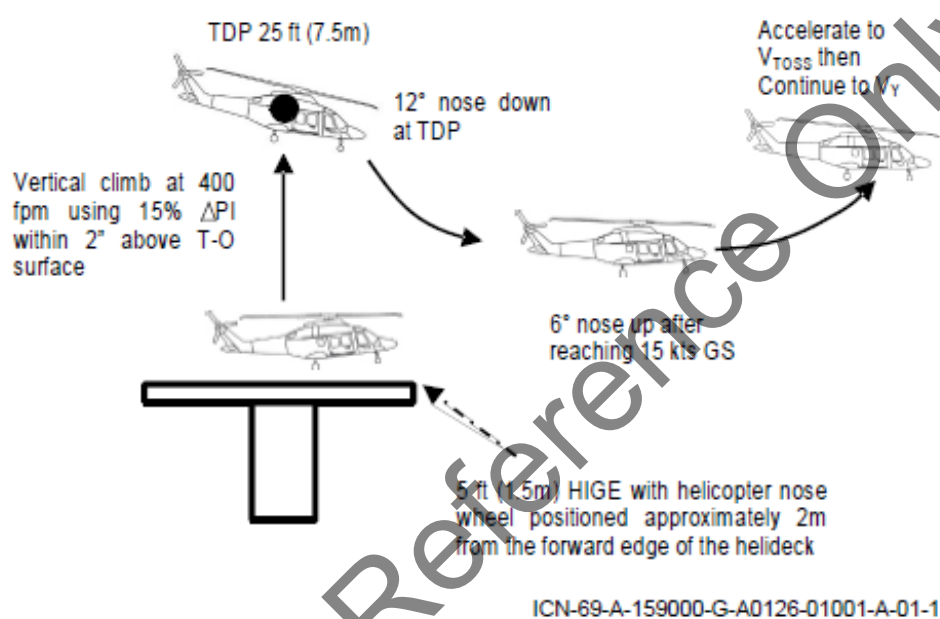


	PF	<i>"Roger, taking-off"</i>	Increases collective to start a slow climb (max 300 ft/min) to TDP (max 400 ft ATS) maintaining the centre of the helipad in sight between yaw pedals
	PM	Checks rate of climb If ROC < 300 ft/min or If ROC = 300 ft/min	<i>"Power available"</i> <i>or</i> <i>"Power set"</i>
At TDP	PM	<i>"TDP"</i>	
	PF	<i>"Continuing"</i>	Rotates 15° nose down with respect to hover attitude within 2 sec. using collective to maintain height
Airspeed indication starts to increase	PM	<i>"Airspeed alive"</i>	
	PF	<i>"Checked"</i>	Reduces pitch down by 5° and continues acceleration to V_Y
At V_{TOSS} and positiv ROC	PM	<i>"V_{TOSS} climbing"</i>	
	PF	<i>"Checked"</i>	
Approaching V_Y	PM	<i>"Approaching V_Y"</i>	
	PF	<i>"Checked"</i>	Adjusts pitch attitude to maintain V_Y and continues climb to 1 000 ft ATS
Passing 1 000 ft	PM	<i>"1 000 feet"</i>	
	PF	Checks altitude / height	<i>"Checked, after take-off checks"</i>
	PM	Performs after take-off checks	



5.1.6.4 Offshore elevated helideck take off

PARAMETERS	
AIRSPEED DEFINITIONS	
Take-Off Safety Speed (V_{Toss})	45 KIAS
Best Rate of Climb Speed (V_Y)	75 KIAS
TAKE-OFF DECISION POINT (TDP)	
Height	25 ft ATS



Standard procedure and phraseology

Event	PF / PM	<i>"Phraseology"</i> and/or Action	
Before lifting	PF	<i>"Lifting"</i>	
	PM	Checks both AP's engaged	<i>"Two AP's"</i>
Hover	PF	Establishes a 5 ft ATS hover with the helicopter nose wheel approximately 2 m from the front edge of the helideck	<i>"Hover checks"</i>
	PM	Performs hover checks by memory	<i>"No warnings, no (further) cautions, T's and P's in the green, PI xx %, target PI xx (hover PI + 15) %"</i>



	PF	<i>"Roger, taking-off"</i>	Increases PI by 15 % within 2 sec. to climb vertically at 400 ft/min or greater to TDP (25 ft ATS) maintaining horizontal position
	PM	Checks PI As long as PI < target PI then If PI = target PI	<i>"Power, power, power, ..."</i> <i>then</i> <i>"Power set"</i>
At TDP (25 ft ATS)	PM	<i>"TDP"</i>	
	PF	<i>"Continuing"</i>	Rotates to 12° nose down within 2 sec.
At 15 kts ground speed	PM	<i>"Ground speed 15"</i>	
	PF	<i>"Checked"</i>	Rotates to 6° nose up and continues acceleration to V_Y
At V_{TOSS} and positiv ROC	PM	<i>"V_{TOSS} climbing"</i>	
	PF	<i>"Checked"</i>	
Approaching V_Y	PM	<i>"Approaching V_Y"</i>	
	PF	<i>"Checked"</i>	Adjusts pitch attitude to maintain V_Y and continues climb to 1 000 ft ATS
Passing 1 000 ft	PM	<i>"1 000 feet"</i>	
	PF	Checks altitude / height	<i>"Checked, after take-off checks"</i>
	PM	Performs after take-off checks	



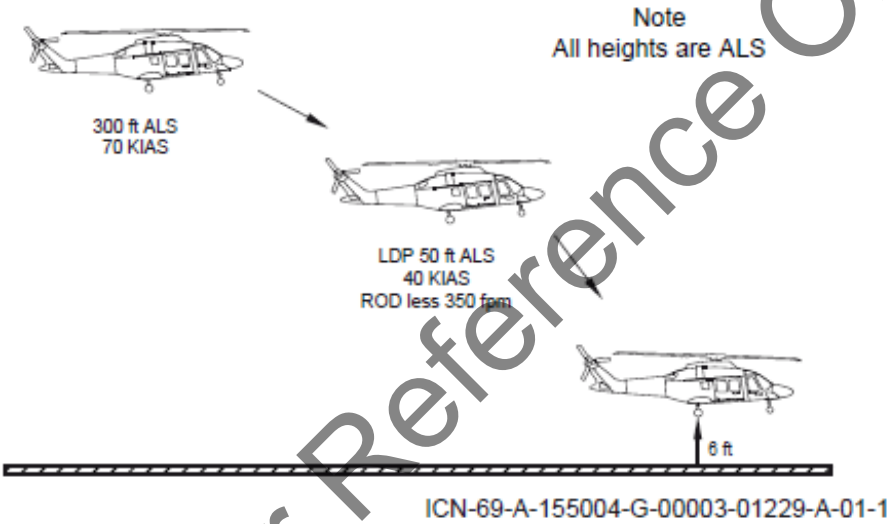
5.1.7 CAT-A Landing

DEFINITIONS	
Landing Decision Point (LDP)	The last point, in the approach landing path, from which it is possible either to land on a predetermined area or accomplish a Balked Landing.
PATH 1	The segment between the end of the BL distance to a height of 200 ft ALS during an OEI 2.5 min power climb at V_{BLSS} and a minimum ROC of 100 ft/min.
PATH 2	The segment between 200 ft ALS and 1 000 ft ALS during a Continuous OEI power climb at V_Y and a minimum ROC of 150 ft/min.
Balked Landing Safety Speed (V_{BLSS})	The airspeed at which the scheduled climb gradient OEI can be achieved.
Landing distance	The horizontal distance required to land and come to a complete stop from a point 50 ft ALS.
Balked landing distance	The horizontal distance from the LDP to the point at least 35 ft ALS where V_{BLSS} and a positive rate of climb are attained following an engine failure before LDP.
Elevated Heliport	A heliport which is at least 10 ft above the surrounding surface.
Helideck	A heliport located on a floating or fixed Offshore structure.
Heliport	A defined area of land or structure intended to be used for the arrival and departure of helicopters.



5.1.7.1 Clear area

PARAMETERS	
AIRSPEED DEFINITIONS	
Balked Landing Safety Speed (V_{BLSS})	45 KIAS
Best Rate of Climb Speed (V_Y)	75 KIAS
LANDING DECISION POINT (LDP)	
Height	50 ft ALS
Airspeed	40 KIAS
Rate of descent	less than 350 ft/min



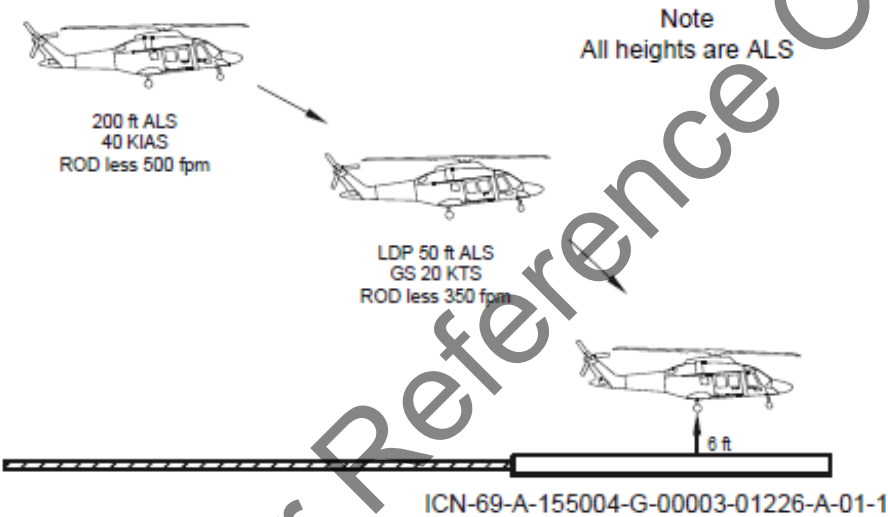
Standard procedure and phraseology

Event	PF / PM	<i>"Phraseology"</i> and/or Action	
Initial point	PF	Stabilizes the aircraft on final approach course at 300 ft ALS / 70 KIAS and initiates a descent and deceleration to pass through LDP with a rate of descent of less than 350 ft/min	<i>"Upper gate"</i>
At LDP	PM	<i>"LDP"</i>	
	PF	<i>"Landing"</i>	Initiates a flare rotating nose up to achieve a hover at 6 ft ALS or to perform a running landing
Passing 20 ft ALS	PM	Checks pitch attitude for a maximum of 15° nose up at touchdown.	<i>"Pitch below 15"</i> or <i>"Check pitch"</i>
	PF	Checks and adjusts pitch attitude if required and lands	<i>"Checked, landing on"</i>
After touchdown	PF	Centralizes controls, reduces collective to MPOG and applies wheel brakes as required	<i>"After landing checks"</i>
	PM	Performs after landing checks	



5.1.7.2 Ground and elevated helipad/helideck

PARAMETERS	
AIRSPEED DEFINITIONS	
Balked Landing Safety Speed (V_{BLSS})	45 KIAS
Best Rate of Climb Speed (V_Y)	75 KIAS
LANDING DECISION POINT (LDP)	
Height	50 ft ALS
Ground speed	20 kts
Rate of descent	less than 350 ft/min



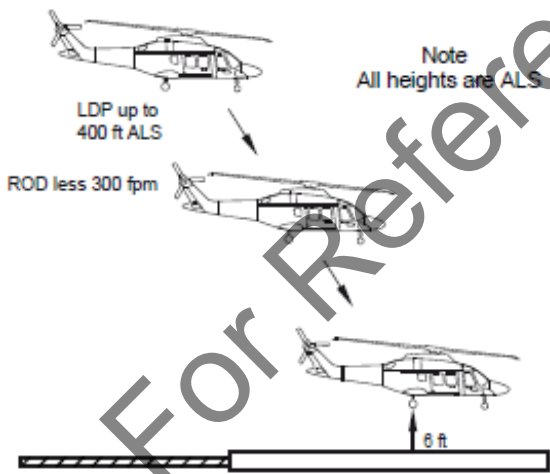
Standard procedure and phraseology

Event	PF / PM	<i>"Phraseology"</i> and/or Action	
Initial point	PF	Establishes an approach to pass through 200 ft ALS at 40 KIAS with a rate of descent of no more than 500 ft/min and initiates a deceleration to pass through LDP with a rate of descent of less than 350 ft/min	
At LDP	PM	<i>"LDP"</i>	
	PF	<i>"Landing"</i>	Initiates a flare rotating nose up to achieve a hover at 6 ft ALS
Passing 20 ft ALS	PM	Checks pitch attitude for a maximum of 15° nose up at touchdown.	<i>"Pitch below 15"</i> or <i>"Check pitch"</i>
	PF	Checks and adjusts pitch attitude if required and lands	<i>"Checked, landing on"</i>
After touchdown	PF	Centralizes controls, reduces collective to MPOG and applies wheel brakes as required	<i>"After landing checks"</i>
	PM	Performs after landing checks	<i>"After landing checks completed"</i>



5.1.7.3 Ground and elevated helipad/helideck variable LDP

PARAMETERS	
AIRSPED DEFINITIONS	
Balked Landing Safety Speed (V_{BLS})	45 KIAS
Best Rate of Climb Speed (V_Y)	75 KIAS
LANDING DECISION POINT (LDP)	
Minimum LDP height, in absence of obstacles	115 ft (100 ft drop down + 15 ft clearance)
Minimum LDP height, with obstacles within BLD	Obstacle height + 135 ft (100 ft drop down + 35 ft clearance)
Maximum LDP height	400 ft ALS
Max groundspeed with PARK BRAKE ON	5 kts
Rate of descent	less than 300 ft/min



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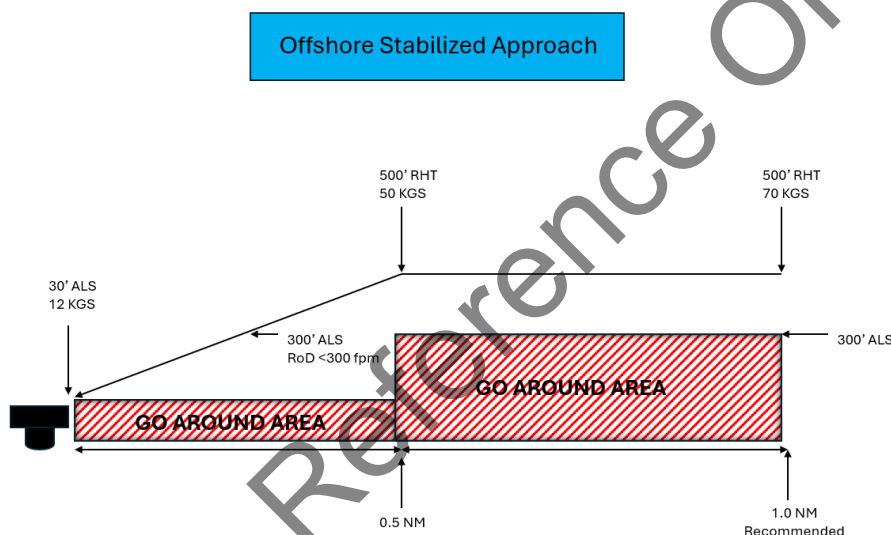
Standard procedure and phraseology

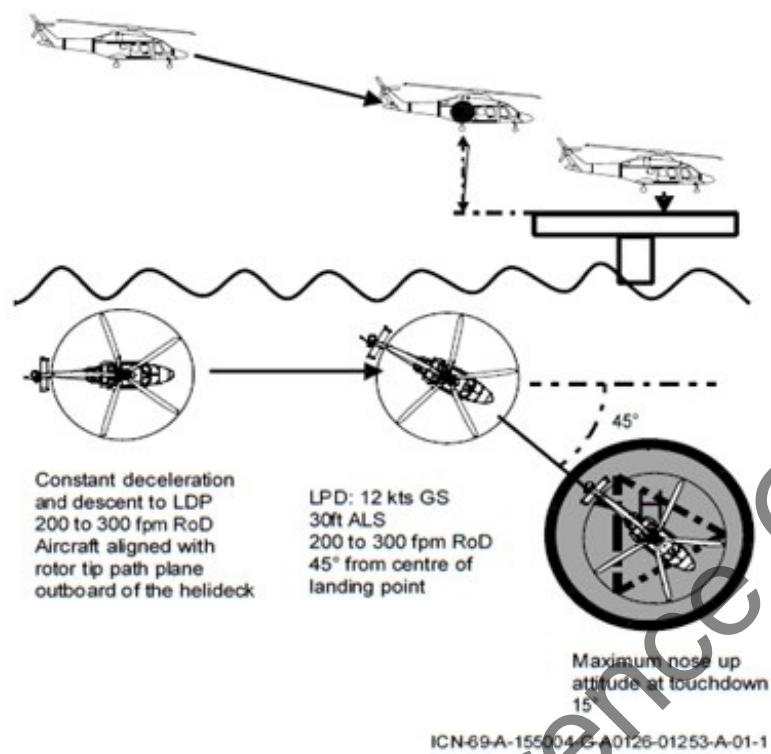
Event	PF / PM	<i>"Phraseology"</i> and/or Action	
Initial point	PF	Establishes an approach at 20 kts GS and 50 ft higher than LDP height and slows down maintaining height to reach a hover followed by a shallow descend at no more than 300 ft/min with the centre of the helipad in sight between the yaw pedals	
At LDP	PM	<i>"LDP"</i>	
	PF	<i>"Landing"</i>	Continues descending with no more than 300 ft/min maintaining the centre of the helipad in sight between the yaw pedals to reach a hover at 6 ft ALS
Passing 20 ft ALS	PM	Checks pitch attitude for a maximum of 15° nose up at touchdown.	<i>"Pitch below 15"</i> or <i>"Check pitch"</i>
	PF	Checks and adjusts pitch attitude if required and lands	<i>"Checked, landing on"</i>
After touchdown	PF	Centralizes controls, reduces collective to MPOG and applies wheel brakes as required	<i>"After landing checks"</i>
	PM	Performs after landing checks	<i>"After landing checks completed"</i>



5.1.8 Offshore / elevated helideck approach

PARAMETERS	
AIRSPEED DEFINITIONS	
Balked Landing Safety Speed (V_{BLS})	45 KIAS
Best Rate of Climb Speed (V_Y)	75 KIAS
LANDING DECISION POINT (LDP)	
Height	30 ft ALS
Ground speed	12 kts
Rate of descent	less than 300 ft/min





5.1.9 Engine Failures

5.1.9.1 Takeoff

5.1.9.1.1 CAT-A

General

The procedures detailed in this chapter only consider the failure of one engine in the take-off path and assume the pilot gives first priority to aircraft control and to achieve a safe flight path.

For all take-off procedures the following shall be observed:

Engine failure prior to TDP	Recognition of engine failure prior to the TDP dictates an OEI REJECTED TAKE-OFF
Engine failure at TDP	Recognition of the failure at the TDP permits either an OEI REJECTED TAKE-OFF or an OEI CONTINUED TAKE-OFF
Engine failure after TDP	Recognition of the failure after TDP dictates an OEI CONTINUED TAKE-OFF

Procedures dealing with subsequent failures / malfunctions, other than engine failure, must be initiated only after the aircraft control and a safe flight / ground condition has been obtained.

The procedures for the emergency / malfunction can be followed according to the appropriate instructions detailed in the QRH.

For Reference Only

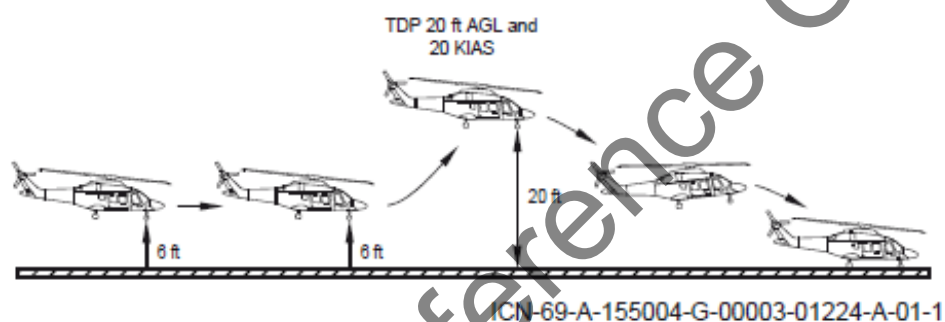


5.1.9.1.1.1 Clear Area

PARAMETERS	
AIRSPEED DEFINITIONS	
Take-Off Safety Speed (V_{Toss})	45 KIAS
Best Rate of Climb Speed (V_Y)	75 KIAS
TAKE-OFF DECISION POINT (TDP)	
Height	20 ft ATS
Airspeed	20 KIAS

5.1.9.1.1.1.1 Before TDP

OEI rejected take-off procedure



Event	PF / PM	<i>"Phraseology"</i> and/or Action	
Engine failure or Master Warning	PM	<i>"Torque split"</i> or <i>"Warning"</i>	
	PF	<i>"Rejecting, call N_R"</i>	Adjusts collective to stop climb and establishes a descent. Adjusts pitch attitude as required to reduce speed.
	PM	Checks and repeatedly reads N_R .	<i>"N_R 103, 102, 101, 101, 101, ..."</i>
Approaching 10 ft ATS	PM	Checks pitch attitude for a maximum of 15° nose up at touchdown.	<i>"Pitch below 15"</i> or <i>"Check pitch"</i>



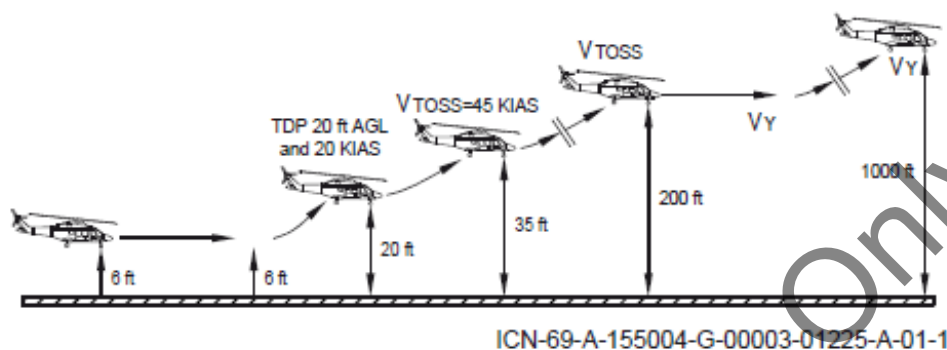
	PF	Checks and adjusts pitch attitude if required and performs a running landing increasing collective to cushion the landing.	<i>"Checked, landing on"</i>
After touchdown	PF	centralizes cyclic, lowers collective to MPOG and applies wheel brakes as required.	

For Reference Only



5.1.9.1.1.2 After TDP OEI continued take-off procedure

Note
All height are ATS



Event	PF / PM	<i>"Phraseology"</i> and/or Action	
Engine failure or Master Warning	PM	<i>"Torque split"</i> or <i>"Warning"</i>	
	PF	<i>"Continuing, call N_R"</i>	Continues to accelerate using collective to maintain N _R close to 93%. Accelerates to V _{TOSS} (45KIAS) and starts to climb.
	PM	Checks and repeatedly reads N _R .	<i>"N_R 95, 94, 93, 93, 93, ..."</i>
V _{TOSS} and positive ROC	PM	<i>"V_{TOSS} climbing"</i>	
	PF	<i>"Checked"</i>	Adjusts pitch attitude to maintain speed and lowers collective to recover N _R to 101%. Continues climb to 200 ft ATS, using 2.5 min power rating.
At 200 ft ATS	PM	<i>"200 ft"</i>	
	PF	<i>"Checked"</i>	Accelerates to V _Y using 2.5 min power rating.



V _Y	PM	"V _Y "	
	PF	"Checked"	Adjusts cyclic to maintain speed and continues climb to 1 000 ft ATS reducing power to OEI MCP (140 % PI) when convenient before expiry of the 2.5 min power rating.

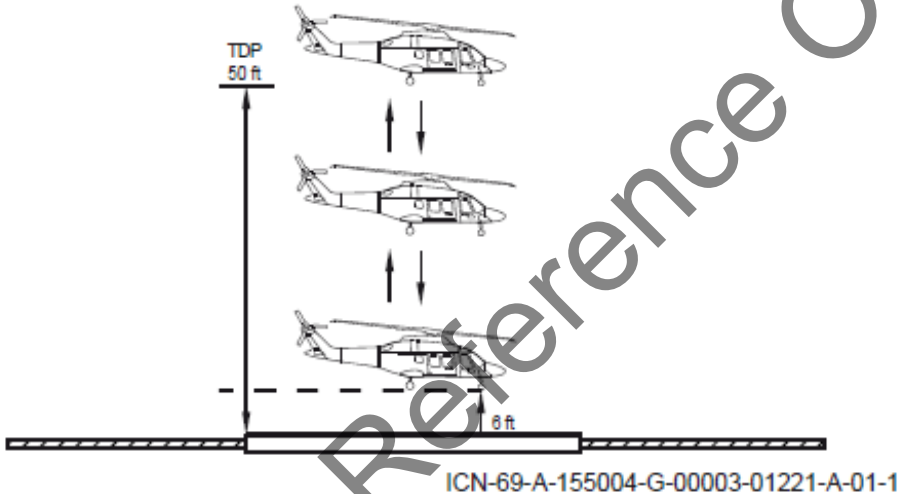
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5.1.9.1.1.2 Ground and elevated heliport/helideck

PARAMETERS	
AIRSPEED DEFINITIONS	
Take-Off Safety Speed (V_{ross})	45 KIAS
Best Rate of Climb Speed (V_Y)	75 KIAS
TAKE-OFF DECISION POINT (TDP)	
Height	50 ft ATS

5.1.9.1.1.2.1 Before TDP
OEI rejected take-off procedure



Event	PF / PM	"Phraseology" and/or Action	
Engine failure or Master Warning	PM	"Torque split" or "Warning"	
	PF	"Rejecting, call N_R "	Adjusts collective to stop climb and establishes a vertical descent while maintaining horizontal position over the take-off point. Maintains N_R close to 100%.
	PM	Checks and repeatedly reads N_R .	" N_R 102, 101, 100, 100, 100, ..."

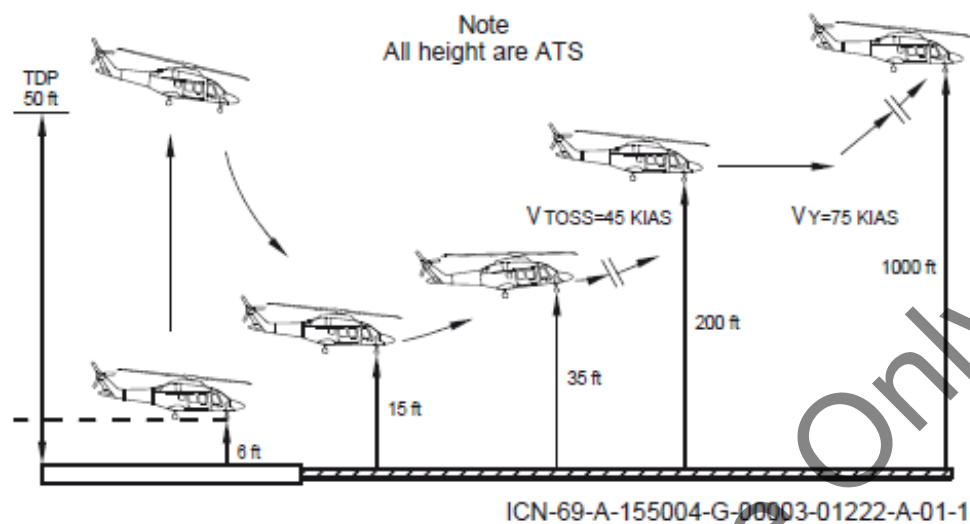


Approaching 10 ft ATS	PM	Checks pitch attitude for a maximum of 15° nose up at touchdown.	<i>"Pitch below 15"</i> or <i>"Check pitch"</i>
	PF	Checks and adjusts pitch attitude if required and increase collective to cushion the landing. Maximum allowed GS at touchdown is 5 kts.	<i>"Checked, landing on"</i>
After touchdown	PF	centralizes cyclic and lowers collective to MPOG.	

For Reference Only



5.1.9.1.1.2.2 After TDP OEI continued take-off procedure



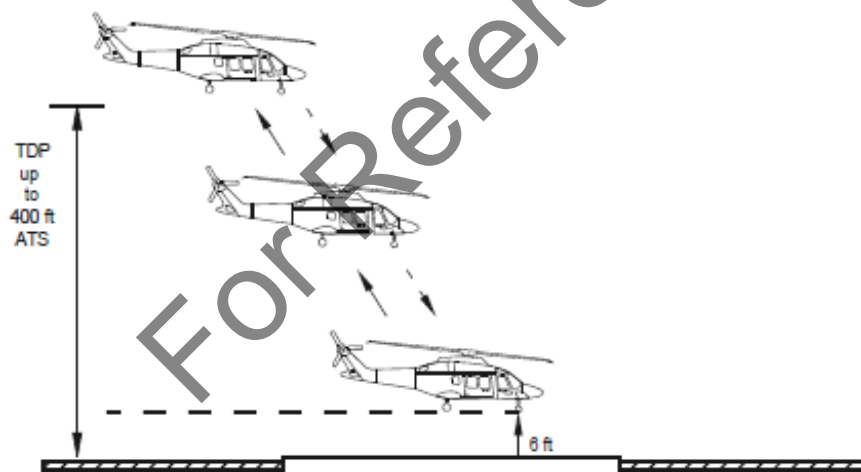
Event	PF / PM	<i>"Phraseology"</i> and/or Action	
Engine failure or Master Warning	PM	<i>"Torque split"</i> or <i>"Warning"</i>	
	PF	<i>"Continuing, call N_R"</i>	Rotates to 15° nose down and maintains this attitude until airspeed indication starts to increase. Adjusts collective to maintain N _R close to 93%.
	PM	Checks and repeatedly reads N _R .	<i>"N_R 95, 94, 93, 93, 93, ..."</i>
When IAS starts to increase	PM	<i>"Airspeed alive"</i>	
	PF	<i>"Checked"</i>	Rotates nose-up to level attitude and maintains for 1 second. After 1 second, continues rotation to 5° nose up and maintains to accelerate to V _{TOSS} (45 KIAS).
V _{TOSS} and positive ROC	PM	<i>"V_{TOSS} climbing"</i>	

	PF	"Checked"	Adjusts pitch attitude to maintain speed and lowers collective to recover N_R to 101%. Continues climb to 200 ft ATS, using 2.5 min power rating.
At 200 ft ATS	PM	"200 ft"	
	PF	"Checked"	Accelerates to V_Y using 2.5 min power rating.
V_Y	PM	" V_Y "	
	PF	"Checked"	Adjusts cyclic to maintain speed and continues climb to 1 000 ft ATS reducing power to OEI MCP (140 % PI) when convenient before expiry of the 2.5 min power rating.



5.1.9.1.1.3 Ground and elevated heliport/helideck variable LDP

PARAMETERS	
WIND LIMITATIONS (in addition to chapter 1.12)	
Maximum cross wind component	10 kts
AIRSPED DEFINITIONS	
Take-Off Safety Speed (V_{TOSS})	45 KIAS
Best Rate of Climb Speed (V_Y)	75 KIAS
TAKE-OFF DECISION POINT (TDP)	
Minimum TDP height, in absence of obstacles	95 ft (80 ft drop down + 15 ft clearance)
Minimum TDP height, with obstacles within CTOD	Obstacle height + 95 ft
Maximum TDP height	400 ft ATS

 5.1.9.1.1.3.1 Before TDP
 OEI rejected take-off procedure


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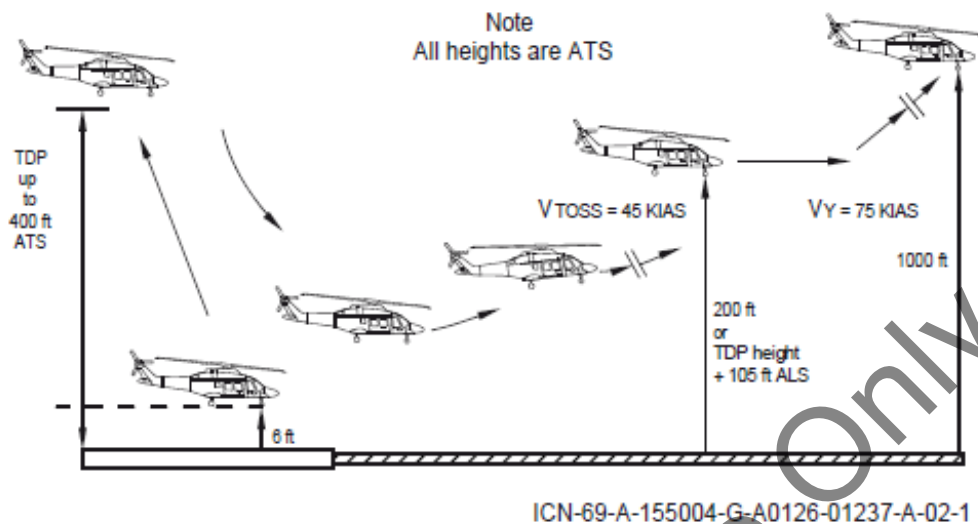
Event	PF / PM	<i>"Phraseology"</i> and/or Action	
Engine failure or Master Warning	PM	<i>"Torque split"</i> or <i>"Warning"</i>	



	PF	<i>"Rejecting, call N_R"</i>	Adjusts collective to stop climb and establishes a descent while maintaining the centre of the helipad in sight between yaw pedals. Maintains N_R close to 100%.
	PM	Checks and repeatedly reads N_R .	<i>"N_R 102, 101, 100, 100, 100, ..."</i>
Approaching 10 ft ATS	PM	Checks pitch attitude for a maximum of 15° nose up at touchdown.	<i>"Pitch below 15"</i> or <i>"Check pitch"</i>
	PF	Checks and adjusts pitch attitude if required and increases collective to cushion the landing. Maximum allowed GS at touchdown is 5 kts.	<i>"Checked, landing on"</i>
After touchdown	PF	centralizes cyclic and lowers collective to MPOG.	



5.1.9.1.1.3.2 After TDP OEI continued take-off procedure



Event	PF / PM	<i>"Phraseology"</i> and/or Action	
Engine failure or Master Warning	PM	<i>"Torque split"</i> or <i>"Warning"</i>	
	PF	<i>"Continuing, call N_R"</i>	Rotates to 15° nose down and maintains this attitude until airspeed indication starts to increase. Adjusts collective to maintain N_R close to 93%.
	PM	Checks and repeatedly reads N_R .	<i>"N_R 95, 94, 93, 93, 93, ..."</i>
When IAS starts to increase	PM	<i>"Airspeed alive"</i>	
	PF	<i>"Checked"</i>	Rotates nose-up to level attitude and maintains for 1 second. After 1 second, continues rotation to 5° nose up and maintains to accelerate to V_{TOSS} (45 KIAS).
V_{TOSS} and positive ROC	PM	<i>"V_{TOSS} climbing"</i>	



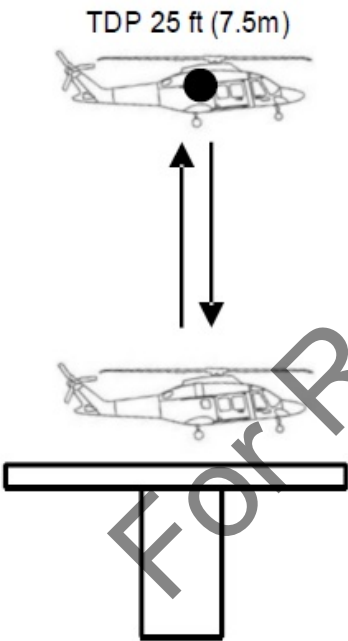
	PF	"Checked"	Adjusts pitch attitude to maintain speed and lowers collective to recover N_R to 101%. Continues climb to 200 ft ATS, using 2.5 min power rating.
At 200 ft ATS	PM	"200 ft"	
	PF	"Checked"	Accelerates to V_Y using 2.5 min power rating.
V_Y	PM	" V_Y "	
	PF	"Checked"	Adjusts cyclic to maintain speed and continues climb to 1 000 ft ATS reducing power to OEI MCP (140 % PI) when convenient before expiry of the 2.5 min power rating.



5.1.9.1.1.4 Offshore elevated helideck take off

PARAMETERS	
WIND LIMITATIONS (in addition to chapter 1.12)	
Maximum cross wind component	10 kts
AIRSPED DEFINITIONS	
Take-Off Safety Speed (V_{ross})	45 KIAS
Best Rate of Climb Speed (V_Y)	75 KIAS
TAKE-OFF DECISION POINT (TDP)	
Height	25 ft ATS

5.1.9.1.1.4.1 Before TDP
OEI rejected take-off procedure



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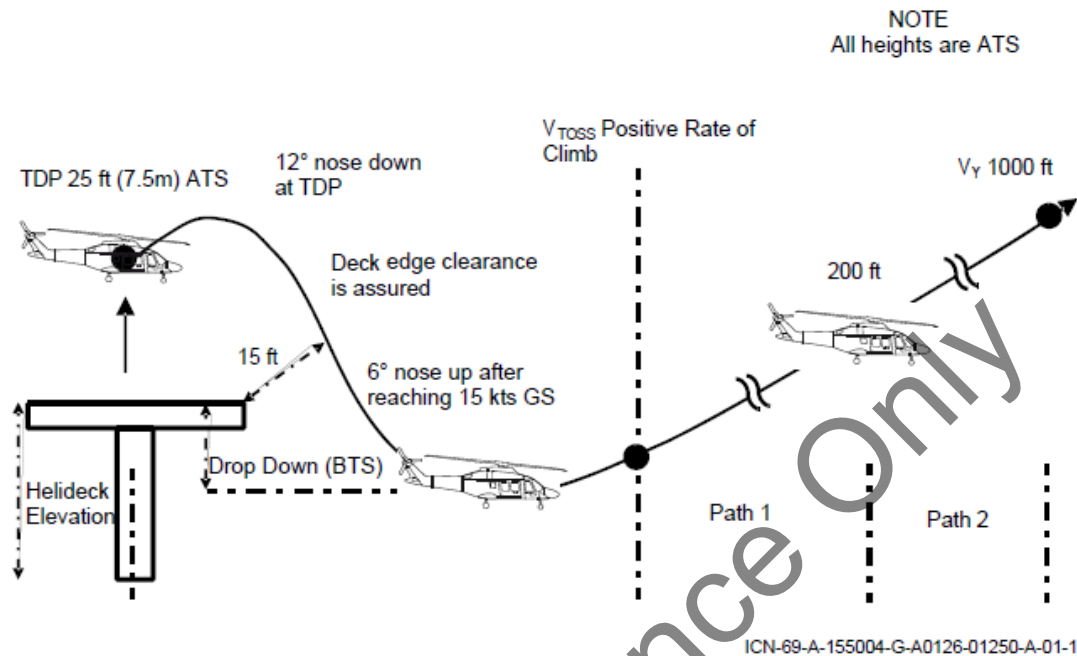
Event	PF / PM	<i>"Phraseology"</i> and/or Action	
Engine failure or Master Warning	PM	<i>"Torque split"</i> or <i>"Warning"</i>	



	PF	<i>"Rejecting, call N_R"</i>	Adjusts collective to stop climb and establishes a vertical descent while maintaining horizontal position over the take-off point. Maintains N_R close to 100%.
	PM	Checks and repeatedly reads N_R .	<i>"N_R 102, 101, 100, 100, 100, ..."</i>
Approaching 10 ft ATS	PM	Checks pitch attitude for a maximum of 15° nose up at touchdown.	<i>"Pitch below 15"</i> or <i>"Check pitch"</i>
	PF	Checks and adjusts pitch attitude if required and increase collective to cushion the landing. Maximum allowed GS at touchdown is 5 kts.	<i>"Checked, landing on"</i>
After touchdown	PF	centralizes cyclic and lowers collective to MPOG.	



5.1.9.1.1.4.2 After TDP OEI continued take-off procedure



Event	PF / PM	<i>"Phraseology"</i> and/or Action	
Engine failure or Master Warning	PM	<i>"Torque split"</i> or <i>"Warning"</i>	
	PF	<i>"Continuing, call N_R"</i>	Rotates to 12° nose down to achieve 15 kts GS. Adjusts collective to maintain N _R close to 93%.
	PM	Checks and repeatedly reads N _R .	<i>"N_R 95, 94, 93, 93, 93, ..."</i>
At 15 kts GS	PM	<i>"Ground speed 15"</i>	
	PF	<i>"Checked"</i>	Rotates to 6° nose up and continues acceleration to V _{TOSS} (45 KIAS).
V _{TOSS} and positive ROC	PM	<i>"V_{TOSS} climbing"</i>	



	PF	"Checked"	Adjusts pitch attitude to maintain speed and lowers collective to recover N_R to 101%. Continues climb to 200 ft ATS, using 2.5 min power rating.
At 200 ft ATS	PM	"200 ft"	
	PF	"Checked"	Accelerates to V_Y using 2.5 min power rating.
V_Y	PM	" V_Y "	
	PF	"Checked"	Adjusts cyclic to maintain speed and continues climb to 1 000 ft ATS reducing power to OEI MCP (140 % PI) when convenient before expiry of the 2.5 min power rating.



5.1.9.2 Landing

5.1.9.2.1 CAT-A

General

The procedures detailed herein only consider the failure of one engine in the landing path and assume the pilot gives first priority to aircraft control and achieve a safe flight path.

For all Landing procedures the following shall be observed:

Engine failure prior to or at LDP	Recognition of engine failure prior to or at LDP permits either an OEI BALKED LANDING or an OEI LANDING
Engine failure after LDP	Recognition of engine failure after the LDP dictates an OEI LANDING

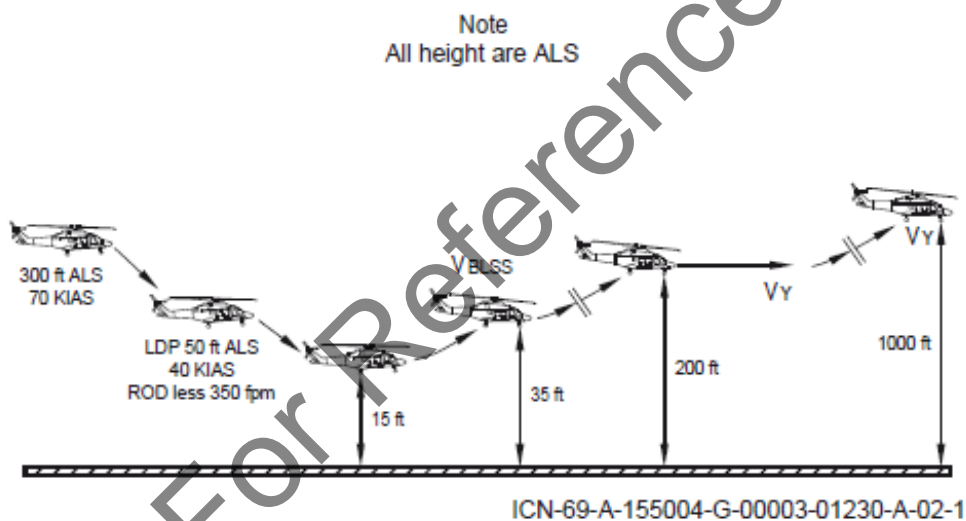
Procedures dealing with subsequent failures / malfunctions, other than engine failure, must be initiated only after the aircraft control and a safe flight / ground condition have been obtained.

The procedures for the emergency / malfunction can be followed according to the appropriate instructions detailed in the QRH.



5.1.9.2.1.1 Clear Area

PARAMETERS	
AIRSPEED DEFINITIONS	
Balked Landing Safety Speed (V_{BLSS})	45 KIAS
Best Rate of Climb Speed (V_Y)	75 KIAS
LANDING DECISION POINT (LDP)	
Height	50 ft ALS
Airspeed	40 KIAS
Rate of descent	less than 350 ft/min

 5.1.9.2.1.1.1 Before TDP
 OEI balked landing procedure


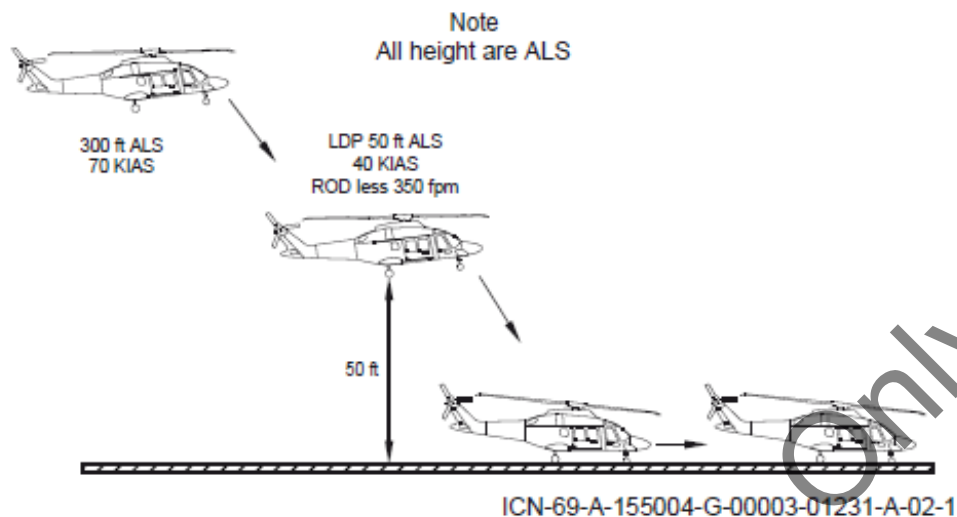
Event	PF / PM	<i>"Phraseology"</i> and/or Action	
Engine failure or Master Warning	PM	<i>"Torque split"</i> or <i>"Warning"</i>	
	PF	<i>"Going-around, call N_R"</i>	Applies collective to set 2.5 minutes power rating and to maintain N_R close to 93%. Adjusts pitch attitude to attain V_{BLSS} (45 KIAS).



	PM	Checks and repeatedly reads N_R .	<i>"N_R 95, 94, 93, 93, 93, ..."</i>
V_{TOSS} and positive ROC	PM	<i>"V_{TOSS} climbing"</i>	
	PF	<i>"Checked"</i>	Adjusts pitch attitude to maintain speed and lowers collective to recover N_R to 101%. Continues climb to 200 ft ALS, using 2.5 min power rating.
At 200 ft ALS	PM	<i>"200 ft"</i>	
	PF	<i>"Checked"</i>	Accelerates to V_Y using 2.5 min power rating.
V_Y	PM	<i>"V_Y"</i>	
	PF	<i>"Checked"</i>	Adjusts cyclic to maintain speed and continues climb to 1 000 ft ALS reducing power to OEI MCP (140 % PI) when convenient before expiry of the 2.5 min power rating.



5.1.9.2.1.1.2 After TDP OEI landing procedure



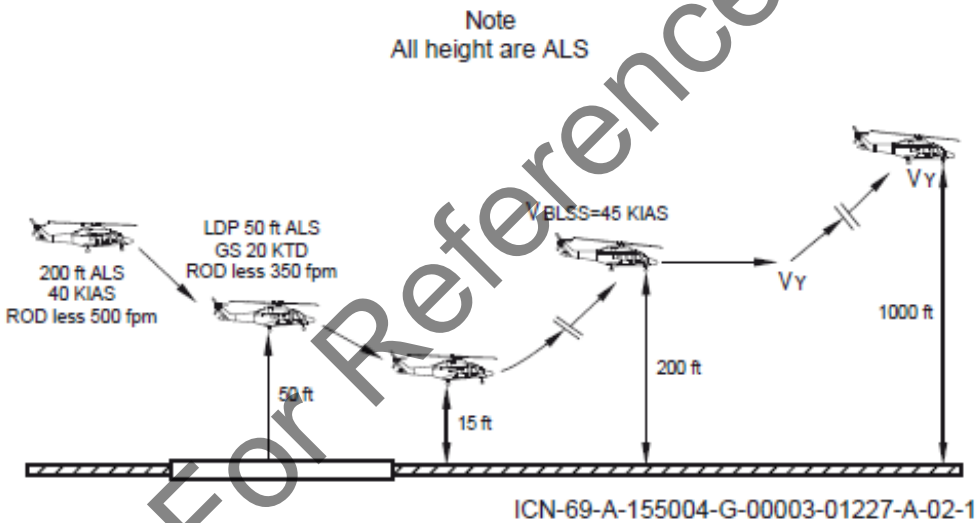
Event	PF / PM	<i>"Phraseology"</i> and/or Action	
Engine failure or Master Warning	PM	<i>"Torque split"</i> or <i>"Warning"</i>	
	PF	<i>"Landing, call N_R"</i>	Adjusts collective to continue the descent. Adjusts pitch attitude as required to reduce speed.
	PM	Checks and repeatedly reads N _R .	<i>"N_R 103, 102, 101, 101, 101, ..."</i>
Approaching 10 ft ALS	PM	Checks pitch attitude for a maximum of 15° nose up at touchdown.	<i>"Pitch below 15"</i> or <i>"Check pitch"</i>
	PF	Checks and adjusts pitch attitude if required and performs a running landing increasing collective to cushion the landing.	<i>"Checked, landing on"</i>
After touchdown	PF	centralizes cyclic, lowers collective to MPOG and applies wheel brakes as required.	



5.1.9.2.1.2 Ground and elevated heliport/helideck

PARAMETERS	
AIRSPEED DEFINITIONS	
Balked Landing Safety Speed (V_{BLSS})	45 KIAS
Best Rate of Climb Speed (V_Y)	75 KIAS
LANDING DECISION POINT (LDP)	
Height	50 ft ALS
Ground speed	20 kts
Rate of descent	less than 350 ft/min

5.1.9.2.1.2.1 Before TDP
OEI balked landing procedure



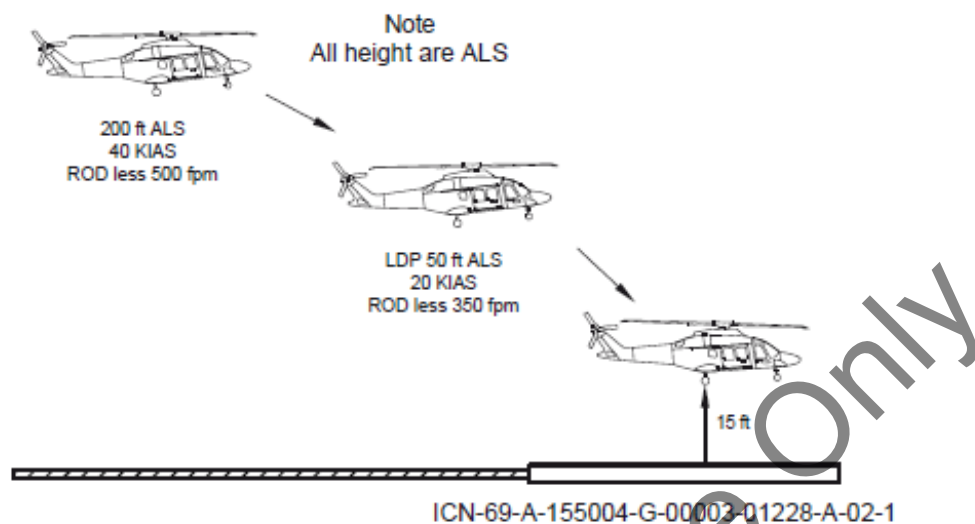
Event	PF / PM	<i>"Phraseology"</i> and/or Action	
Engine failure or Master Warning	PM	<i>"Torque split"</i> or <i>"Warning"</i>	
	PF	<i>"Going-around, call N_R"</i>	Applies collective to set 2.5 minutes power rating and to maintain N_R close to 93%. Adjusts pitch attitude to attain V_{BLSS} (45 KIAS).



	PM	Checks and repeatedly reads N_R .	<i>"N_R 95, 94, 93, 93, 93, ..."</i>
V_{BLSS} and positive ROC	PM	<i>"V_{TOSS} climbing"</i>	
	PF	<i>"Checked"</i>	Adjusts pitch attitude to maintain speed and lowers collective to recover N_R to 101%. Continues climb to 200 ft ALS, using 2.5 min power rating.
At 200 ft ALS	PM	<i>"200 ft"</i>	
	PF	<i>"Checked"</i>	Accelerates to V_Y using 2.5 min power rating.
V_Y	PM	<i>"V_Y"</i>	
	PF	<i>"Checked"</i>	Adjusts cyclic to maintain speed and continues climb to 1 000 ft ALS reducing power to OEI MCP (140 % PI) when convenient before expiry of the 2.5 min power rating.



5.1.9.2.1.2.2 After TDP OEI landing procedure



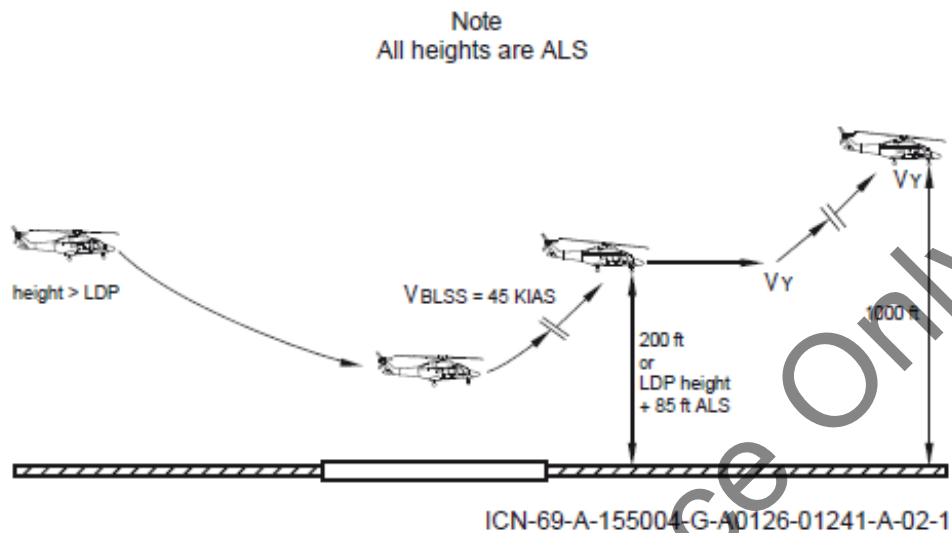
Event	PF / PM	<i>"Phraseology"</i> and/or Action	
Engine failure or Master Warning	PM	<i>"Torque split"</i> or <i>"Warning"</i>	
	PF	<i>"Landing, call N_R"</i>	Adjusts collective continue the descent while reducing rate of descent. Adjusts pitch attitude as required to reduce speed.
	PM	Checks and repeatedly reads N _R .	<i>"N_R 102, 101, 100, 100, 100, ..."</i>
Approaching 10 ft ALS	PM	Checks pitch attitude for a maximum of 15° nose up at touchdown.	<i>"Pitch below 15"</i> or <i>"Check pitch"</i>
	PF	Checks and adjusts pitch attitude if required and increase collective to cushion the landing. Maximum allowed GS at touchdown is 5 kts.	<i>"Checked, landing on"</i>
After touchdown	PF	centralizes cyclic and lowers collective to MPOG.	



5.1.9.2.1.3 Ground and elevated heliport/helideck variable LDP

5.1.9.2.1.3.1 Before TDP

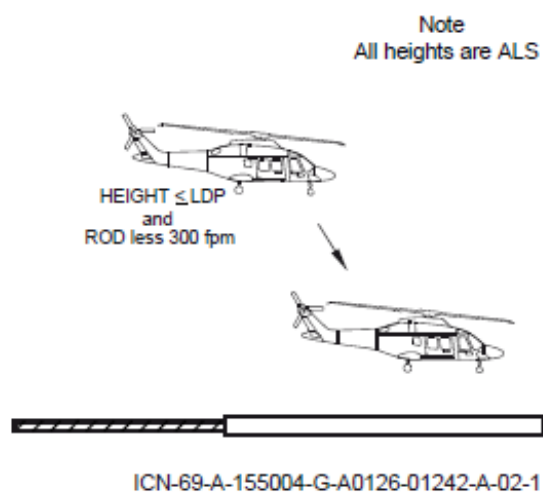
OEI bailed landing procedure



Event	PF / PM	<i>"Phraseology"</i> and/or Action	
Engine failure or Master Warning	PM	<i>"Torque split"</i> or <i>"Warning"</i>	
	PF	<i>"Going-around, call N_R"</i>	Rotates to 15° nose down and maintains this attitude until airspeed indication starts to increase. Adjusts collective to maintain N _R close to 93%.
	PM	Checks and repeatedly reads N _R .	<i>"N_R 95, 94, 93, 93, 93, ..."</i>
When IAS starts to increase	PM	<i>"Airspeed alive"</i>	
	PF	<i>"Checked"</i>	Rotates nose-up to level attitude and maintains for 1 second. After 1 second, continues rotation to 5° nose up and maintains to accelerate to V _{BLSS} (45 KIAS).
V _{BLSS} and positive ROC	PM	<i>"V_{TOSS} climbing"</i>	
	PF	<i>"Checked"</i>	Adjusts pitch attitude to maintain speed and lowers collective to recover N _R to 101%. Continues climb to 200 ft ALS, using 2.5 min power rating.
At 200 ft ALS	PM	<i>"200 ft"</i>	
	PF	<i>"Checked"</i>	Accelerates to V _Y using 2.5 min power rating.
V _Y	PM	<i>"V_Y"</i>	
	PF	<i>"Checked"</i>	Adjusts cyclic to maintain speed and continues climb to 1 000 ft ALS reducing power to OEI MCP (140 % PI) when convenient before expiry of the 2.5 min power rating.



5.1.9.2.1.3.2 After TDP OEI landing procedure

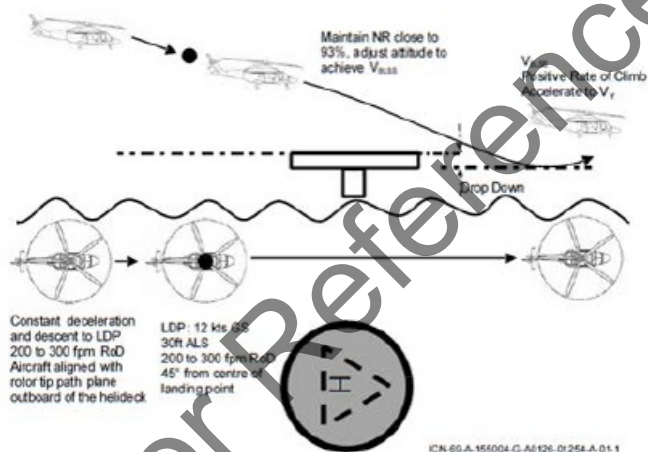


Event	PF / PM	<i>"Phraseology"</i> and/or Action	
Engine failure or Master Warning	PM	<i>"Torque split"</i> or <i>"Warning"</i>	
	PF	<i>"Landing, call N_R"</i>	Adjusts collective to establish a descent while maintaining the centre of the helipad in sight between yaw pedals. Maintains N _R close to 100%.
	PM	Checks and repeatedly reads N _R .	<i>"N_R 102, 101, 100, 100, 100, ..."</i>
Approaching 10 ft ALS	PM	Checks pitch attitude for a maximum of 15° nose up at touchdown.	<i>"Pitch below 15"</i> or <i>"Check pitch"</i>
	PF	Checks and adjusts pitch attitude if required and increases collective to cushion the landing. Maximum allowed GS at touchdown is 5 kts.	<i>"Checked, landing on"</i>
After touchdown	PF	centralizes cyclic and lowers collective to MPOG.	



5.1.9.2.1.4 Offshore / elevated helideck

PARAMETERS	
AIRSPEED DEFINITIONS	
Balked Landing Safety Speed (V_{BLSS})	45 KIAS
Best Rate of Climb Speed (V_Y)	75 KIAS
LANDING DECISION POINT (LDP)	
Height	30 ft ALS
Ground speed	12 kts
Rate of descent	less than 300 ft/min

5.1.9.2.1.4.1 Before TDP
OEI bailed landing procedure

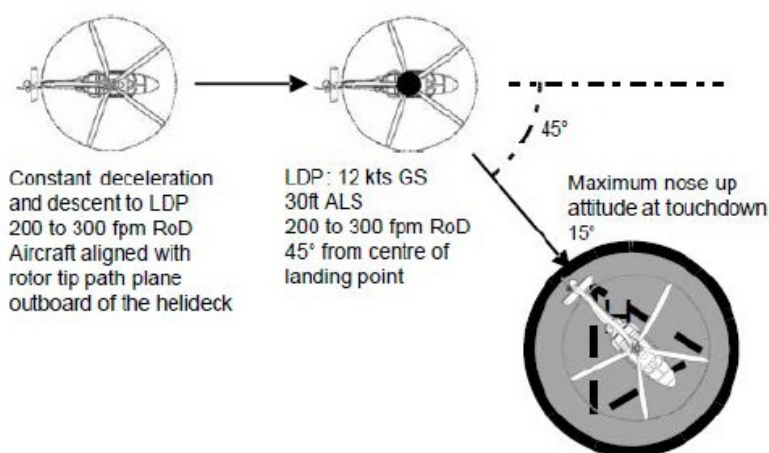
Event	PF / PM	"Phraseology" and/or Action	
Engine failure or Master Warning	PM	"Torque split" or "Warning"	
	PF	"Going-around, call N_R "	Applies collective to set 2.5 minutes power rating and to maintain N_R close to 93%. Adjusts pitch attitude to attain V_{BLSS} (45 KIAS).

	PM	Checks and repeatedly reads N_R .	<i>"N_R 95, 94, 93, 93, 93, ..."</i>
V_{BLSS} and positive ROC	PM	<i>"V_{ross} climbing"</i>	
	PF	<i>"Checked"</i>	Adjusts pitch attitude to maintain speed and lowers collective to recover N_R to 101%. Continues climb to 200 ft ALS, using 2.5 min power rating.
At 200 ft ALS	PM	<i>"200 ft"</i>	
	PF	<i>"Checked"</i>	Accelerates to V_Y using 2.5 min power rating.
V_Y	PM	<i>"V_Y"</i>	
	PF	<i>"Checked"</i>	Adjusts cyclic to maintain speed and continues climb to 1 000 ft ALS reducing power to OEI MCP (140 % PI) when convenient before expiry of the 2.5 min power rating.

5.1.9.2.1.4.2 After TDP



OEI landing procedure



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Event	PF / PM	<i>"Phraseology"</i> and/or Action	
Engine failure or Master Warning	PM	<i>"Torque split"</i> or <i>"Warning"</i>	
	PF	<i>"Landing, call N_R"</i>	Adjusts collective continue the descent while reducing rate of descent. Adjusts pitch attitude as required to reduce speed.
	PM	Checks and repeatedly reads N _R .	<i>"N_R 102, 101, 100, 100, 100, ..."</i>
Approaching 10 ft ALS	PM	Checks pitch attitude for a maximum of 15° nose up at touchdown.	<i>"Pitch below 15"</i> or <i>"Check pitch"</i>
	PF	Checks and adjusts pitch attitude if required and increase collective to cushion the landing. Maximum allowed GS at touchdown is 5 kts.	<i>"Checked, landing on"</i>
After touchdown	PF	centralizes cyclic and lowers collective to MPOG.	

5.1.10 IFR Approaches

5.1.10.1 Stabilized Approach Concept

For all IFR approaches, stabilization shall be achieved at the latest when passing 1,000 ft above runway threshold elevation. At this point, the following criteria shall be met:



- Helicopter is in the landing configuration with before landing checks completed.
- Helicopter speed is stable at the briefed approach speed and on the correct flight path.
(The final approach speed may be reduced, but not below speed limitations.)
- Only small changes in heading (+/- 30°) and power should be required to maintain the flight path.
- RoD <500 ft/min or as required to maintain the briefed approach procedure (+/- 200 ft/min)

If an approach is 'not stable' or becomes 'unstable' outside the stabilized approach criteria, a missed approach is to be initiated. Either pilot may call for a go-around if he deems the approach to be unstable.

For Reference Only



Standard procedure and phraseology

Event	PF / PM	<i>"Phraseology"</i> and/or Action	
Passing 1 000 ft above runway threshold	PM	Checks if approach is stable	
unstable -> go-around			
If approach is unstable	PM	<i>"Passing 1 000, unstable, go-around"</i>	
	PF	<i>"Roger, going-around, go-around checks"</i>	Presses the "GA/TU" button on the collective
	PM	Performs go-around checks	<i>"Go-around checks completed"</i>
stabilized -> continue			
If approach is stabilized	PM	<i>"Passing 1 000, stabilized, continue"</i>	
	PF	<i>"Roger, continuing"</i>	
At 500 ft above MDA / DH	PM	<i>"500 to the minimum"</i>	
	PF	<i>"Checked"</i>	
At 100 ft above MDA / DH	PM	<i>"100 to the minimum, looking out"</i>	Checks landing environment in sight
50 ft above MDA / 20-30 ft above DH	PF	<i>"Approaching minimum"</i>	
not visual -> go-around			
Landing environment not visual	PM	<i>"Negative visual, go-around!"</i>	
	PF	<i>"Roger, going-around"</i>	Presses the "GA/TU" button on the collective



Passing 1 000 ft AGL	PM	<i>"1 000 ft"</i>	
	PF	Checks altitude / height	<i>"Checked, go-around checks"</i>
	PM	Performs go-around checks	<i>"Go-around checks completed"</i>
visual -> continue			
Landing environment in sight	PM	<i>"Visual, look up for sight xx o'clock"</i>	
	PF	Checks landing environment in sight	<i>"Visual, continuing, final checks"</i>
	PM	Performs final checks	<i>"Final checks completed"</i>
At LDP / CAT B Gate	PM	<i>"LDP" / "Gate"</i>	
	PF	<i>"Landing"</i>	

5.1.10.2 Precision Approach

While being vectored to the final leg, set the final approach track on the course pointer. When cleared for the approach the APP mode shall be armed when flying coupled.

When the APP mode is armed the HDG mode will automatically engage at the current heading (if not already engaged) indicated by the green HDG caption on the PFD and the LOC and GS modes will be armed, as shown by the white LOC and GS indications on the PFD.

As the LOC mode is captured a chime will sound to indicate that the HDG mode is no longer captured and the LOC indication will change from white to green. As the GS mode is captured a chime will sound to indicate that the ALT mode is no longer captured and the GS indication will change from white to green.

Auto level function

During a coupled ILS approach pilots are encouraged to keep the coupler engaged below decision altitude until PF is fully satisfied with the visual references, particularly in marginal conditions, provided



that:

- PNF had the required references in sight to continue the approach
- PF keeps his hands and feet on the controls at all times when the aircraft is below decision altitude
- With the FD still coupled to the autopilot, the auto-level feature will automatically flare the aircraft at 200 feet to a radar altitude of 75 feet radar altitude and hold it. At the start of the flare manoeuvre, the GS in the vertical capture modes field of the PFD is replaced by ALVL.

5.1.10.3 GPS Approach

General

The RNP APCH with LP or LPV minima (charted as RNAV(GPS) or RNAV(GNSS)) can be selected on the EDCU approach page as one of the procedures for the selected runway displayed as RNAV.

Once the LPV (or LP) approach has been selected, the channel number and reference path ID is displayed for cross-check with the approach chart.

In case of SBAS unavailability at destination, the LPV approach page provides minima type selection to downgrade from LPV minima to Baro-VNAV minima (LNAV/VNAV or LNAV minima).

In case of degradation verify that the "LPV" annunciation is not displayed on the PFD leaving the "VGP" annunciation (white/green) as confirmation of Baro- VNAV guidance on vertical axis.

Prior to reaching the Final Approach Fix (FAF)

- | | |
|------------------------|--|
| 1. PFD/EDCU/MFD | <p>Once the approach has been activated verify:</p> <ul style="list-style-type: none"> • Advisory message "LPV" or "LP" (white) displayed on PFD • DA(H) accordingly set with approach chart. • PERF INIT executed with CRUISE altitude required before the approach • VNAV phase equal to CRUISE before approach VPATH capturing. • TOD correctly computed on PROGRESS - BARO VNAV EDCU page. • TOD symbol displayed on FPLN page and on VSD. |
| 2. EDCU APPROACH SPEED | If deceleration profile is required verify/set the INITIAL and FINAL speeds and relative distances with respect to the MAP waypoint. |
| 3. AFCS panel | <p>Press APP or DCL pushbutton to arm NAPP modes on collective and cyclic lateral, as required.</p> <p>Confirm NDCL armed on cyclic longitudinal if DCL mode selected.</p> |
| 4. Altimeters | Verify correct QNH at destination. |



- | | |
|--------------------------|--|
| 5. Approach path | Adjust ground speed to comply with the approach plate and table "GPA AND GS COMBINATIONS" in chapter 1.29 of this manual. |
| 6. Before landing checks | Complete |
| 7. Approach speed | If automatic FMS deceleration mode is not required, adjust the ground speed as required to maintain ROD equal or less than 900 ft/min while approaching the FAF. |

If the aircraft is above the FAF altitude, the FMS will capture the VPATH in Baro-VNAV to FAF altitude transitioning to NLOC and NGS guidance at a distance equal to or below 2 NM from the FAF waypoint.

During final approach segment

- | | |
|---------|--|
| 1. PFD | Once APP is displayed in green, verify the "LPV or "LP" annunciation is green with automatic transition on cyclic lateral axis from NAPP mode to NLOC mode. |
| 2. AFCS | Confirm: <ul style="list-style-type: none"> • NGS mode engaged. • NDCL or NIAS mode engaged, if required, in accordance with APPROACH SPEED profile. |

Missed approach procedure

- | | |
|---------------|----------------------|
| 1. Collective | Press GA pushbutton. |
|---------------|----------------------|

5.1.10.4 VOR Approach

1. When the VOR approach exists in the FMS database the approach may be flown coupled to the FMS provided raw data can be monitored and is used primarily from the FAF inbound.
2. All crossing fixes may be identified with the FMS.
3. If the approach does not exist in the database or if the FMS is unserviceable proceed as depicted below:
 - a. When cleared for the approach the APP mode shall be armed when flying coupled. When the APP mode is armed the HDG mode will automatically engage at the current heading (if not already engaged) indicated by the green HDG caption on the PFD and the VAPP mode will be armed as shown by the white VAPP indication on the PFD. As the VAPP mode is captured a chime will sound to indicate that the HDG mode is no longer captured and the VAPP indication will change from white to green.



- b. Use the ALT SEL knob to set an altitude between MDA and MDA **plus 50 ft** to avoid inadvertent descent through the MDA. Use the collective beep trim switch, as required, to set the required rate of descent in order to pass the minimum crossing altitudes at the correct altitude. As the ALT mode engages use the collective beep trim switch to descent to the MDA.

5.1.10.5 Circling to Land

Circle to land

If at the end of a non-precision approach, a 'circle to land' is required, it shall be conducted according to the respective IAP chart.

For Reference Only



6 **Forms**

6.1 **Forms**

6.1.1 **TF 01 - Initial Training Log**

For Reference Only



Pilot Information:		
Last Name:	First Name:	Certificate:
Training type: Initial	A/C type:	
Grading: Instructor will initial each applicable module when completed. Check Pilots will verify each module has been completed prior to the students checkride.		

	Basic Indoc	General Emerger	Hazmat & CTS	CRM & CTS	Acft Gnd - Sys	Acft Gnd - Gen	Acft Sim - Mod	Qualification
Total Hours Planned:	24	4	4	4 (8*)	20	4	16	As req.
Module 1								
Module 2								
Module 3								
Module 4								
Module 5								
Module 6								
Module 7								
Module 8								
Module 9								
Module 10								
Module 11								
Module 12								
Module 13								
Module 14								
Module 15								
Module HAA/SAR Crews* 16								
Module HAA/SAR Crews* 17								
Module Offshore IFR Crews** 18								
Module 19								
Module 20								
Module 21								
Module 22								
Module 23								
Module 24								
Module 25								
Module 26								
Module 27								
Module 28								
Module 29								
Module 30								
Verify CTS Completed								
Total Hours Planned:	24	4	4	4 (8*)	20	4	16	As req.
Total Hours Completed:								
Date all modules completed								
Instructor Initials:								

I certify that the above named pilot has satisfactorily completed all required training in accordance with FAA Regulations and HeliService USA Training syllabus.

Check Pilot's Signature:

Date:



6.1.2 TF 02 - Initial Flight Training

For Reference Only



Pilot						
Last Name:		First Name:		Certificate:		
Aircraft Type:		Grading: All items must be completed to a "3" to be considered satisfactory				
Training Type:		1 - Below Acceptable Standards	2 - Below Average	3 - Required Average	4 - Above Average	5 Excellent
☐ Initial ☐ Transition						

Maneuvers	Sim 1	Sim 2	Sim 3	Sim 4	Sim 5	Sim 6	Sim 7	Sim 8	Add 1	Add 2
Pre-Flight Procedures										
Trip planning										
Weight and balance										
Aircraft performance										
Human factors										
Preflight inspection										
Emergency equipment										
Engine starts - Normal/abnormal										
Pre-takeoff checks										
Lift to hover										
Hovering maneuvers										
Taxiing - Ground/hover/air										
Slopes										
Hot weather operations (optional)										
Cold weather operations (optional)										
Helicopter operational hazards										
Dynamic rollover										
Ground resonance										
Retreating blade stall										
Loss of tail rotor effectiveness										
Takeoff and Departure Phase										
Normal takeoff										
Crosswind takeoff										
Running/rolling takeoff										
CAT B takeoff										
CAT A clear area takeoff										
CAT A G/E helipad takeoff										
CAT A Variable TDP takeoff										
Offshore helideck takeoff										
Rejected takeoff - Quick stop										
In Flight Maneuvers										
CFIT avoidance - Use of TAWS										
CFIT avoidance - Use of WX radar										
Navigation and communication										
Basic attitude flying										
Traffic pattern work										
Standard rate turns										



Maneuvers	Sim 1	Sim 2	Sim 3	Sim 4	Sim 5	Sim 6	Sim 7	Sim 8	Add 1	Add 2
Steep turns										
Rapid decelerations										
Recovery from settling with power										
Recovery from unusual attitudes										
Use of AP modes										
Use of CCD and CCJ										
Demonstration of Collective safety functions										
Landings & Approaches to Landing										
Normal approach										
Confined areas or pinnacle operations										
Shallow approach to a running landing										
CAT B landing										
CAT A clear area landing										
CAT A G/E helipad landing										
CAT A Variable TDP landing										
Offshore Helideck Landing										
Rejected landing - Go around										
Emergency Procedures										
Autorotation (Demo)										
Ditching/emergency equipment										
In flight fire and smoke removal										
Tail rotor malfunctions										
Fuel systems malfunctions										
Hydraulic malfunctions										
Powertrain (MR & TR drive malfunctions)										
Electrical system malfunctions										
Powerplant malfunctions										
Auxiliary power unit malfunctions										
Landing gear malfunctions										
Flight controls / AP systems malfunctions										
Avionics and GPS malfunctions										
Engine control malfunctions										
Pitot-static system malfunctions										
AHRS malfunctions										
Display/EDCU malfunctions										
Fire after shutdown & dry motor procedure										
IFR Procedures										
Straight and level flight										
Standard rate turns										
Steep turns										
Climbing and descend turns										
Recovery from unusual attitudes										
Emergency descent										
Instrument departures (ODP)										
SID/STAR										
Holding procedures										
Enroute and Airways										



Maneuvers	Sim 1	Sim 2	Sim 3	Sim 4	Sim 5	Sim 6	Sim 7	Sim 8	Add 1	Add 2
Radio navigation (NDB, GPS & VOR)										
ILS Approach coupled										
ILS Approach hand flown										
RNAV approach LPV coupled										
RNAV approach BARO/VNAV coupled										
RNAV approach LNAV coupled										
VOR approach coupled										
HEDA (Helicopter Enroute Descent Area)										
OSAP (Offshore Standard Approach Procedure)										
Missed approach with GA mode										
Missed-approach hand flown										
IFR Abnormal Procedures										
Avionic failures										
Pitot Static failures										
Hydraulic failures										
Auto Pilot failures										
Engine failures										
During coupled precision approach										
During hand flown precision approach										
During coupled non precision approach										
During hand flown non precision approach										
Post Flight Procedures										
After landing procedures										
Parking and securing										

	Sim 1	Sim 2	Sim 3	Sim 4	Sim 5	Sim 6	Sim 7	Sim 8	Add 1	Add 2	Total
Date:											
Tail or Sim #:											
Inst Initial:											
Stud Initial:											
Times:											
Simulator:											
Aircraft:											
PF:											
PM:											
Day:											
Night:											
IFR:											



Comments / Evaluation	
#1	1
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#2	1
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#3	1
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#4	1
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	5
#5	1
	2
	3
	4
	5
#6	1
	2
	3
	4
	5
#7	1
	2
	3
	4
	5
#8	1
	2
	3
	4
	5
Additional Flights	1
	2
	3
	4
	5

I certify that the above named pilot has satisfactorily completed all required training in accordance with FAA Regulations and HeliService USA Training syllabus.

Check Pilot's Signature:

Date:



6.1.3

TF 03 - Evaluation Form

For Reference Only



NAME OF AIRMAN:		DATE OF CHECK:		LOCATION:	
BASE MONTH:		NAME OF CHECK AIRMAN:		CERTIFICATE TYPE/NUMBER:	
TYPE:		FLIGHT TIME:		AIRCRAFT NUMBER:	
TYPE OF COMPETENCY / PROFICIENCY CHECK					
☐ FAR 135.293 (a)	EXPIRES:	☐ DAY ☐ NIGHT	☐ INITIAL ☐ RECURRENT ☐ UPGRADE ☐ REQUALIFICATION		
☐ FAR 135.293 (b)	EXPIRES:				
☐ FAR 135.297	EXPIRES:				
☐ FAR 135.297 (g)	EXPIRES:				
☐ FAR 135.299	EXPIRES:				
☐ FAR 135.337(d) (LEM)	EXPIRES:				
☐ NVG PROFICIENCY CHECK	EXPIRES:				
REQUIRED FOR QUALIFICATION (A) .293 (I) .297 (N) NVG / FLIGHT MANEUVERS GRADE (U – UNSATISFACTORY) (S – SATISFACTORY) (/ – NOT COMPLETED)					
GROUND OPERATIONS		U	S	LANDINGS AND APPROACHES TO LANDINGS	
1	A/N/I PREFLIGHT PREPARATION ☐ ORAL ☐ WRITTEN			28	A/N/I NORMAL
2	A/N/I PREFLIGHT INSPECTION			29	I LANDING FROM ILS
3	A/N/I START PROCEDURES			30	A/N/I LANDING WITH ENGINE OUT
4	A/N/I TAXIING AND GROUND HOVER			31	I LANDING FROM CIRCLING APPROACH
5	A/N/I PRE-TAKEOFF CHECKS			32	I OFFSHORE FACILITY *
TAKEOFF AND DEPARTURES				NON-NORMAL AND EMERGENCY PROCEDURES	
6	A/N/I NORMAL			33	A/N/I SYSTEM MALFUNCTION
7	A/I INSTRUMENT			34	A/N/I RECOVERY FROM IIMC
8	A/N/I RAPID DECELERATION			35	A/I MANEUVER BY PARTIAL PANEL
9	I OFFSHORE FACILITY *			36	I INSTRUMENT APPROACH
10	A/N/I WITH SIMULATED POWERPLANT FAILURE			37	A/N CONFINED AREA
INFIGHT MANEUVERS				38	A/N PINNACLE OPERATIONS
11	I STEEP TURNS			39	A/N SLOPE OPERATIONS
12	A/N/I SETTling WITH POWER			40	N GROUND HAZARD RECOGNITION
13	A/N/I UNUSUAL ATTITUDE RECOVERY			41	A/N/I BROWNOUT / WHITEOUT / FLAT-LIGHT OPERATIONS
INSTRUMENT PROCEDURES				42	N USE OF EXTERNAL LIGHTING
14	I AUTOPILOT CHECK / USE OF AUTOPILOT			43	N SIMULATED NIGHT VISION GOGGLE FAILURE
15	I AREA DEPARTURE			ORAL OR SIMULATOR ONLY	
16	I AREA ARRIVAL			44	A/N TAIL ROTOR FAILURE
17	I HOLDING			45	A/N DYNAMIC ROLLOVER
18	A/I NORMAL ILS			46	A/N LOW ROTOR RPM
19	A/I ENGINE-OUT ILS			47	A/N ANTITORQUE SYSTEM FAILURE
20	A/I COUPLED APPROACH			GENERAL	
21	A/I NON-PRECISION APPROACH TYPE:			48	A/N/I JUDGEMENT
22	I SECOND NON-PRECISION APPROACH			49	A/N/I CRM
23	I MISSED APPROACH FROM AN ILS			* OFFSHORE FACILITY / HEDA / OSAP PROCEDURES ARE NOT REQUIRED. **HEDA / OSAP APPROACHES MAY BE USED IN LIEU OF 1 NON-PRECISION APPROACH.	
24	I SECOND MISSED APPROACH				
25	I CIRCLING APPROACH				
26	I HEDA (Helicopter Enroute Descent Area) **				
27	I OSAP (Offshore Standard Approach Procedure) **				
REMARKS:					
FAR 135.299 ROUTE OF FLIGHT:				MEDICAL	
				1. CLASS _____	
				2. DATE _____	
RESULT OF CHECK: AIRMAN	☐ SATISFACTORY ☐ UNSATISFACTORY		CHECK AIRMAN'S SIGNATURE:		
RESULT OF CHECK: CHECK AIRMAN	☐ SATISFACTORY ☐ UNSATISFACTORY		FAA INSPECTOR'S SIGNATURE:		




**(TF 04) General Emergency Situation
& Drill Training**

 Version 1
 Page 1 of 1

Pilot Information		
Last name:	Date:	Certificate:
First name:	Last EMER expiry:	Place of check:
A/C type:	New EMER expiry:	

Skills and knowledge to be checked	S	U
Emergency Drill		
Ditching (If Applicable)		
Emergency Evacuation		
Fire Extinguisher and Smoke Control		
Operation and Use of Emergency Exits		
Crew and Passenger Oxygen – If Applicable		
Removal of Life Rafts from the Aircraft, Inflation of the Life Rafts – If Applicable		
Donning and Inflation of Life Vests – If Applicable		
Emergency Situation		
Crewmember Duties and Responsibilities		
Crew Coordination		
Aircraft Fires		
First Aid Equipment		
Illness, Injury and Basic First Aid		
Ground Evacuation		
Ditching		
Previous Aircraft Accidents/Incidents		
Crewmember Incapacitation		
Basic Survival – As Required		
Hijacking/Bomb Threat and Other Unusual Situations		

Result: ☐ Pass ☐ Fail
--

Instructor/Check Airman	
Last name:	First name:
Place:	Signature:

Notes:



6.1.5 TF 05 - Pilot Line Training

Pilot												
Last Name:						First Name:						
Crew Status: ☐ PIC ☐ SIC			A/C type:				Certificate:					
						First Officer ↓				Captain ↓		
Flight / Leg Nr.:	M	1	2	3	4	5	6	7	8	9	10	CKD?
Date:												
Instructor: (3LC)												
Destination:												
Mission: (HHO or CC)												
Seat: (LHS / RHS)												
Complete Flight & Digital AirWare	5											
HHO performance	3											
Power assurance check	1											
CAT-B clear area T/O	1											
CAT-A ground level helipad T/O	1											
CAT-A variable TDP T/O	1											
OPD Departure	1											
Stabilized approach offshore (PF or PM)	2											
User approach (AW169)	1											
HHO (PF/PM) IF Req	2/4											
CAT-A offshore helideck LDG	2/4											
CAT-A offshore helideck T/O	1											
HEDA (Helicopter Enroute Descent Area) If Req	1											
OSAP (Offshore Standard Approach Procedure) If Req	1											
CAT-A ground level helipad LDG	1											
Shut down into/from APU mode	1											
Complete Flight, Tech Log	5											

I certify that the above named pilot has satisfactorily completed all required training in accordance with FAA Regulations and HeliService USA Training syllabus.

Instructor's Signature: _____

Date: _____



Comments / Evaluation	
#1	1
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#2	1
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#3	1
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#4	1
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#9	1
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#10	1
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6.1.6 TF 06 - Pilot Duty Assignment Form

For Reference Only



Pilot Information		
Last Name:	First Name:	Certificate:

CURRENT DUTY & DATE OF ASSIGNMENT / AUTHORIZATION

DUTY POSITION & ASSIGNMENT	QUALIFICATION	AIRCRAFT	DATE ASSIGNED	PILOT INITIALS	CHIEF PILOT / DO INITIALS

TEMPORARILY REMOVED FROM DUTY

	DATE FROM	DATE TO	REASON FOR REMOVAL
1			
2			
3			
4			
5			
6			
7			
8			

RELEASE FROM EMPLOYMENT

REASON / ACTION TAKEN:

Signature (DO/CP):

Date:

Title:

6.1.7 TF-07 - Single Line Entry



6.1.8 TF 08 - Recurrent Training Log

For Reference Only



Pilot Information:		
Last Name:	First Name:	Certificate:
Training Type: Recurrent	A/C type:	Rotation A/B/C:
Grading: Instructor will initial each applicable module when completed. Check Pilots will verify each module has been completed prior to the students checkride.		

	General Subjects & CTS	General Emer	HAZMAT & CTS	CRM & CTS	Acft Gnd - Sys	Acft Sim - Mod	Qualification
Total Hours Planned:	8	1 (3*)	4	2 (8*)	8	4	As req.
Module 1							
Module 2							
Module 3							
Module 4							
Module 5							
Module 6							
Module 7							
Module 8							
Module 9							
Module 10							
Module 11							
Module 12							
Module 13							
Module 14							
Module 15							
Module Medical SAR/Crews*							
Module Medical SAR/Crews*							
Module Offshore IFR Crews**							
Module 19							
Module 20							
Module 21							
Module 22							
Module 23							
Module 24							
Module 25							
Module 26							
Module 27							
Module 28							
Module 29							
Module 30							
Verify CTS Complet							
Total Hours Planned:	8	1	4	2	8	4	As req.
Total Hours Completed:							
Check Pilot Initials:							
Student Initials:							

I certify that the above named pilot has satisfactorily completed all required training in accordance with FAA Regulations and HeliService USA Training syllabus.

Check Pilot's Signature:

Date:



6.1.9 TF 09 - Recurrent Flight Training

For Reference Only



Pilot					
Last Name:	First Name:		Certificate:		
Aircraft Type:	Grading: All items must be completed to a "3" to be considered satisfactory				
Training type: <i>Recurrent</i>	1 - Below Acceptable Standards	2 - Below Average	3 - Required Average	4 - Above Average	5 Excellent

Notes	Instructors will complete each required maneuver per the pilot's rotational year (ex. Year 1 – A, Year 2- B, Year 3, C). Each maneuver will be completed to minimum grade level of three. Recurrent training will consist of a minimum of two sessions totaling 4 hours, split between Pilot Flying and Pilot Monitoring duties. Additional simulator sessions can be added if required.
--------------	--

1. Pre-Flight Procedures	Sim 1	Sim 2	Add.	Add.
Trip planning (A/B/C)				
Weight and balance (A/B/C)				
Aircraft performance (A/B/C)				
Human factors (A/B/C)				
Preflight inspection (If Req.) (A/B/C)				
Emergency equipment (A/B/C)				
Engine starts - Normal/abnormal (A/B/C)				
Pre-takeoff checks (A/B/C)				
Lift to hover (A/B/C)				
Taxiing - Ground/hover/air (A/B/C)				
Slopes (A/B/C)				
Dynamic rollover (A/B/C)				
Ground resonance (A/B/C)				
Retreating blade stall (A/B/C)				
Loss of tail rotor effectiveness (A/B/C)				
2. VFR Takeoff and Departure Phase	Sim 1	Sim 2	Add.	Add.
CAT B takeoff (A/B/C)				
CAT A G/E helipad takeoff (A/B/C)				
CAT A Variable TDP takeoff (A/B/C)				
Rejected takeoff - Quick stop (A/B/C)				
3. VFR In Flight Maneuvers	Sim 1	Sim 2	Add.	Add.
Basic attitude flying (A/B/C)				
Traffic pattern work and helicopter familiarization (A/B/C)				
Steep turns (A/B/C)				
Settling with power & vortex ring state (C)				
Recovery from unusual attitudes (A/B/C)				
Use of AP modes (A/B/C)				
CFIT Avoidance – Use of TAWS (A/B/C)				
CFIT Avoidance – Use of WX Radar (A/B/C)				
Navigation and Communication (A/B/C)				



4. VFR Landings and Approaches to Landing	Sim 1	Sim 2	Add.	Add.
CAT B landing (A/B/C)				
Confined areas or pinnacle operations (A/B/C)				
Shallow approach to a running landing (A/B/C)				
CAT A G/E helipad landing (A/B/C)				
CAT A Variable TDP landing (A/B/C)				
5. Emergency Procedures	Sim 1	Sim 2	Add.	Add.
Engine failure before/after TDP, CAT A clear area & G/E Helipad (A/B/C)				
Engine failure before/after LDP, CAT A clear area & G/E Helipad (A/B/C)				
Engine failure during hover (A/B/C)				
Dual engine failure during hover (A/B/C)				
Straight in autorotation VMC and IMC (A/B/C)				
Engine failure, relight in flight (A,C)				
Engine failure before/after TDP, offshore CAT A elevated helipad (If Req. A/B/C)				
Engine failure before/after LDP, offshore CAT A elevated helipad (If Req. A/B/C)				
Ditching/emergency equipment (A/B/C)				
Engine fire and fuselage fire (A/B/C)				
Tail rotor control failure (B/C)				
Tail rotor driveshaft failure (B/C)				
Hydraulic abnormal procedures (A)				
Electrical malfunctions (B)				
Powerplant abnormal procedures (C)				
Landing gear malfunctions (If Req. A/B/C)				
Avionics malfunctions (B)				
Engine malfunctions (C)				
AHRS malfunctions (A/B/C)				
Display/EDCU malfunctions (A/B/C)				
Exhaust fire after shutdown/ dry motor procedure (B)				



6. IFR Procedures	Sim 1	Sim 2	Add.	Add.
IFR Takeoff with low visibility (1/2sm) (A/C)				
IFR Takeoff with low visibility (400Meters) (B)				
Straight and level flight (A/B/C)				
Recovery from unusual attitudes (A/B/C)				
EGPWS Procedures (A/B/C)				
Precision approach to minima with OEI Landing (A/B/C)				
Go around on instruments from minima on OEI (A/B/C)				
LPV approach to minima with downgrade to LNAV/VNAV (A/B/C)				
Autopilot and flight director malfunctions in turbulence (A/B/C)				
Instrument departures (ODP) (A/B/C)				
Steep Turns, timed standard rate turns, and timed standard rate climbing turns (A)				
Offshore approach to a ship (If Req. B)				
Holding procedures (B)				
Enroute and Airways (A/B/C)				
ILS Approach coupled (A/B/C)				
ILS Approach hand flown (A/B/C)				
RNAV approach LPV coupled (A/B/C)				
RNAV approach BARO/VNAV coupled (A/B/C)				
RNAV approach LNAV coupled (A/B/C)				
VOR approach coupled (A/B/C)				
HEDA (If Req. A/B/C)				
OSAP (If Req. A/B/C)				
Missed approach with GA mode (A/B/C)				
Missed-approach hand flown (A/B/C)				
TCAS II Procedures (A/B/C)				
7. IFR Abnormal Procedures	Sim 1	Sim 2	Add.	Add.
Navaid failure during approach (B)				
Hydraulic system #2 failure (A)				
Fuel System Failure (C)				
IFR Autorotation with cloud-base 500' (C)				
Engine Chip & Fail Fixed (C)				
Engine failure during coupled non precision approach (A/B/C)				
Pitot Static failure (A)				
Auto pilot failures (A/B/C)				
Pilot Incapacitation (A/B/C)				
Emergency Descent (A/B/C)				
8. Post Flight Procedures	Sim 1	Sim 2	Add.	Add.
After landing procedures (A/B/C)				
Parking and securing (A/B/C)				



	Sim 1	Sim 2	Add 1	Add 2	Add 3	Totals
Date:						
Tail or Sim #:						
Inst Initial:						
Stud Initial:						
Times:						
Simulator:						
Aircraft:						
PF:						
PM:						
Day:						
Night:						
IFR:						

Comments / Evaluation	
#1	1
	2
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	4
	5
#2	1
	2
	3
	4
	5
Additional Flights	1
	2
	3
	4
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I certify that the above named pilot has satisfactorily completed all required training in accordance with FAA Regulations and HeliService USA Training syllabus.

Instructor's Signature:  Date: 



For Reference Only

